

Daily Briefing - October 23, 2026

Your Daily Tech & Programming Digest

Friday, October 23, 2026

1000	128261	1173	40
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

Easily download files from the Open Science Framework with Papercheck

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-07-22 📖 3 min 📝 765 words

TWENTY PERCENT STATISTICIAN

> **Summary:** <p>Researchers increasingly use the Open Science Framework (OSF) to share files, such as data and code underlying scientific publications, or presentations and materials for scientific workshops. The OSF is an amazing service that has contributed immensely to a changed ...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/easily-download-files-from-open-science.html

How I use AI in my work

noreply@blogger.com (Daniel Lakens)
2026-10-11 21 min 4292 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Tomorrow we will have a discussion in our lab about how to use AI as a scientist. I have increasingly started to use AI, having moved from Copilot Chat that our university provides last year, to some free credit on GitHub Copilot under their educational program, to Claude Pro integrated in Visual...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/how-i-use-ai-in-my-work.html

Active Lubrication Brakes in Tabletop and Hand-Held Haptic Surgical Drilling Simulators

2026-06-29 1 min 222 words

TRANSACTIONS HAPTICS

Summary: Haptic user interfaces that provide both kinesthetic and vibrotactile feedback have strong potential to enhance a wide range of medical training simulations. Yet, designing such devices remains challenging due to constraints related to integration, workspace, cost, and the need for diverse feedback ...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11585967

Reconciling Reduced the Sense of Ownership and Agency Toward a Virtual Keypad With Enhanced Haptic Experience Using Visually Adjusted Self-Haptics for Key-Input Interaction

📅 2026-07-06 📖 1 min 📝 250 words TRANSACTIONS HAPTICS

Summary: Self-Haptics is a tactile feedback technique where users utilize their own body to generate tactile sensations. This method allows the body to act purely as a prop, making it useful for virtual objects and user interfaces not directly related to the body. However, any method for maintaining the sens...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11595768

VibGrasp: Spatiotemporal Vibration Based Multimodal Haptic Rendering With a Lightweight Exo-Glove for 3D Shape Perception

📅 2026-04-20 📖 1 min 📝 173 words TRANSACTIONS HAPTICS

Summary: Haptic feedback has been employed to enhance shape perception during grasp interactions, improving immersion and task performance in extended reality (XR). Although wearable devices offer precise multi-phalangeal haptic feedback, their bulkiness and complexity often hinder practicality. In addition,...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11488477

Propeller-Based Handheld Force-Feedback Device for Navigation

📅 2026-04-22 📖 1 min 📝 168 words TRANSACTIONS HAPTICS

Summary: This article presents an evaluation of CrazyJoystick, a propeller-based handheld force-feedback device, for spatial navigation tasks in real-world and virtual reality environments. Building upon prior work validating the device's on-demand aerial deployment and directional cue discriminability, we i...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11493624

A Wearable Shoulder Skin-Deformation Device Generating Illusory Forces for Posture Correction

📅 2026-04-22 📖 1 min 📝 208 words TRANSACTIONS HAPTICS

Summary: The hanger reflex is a haptic illusion in which skin-shear deformation elicits a directional force sensation. This study presents a wearable haptic device that uses four pneumatic airbags to apply controlled skin shear to both shoulders for posture guidance. Although previous work has shown that sho...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11489488

Sensitivity at the Edges of the Finger Enables Mixed Reality Force Feedback

📅 2026-04-06 📖 1 min 📝 155 words TRANSACTIONS HAPTICS

Summary: We introduce a new paradigm for rendering high-force sensations by stimulating the edges of the fingerpad rather than the central contact surface. This non-blocking approach preserves natural finger use, enabling simultaneous physical and virtual interactions. To investigate its feasibility, we prop...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11475177

ThermalTexture: Temperature-Integrated Surface Rendering Using Thermal Referral Induced by Texture Slip-Feedback

📅 2026-03-23 📖 1 min 📝 204 words TRANSACTIONS HAPTICS

Summary: Thermal–vibration feedback has been reported to elicit the thermal referral phenomenon, offering new possibilities for enhancing user experience. Yet, little research has explored whether thermal referral occurs in the presence of texture-slip feedback. To address this, we developed a custom haptic ...

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http://ieeexplore.ieee.org/document/11454471

Guest Editorial: World Haptics Conference 2025

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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Front Cover

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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http://ieeexplore.ieee.org/document/11716326

Neural Distinctions between Social and Nonsocial Reward Anticipation: An Examination of Activation Magnitude, Representational Patterns, Individual Differences, and Functional Connectivity

📅 2026-10-19 📖 1 min 📝 235 words COGNITIVE NEUROSCIENCE

Summary: Prevailing theories emphasize a “common neural currency” for social and nonsocial rewards, particularly during reward anticipation. However, direct comparisons of social and nonsocial reward anticipation remain limited, and clinical observations suggest that social reward processing can be selective...

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http://ieeexplore.ieee.org/document/11741003

Exploring Neural Mechanisms Underlying Error-related Impairments in Active Working Memory Suggests an Adaptive Shielding of Contents during Cognitive Control

📅 2026-10-19 📖 1 min 📝 302 words COGNITIVE NEUROSCIENCE

Summary: Goal-directed behavior relies on cognitive flexibility—the ability to rapidly adapt ongoing thoughts and behaviors while preserving task-relevant information. The performance monitoring system optimizes such behavior by detecting and evaluating errors, whereas the working memory (WM) system maintain...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11741004

Slow Oscillations Modulate Overnight Brain Changes in Working Memory Function

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM), the transient storage and manipulation of information, is a cognitive function that can improve with training. Accumulating evidence suggests that sleep, particularly slow oscillations (SOs) that occur during nonrapid eye movement sleep, supports this improvement. Yet, how sleep...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11741002

Visual Perspective Shapes Subjective Experience: Dissociable Parietal Contributions to the Constructive Nature of Memory

📅 2026-10-19 📖 1 min 📝 223 words COGNITIVE NEUROSCIENCE

Summary: Subjective features of memory are often treated as secondary to the objective content of remembered events. However, growing evidence suggests that these features actively shape how memories are constructed, experienced, and used. Rather than treating visual perspective as a peripheral correlate of ...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11740941

Memory Traces Reflect How They Were Last Accessed

📅 2026-10-19 📖 1 min 📝 216 words COGNITIVE NEUROSCIENCE

Summary: Episodic memories are known to change with each act of retrieval. We hypothesize that accessing semantic knowledge in different ways during episodic retrieval—from unique perceptual features to taxonomic and thematic context—shapes how those memories are subsequently represented during recognition. ...

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http://ieeexplore.ieee.org/document/11740999

The Temporal Dynamics of Attentional Allocation during Counterfactual Learning

📅 2026-10-19 📖 1 min 📝 194 words COGNITIVE NEUROSCIENCE

Summary: Counterfactual learning, the ability to learn from what could have happened under different circumstances, is a key cognitive mechanism supporting behavioral adaptation. While its neural and computational underpinnings are increasingly understood, the temporal dynamics of attention toward factual an...

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http://ieeexplore.ieee.org/document/11741005

Theta and High-alpha Phase-locking Dynamics Unveil Face Perception in Ambiguous Stimuli

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Previous studies emphasize phase synchronization as a fundamental mechanism for integrating local features into coherent percepts. We employed a novel paradigm and dynamic graph analysis based on EEG to track neural dynamics associated with perceiving a face (or not) while keeping the stimuli identi...

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http://ieeexplore.ieee.org/document/11741000

Capturing the Attentional Trade-off between Speech Planning and Comprehension

📅 2026-10-19 📖 1 min 📝 255 words COGNITIVE NEUROSCIENCE

Summary: In conversation, future speakers often plan speech simultaneously with comprehension, which means that they must divide attentional resources between these processes. In this EEG study, we used responses to linguistic attention probes (i.e., syllable “BA” presented during spoken sentences) to track ...

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Therapeutic potential of proline-rich peptide-1 (PRP-1) in counteracting amyloid-β-mediated cortical and hippocampal neurodegeneration in a rat model

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Harutyun Stepanyan, Hamzeh Ghorbani, Mikhail Poghosyan, Margarita Danielyan, Arsen Minasyan, Arsen Hunanyan, Naira Hunanyan

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006615?dgcid=rss_sd_all

Decoding spatial attention in the cocktail party problem using whole-head high-density fNIRS

📖 1 min 📝 47 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Sudan Duwadi, De'Ja Rogers, Alex D. Boyd, Laura B. Carlton, Yiwen Zhang, Anna Kawai Gaona, Aneesa Diya Pathiyaparambath, Ravin Chaudhury, Grace Magee, Bernhard B Zimmermann, Walker J O'Brien, Alexander...

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https://www.sciencedirect.com/science/article/pii/S1053811926006075?dgcid=rss_sd_all

Subjective emotional ratings dissociate brain activation and network patterns during counterfactual outcome processing

📖 1 min 📝 24 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Tingting Zhang, Denghui Bai, Xiaotong Wei, Lei Zhuo, Qiuzhu Zhang, Junjun Zhang, Zhenlan Jin, Ling Li

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600618X?dgcid=rss_sd_all

From Arm to Leg: A Shared Primitive Framework for Solving Inverse Kinematics in Human Movement

👤 Koren, Y., Flash, T. 📅 2026-10-23 📖 1 min 📝 180 words BIORXIV NEUROSCIENCE

Summary: In motor control, the idea of a modular control architecture has been repeatedly discussed. This framework has been widely applied to kinematic and kinetic data, primarily to examine how the central nervous system (CNS) controls multiple elements simultaneously and how it handles redundant or excess...

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https://www.biorxiv.org/content/10.64898/2026.10.22.760905v1?rss=1

Physiological learning involves intrasynaptic reorganization of GluA1-containing AMPA receptors

Jevtic, M., Hollergschwandtner, E., Kartono, B., Sugiyama, Y., Strnadova, L., Ramesh, R., Fukazawa, Y., Itakura, M., Shigemoto, R.

2026-10-23 1 min 245 words BIORXIV NEUROSCIENCE

Summary: Learning and memory formation depend on persistent modifications of excitatory synaptic transmission in the hippocampus through changes in AMPA receptor (AMPAR) content and properties at potentiated synapses. However, directly demonstrating such changes after physiological, behavior-driven learning ...

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https://www.biorxiv.org/content/10.64898/2026.10.22.760928v1?rss=1

Learning distributed representations with complementary components - A generative approximation framework for episodic memory encoding

Liu, H., Mattar, M., Michelmann, S.

2026-10-23 1 min 192 words BIORXIV NEUROSCIENCE

Summary: Episodic memory must preserve detail while storing memories efficiently. We propose a framework of memory encoding in which generative neural network models approximate input through their frozen weights, yielding starting points from which the same weights can reinstate the input. This removes the ...

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https://www.biorxiv.org/content/10.64898/2026.10.18.759853v1?rss=1

The anger-insula-body link in heart arrhythmias

👤 Mazzola, V., Martins-Dominati, E., Bondolfi, G., Arciero, G.

📅 2026-10-23 📖 1 min 📝 242 words

BIORXIV NEUROSCIENCE

Summary: Anger is a well-known risk factor in cardiovascular diseases (CVDs) since it triggers a sympathetic hyperactivity of peripheral autonomic nervous system (ANS). As our previous findings showed, facing an angry situation instead of a joyful one elicits in healthy participants an affective engagement t...

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https://www.biorxiv.org/content/10.64898/2026.10.19.760035v1?rss=1

Generalizable neural decoding for clinically scalable brain-computer interfaces

📅 2026-10-23 📖 1 min 📝 0 words

NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s44385-026-00119-0

Central sensitization-related measures and peripheral treatment outcomes in patients with chronic pain: a systematic review and meta-analysis

📅 2026-10-23 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-75274-w

A deep learning framework for automated cervical spinal cord vertebral labeling and morphometric extraction from brain MRI with large-scale multicenter validation

📅 2026-10-23 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41746-026-03403-w

Orally bioavailable CB2 receptor agonists attenuate neuroinflammation and stabilize the blood–brain barrier

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🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-77296-w

Effects of a cognitive training program on cancer-related cognitive impairment in breast cancer patients in active treatment: a randomized controlled trial

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🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76536-3

The role of long-term memory, flexible templates and learning in attentional guidance

📅 2026-10-23 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s44159-026-00626-2

Role of CB1 receptors in the gastrointestinal tract after microbiota depletion and natural recolonization

📅 2026-10-23 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-77334-7

Privileged representational axes in biological and artificial neural networks

📅 2026-10-23 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41562-026-02515-3

Unveiling the functional specialisation of human white matter circuits with naturalistic stimuli

📅 2026-10-23 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78839-5

Relationship between X chromosome mosaicism, neuroanatomy and cognitive performance in females

📅 2026-10-23 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11171-2

Early lineage divergence segregates sensory and motor-associative identities in the thalamus

👤 Guillermina López-Bendito 📅 2026-10-20 📖 1 min 📝 35 words NATURE NEUROSCIENCE

> **Summary:** Nature Neuroscience, Published online: 20 October 2026; doi:10.1038/s41593-026-02485-9 Sensory and motor-associative identities in the thalamus arise from two early-diverging lineages, with sensory neurons progressing through sha...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02485-9

Dysregulated cholesterol trafficking through ABCA1 drives neurodegeneration in Parkinson’s disease

👤 Lei Cao 📅 2026-10-20 📖 1 min 📝 39 words NATURE NEUROSCIENCE

> **Summary:** Nature Neuroscience, Published online: 20 October 2026; doi:10.1038/s41593-026-02487-7 This study identifies ABCA1 as a driver of dysregulated cholesterol trafficking in Parkinson’s disease, showing that reducing ABCA1 in astrocy...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02487-7

Sustained antidepressant-like effects of psychedelics through 5-HT_{2A}R–β-arrestin signaling

Sheng Wang 2026-10-23 1 min 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 23 October 2026; doi:10.1038/s41593-026-02497-5 This study shows that the sustained antidepressant-like effect of psychedelics is driven by 5-HT2A receptor–β-arrestin signaling. Repeated low...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02497-5

Identification of candidate markers for the isolation of neuron-specific extracellular vesicles

David R. Walt 2026-10-23 1 min 41 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 23 October 2026; doi:10.1038/s41593-026-02466-y The authors developed methods to identify extracellular vesicles (EVs) in the blood that come from the brain by capturing EV surface proteins....

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02466-y

Neuropharmacology of language: ERP evidence for catecholaminergic modulation of syntactic processing

👤 Yingying TanAshley G. LewisIris C. SchmitsRobert-Jan VerkesRoshan CoolsPeter Hagoortahttps://ror.org/01bn89z48Shanghai International Studies University, Institute of Language Sciences, Shanghai 201620, Chinabhttps://ror.org/00671me87Max Planck Institute for Psycholinguistics, Neurobiology of Language Department, Nijmegen 6525 XD, the Netherlandschttps://ror.org/03q8dnn23City University of Hong Kong, Department of Linguistics and Translation, Kowloon, Hong Kong Special Administrative Region, Chinadhttps://ror.org/016xsfp80Radboud University, Donders Institute for Brain, Cognition and Behaviour, Center for Cognitive Neuroimaging, Nijmegen 6525 EN, the Netherlandsehttps://ror.org/05wg1m734Radboud University Medical Center, Department of Psychiatry, Nijmegen 6525 GA, the Netherlands

📅 2026-10-22 📖 1 min 📝 42 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceHow the brain’s neuromodulatory systems influence language, a defining human capacity, remains poorly understood. Here we show that catecholaminergic neurotransmission—primarily involving dopamine ...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2606806123?af=R

Safety of lecanemab in real-world patients with Alzheimer’s disease: a prospective, observational case series supported by FAERS signal analysis

Erhao Bao 2026-10-02 1 min 354 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveLecanemab, a humanized immunoglobulin monoclonal antibody, has been approved for the treatment of Alzheimer’s disease (AD). Although clinical trials have demonstrated its favorable safety profile, real-world studies are crucial for identifying rare adverse events (AEs) and evaluating post-m...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1977226

EEG emotion-recognition datasets for neuroarchitecture: an appraisal of openness, AI-readiness and ontology maturity

Han Jong Jun 2026-10-20 1 min 341 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionNeuroarchitecture studies how designed space shapes emotion, often with electroencephalography (EEG), and computational models that decode emotion from EEG must be trained on labelled recordings. Yet none of the widely shared EEG emotion datasets was recorded while people moved through d...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1971441

A systematic review of machine learning approaches for MRI-based brain tumor analysis

Almira Melsovna Kustubayeva
2026-10-12 1 min 213 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Machine learning is now widely applied to magnetic resonance imaging-based brain tumor analysis, with task-specific performance metrics: the Dice similarity coefficient (DSC) quantifies segmentation overlap, while the area under the curve (AUC) and accuracy measure classification performance. A PRIS...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1937002

Are cortical microcircuits optimized for information flux?—A simulation-based reverse engineering study

Patrick Krauss
2026-10-20 1 min 271 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionA sufficiently large information flux in recurrent neural networks, quantified by the mutual information between successive network states, has been proposed as a marker of rich information-processing capabilities. This raises the question of whether biological neural networks, such as c...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1886601

Artificial intelligence-generated and algorithmically personalized music interventions and their effects on brain function, behavior, and health: a critical review

Jingting Jin 2026-10-23 1 min 310 words FRONTIERS HUMAN NEUROSCIENCE

Summary: This critical review synthesizes empirical studies of algorithmically personalized and artificial-intelligence-generated music interventions reporting neurophysiological, neuroimaging, or objective behavioral outcomes. Literature was identified through structured searches of PubMed, Scopus, Google S...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1967909

Effect of robotic assistance on upper-limb sensorimotor recovery: a systematic review and meta-analysis

Sivakumar Balasubramanian 2026-10-20 1 min 359 words FRONTIERS NEUROROBOTICS

Summary: IntroductionRobot-assisted upper-limb neurorehabilitation enables several beneficial therapy components, including repetitive task-specific practice, high training intensity, real-time feedback, gamification, and continuous physical interaction with the patient’s limb. However, conventional comparis...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1860365

Research progress on the mechanism of microglia–astrocyte signal crosstalk in central sensitization of vestibular migraine

Huailiang Zhang
2026-10-23 1 min 127 words

FRONTIERS NEUROSCIENCE

Summary: Vestibular migraine (VM) is a vestibular disorder characterized by recurrent episodes of vertigo, frequently accompanied by headache, nausea, and vomiting, which substantially reduce patients' quality of life. The pathogenesis of VM remains incompletely understood. Current studies mainly focus on io...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1844932

From confidence to caution: conformal gating for cross-session motor imagery brain–computer interfaces

Sixue Xing and Aarthy Nagarajan
2026-10-22 1 min 278 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Erroneous motor imagery brain–computer interface (MI-BCI) commands can cause unintended device actions, yet standard classifiers provide no mechanism to withhold uncertain predictions. We evaluate whether conformal prediction (CP) can serve as a command gate under cross-session EEG shift,...

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https://iopscience.iop.org/article/10.1088/1741-2552/aead26

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261023075728&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261023075728&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Cost-effective long-range measurement of structural displacement using a two-stage method integrating super-resolution reconstruction and sub-pixel registration

Xiaosong Li 2026-10-20 1 min 61 words LOW VISION

Summary: Displacement is an imperative metric for assessing structural performance. However, the accuracy of structural displacement measurement is constrained by the physical resolution of the camera, particularly in long-range monitoring scenarios. To address these constraints, a cost-effective long-range ...

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Physics-aware restoration and calibrated soft demodulation for turbulence-degraded OAM mode recognition

Jun Zhu 2026-10-20 1 min 58 words LOW VISION

Summary: Turbulence-degraded orbital angular momentum (OAM) mode recognition is critical for free-space optical (FSO) communication but is limited by atmospheric distortion. We propose a cascaded receiver integrating a vision transformer-based Uformer-OAM intensity-restoration network, an SE-ResNet-18 neural...

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Image-free ghost edge detection at ultra-low sampling rates via a Vision Mamba network

👤 Rui Wang 📅 2026-10-20 📖 1 min 📝 66 words LOW VISION

Summary: Edge detection is essential for extracting structural boundaries in visual scenes, but conventional deep-learning-based methods remain sensitive to degradations in the input images and involve redundant full-image acquisition. Image-free ghost edge detection provides an efficient alternative by dire...

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Ex vivo evaluation of corneal permeation and irritation potential associated with proparacaine hydrochloride misuse

👤 Yolima Baena 📅 2026-10-20 📖 1 min 📝 35 words LOW VISION

Summary: CONCLUSION: Proparacaine demonstrates dose-dependent permeation and histopathological changes, emphasizing the risks associated with misuse. The ex vivo model provides relevant and ethically aligned evidence to support ocular hazard assessment through the characterizing of histopathological changes.

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Shared variance across FrACT visual tasks: Robust positive associations but inconclusive latent structure

👤 Luca Battaglini 📅 2026-10-21 📖 1 min 📝 68 words LOW VISION

Summary: A long-standing question in vision science is whether individual differences in visual performance reflect a general visual ability or a set of largely task-specific processes. Previous studies have often reported weak associations across heterogeneous perceptual tasks, challenging the idea of a com...

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New challenges in horticultural IPM: artificial intelligence can enhance the use of biocontrol agents

👤 Andrea Lovazzano 📅 2026-10-22 📖 1 min 📝 59 words LOW VISION

Summary: Modern horticulture faces the dual imperatives of increasing global food production while mitigating the severe environmental and health impacts of conventional chemical pesticides. Biological control agents (BCAs), which utilize beneficial microorganisms, macroorganisms, semiochemical and botanical...

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Bridging the performance-efficiency gap: a resource-efficient hybrid DL-ML framework for jackfruit disease detection

👤 Hezerul Bin Abdul Karim 📅 2026-10-22 📖 1 min 📝 62 words LOW VISION

Summary: Complex environment variability and limited computational resources in agricultural settings create challenges in automated detection in tropical regions, where jackfruit diseases pose major threats to harvest productivity and food security. Conventional machine learning models often fail to general...

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Atomic Defect-Mediated Charge Trapping Enables Multimodal Plasticity in van der Waals Heterostructures for In-Sensor Neuromorphic Vision

👤 Zhiqun Lin 📅 2026-10-22 📖 1 min 📝 60 words LOW VISION

Summary: Neuromorphic hardware that unifies sensing, memory, and learning at the device level remains challenging due to limited optoelectronic comodulation and rigid architectures. Here, we report a flexible, van der Waals (vdW)-integrated MoS2/hexagonal boron nitride (h-BN)/graphene memtransistor in which ...

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Low-Dimensional Chiral Perovskites for Polarization-Sensitive Photodetection

Zhiyong Fan 2026-10-22 1 min 57 words LOW VISION

Summary: Emerging low-dimensional chiral perovskites have recently demonstrated significant application potential in circularly polarized light detection, spintronics, and quantum information technology due to their unique chiroptical activity and spin-selective properties. This review systematically summari...

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Visual context influences lateral balance while walking on winding paths

Jonathan B Dingwell 2026-10-22 1 min 28 words LOW VISION

Summary: BACKGROUND: Navigating daily environments necessitates adapting to ever-changing conditions while maintaining stability. This requires integrating sensory feedback, primarily from vision, to maintain balance and maneuverability as we walk.

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

👤 Béatrice Berçot 📅 2026-09-04 📖 1 min 📝 60 words BRAILLE

> **Summary:** CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan 📅 2026-09-09 📖 1 min 📝 57 words BRAILLE

> **Summary:** Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

> **Summary:** CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Home Transcranial Direct Current Stimulation With Gamified Cognitive Control Training for Depression: A Randomized Clinical Trial

👤 Mor Nahum 📅 2026-10-19 📖 1 min 📝 45 words TDCS TACS TRNS

> **Summary:** CONCLUSIONS AND RELEVANCE: This RCT of home-based tDCS combined with CCT in adults with MDD was feasible and safe but did not demonstrate superior efficacy over the control condition. Future studies should refine the mechanistic alignment between stimulation and cognitive interventions to optimize t...

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Comparative chloroplast genomics of Ajania sericea and sympatric Asteraceae species identifies shared candidate molecular features associated with alpine adaptation

👤 Xi Chen 📅 2026-10-20 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study showed that the chloroplast genomes of four sympatric Asteraceae species were highly conserved in overall structure, but still exhibited certain differences in gene loss, IR boundary variation, repeat sequence composition, and local hypervariable regions. These features provi...

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Brain Stimulation as a Treatment for Apathy: A Scoping Review of Effects and Safety

👤 Lisa M McTeague 📅 2026-10-20 📖 1 min 📝 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: Investigations into brain stimulation as a treatment for apathy show some promise. Conclusions were limited by small sample sizes, heterogeneous study designs, and variable outcome measures. Randomized trials with common validated apathy instruments, standardized stimulation parameters,...

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Effect of Different Cortical Targets of Transcranial Magnetic Stimulation on Freezing of Gait in Parkinsonism: A Systematic Review

Aliaa Elabd
2026-10-20 1 min 68 words TDCS TACS TRNS

Summary: BACKGROUND AND PURPOSE: Freezing of gait (FOG) is a disabling and episodic gait disturbance in patients with Parkinson's, with varying responses to noninvasive brain stimulation. Transcranial magnetic stimulation (TMS) has been explored as a therapeutic option; however, the optimal cortical target f...

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Sleep Spindles and Slow Oscillations as Potential Biomarkers and Therapeutic Targets in Mild-to-Moderate Alzheimer's Disease

👤 Thien Thanh Dang-Vu 📅 2026-10-20 📖 1 min 📝 35 words TDCS TACS TRNS

Summary: PURPOSE OF REVIEW: This review critically evaluates evidence on sleep spindles (SP), slow oscillations (SOs), and their coordination in mild-to-moderate Alzheimer's disease (AD), focusing on their relationships with cognition, AD biomarkers, and their therapeutic potential.

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

👤 Xin Zhou 📅 2026-10-20 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiple Sessions of Transcranial Direct Current Stimulation Combined With Squat Training Improve Lower-Limb Maximum Strength in Healthy Adult Men

👤 Fengxue Qi
📅 2026-10-22 📖 1 min 📝 79 words

TDCS TACS TRNS

Summary: Yan, X, Zuo, Z, Nitsche, MA, Zhang, X, and Qi, F. Multiple sessions of transcranial direct current stimulation combined with squat training improve lower-limb maximum strength in healthy adult men. J Strength Cond Res XX(X): 000-000, 2026-This study aimed to investigate the effects of transcranial d...

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Time-aware frame-aligned diffusion-inspired model for late dynamic PET frame prediction from early [¹¹C] raclopride

👤 Jae Sung Lee
📅 2026-10-22 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: OBJECTIVE: To reduce the burden of dynamic PET acquisition while preserving quantitative analysis, we propose a time-aware frame-aligned diffusion-inspired residual denoising framework for predicting late dynamic PET frames from a short early acquisition window (10 min in this study) of [¹¹C] raclop...

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Effects of transcranial direct current stimulation on patient-reported outcomes in chronic low back pain: a systematic review and three-level meta-analysis

👤 Peng Chen 📅 2026-10-23 📖 1 min 📝 34 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence suggests that tDCS may reduce pain and produce a small improvement in disability in individuals with CLBP. Given the low-to-moderate certainty of the available evidence, these findings should be interpreted cautiously.

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang 📅 2026-10-19 📖 1 min 📝 64 words FNIRS

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Neural dysfunction underlying divergent thinking and its shared mechanisms with protracted withdrawal symptoms in heroin use disorder

👤 Haijun Duan
📅 2026-10-19 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSIONS: These findings reveal a sustained compensatory pathological mechanism and an imbalance in neural resource allocation during complex cognition in HUD individuals. The identified signatures of cortical activity by fNIRS offer reliable biomarkers for the rapid detection of both higher-orde...

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Functional near-infrared spectroscopy as a tool for monitoring treatment-related neurofunctional changes in posttraumatic stress disorder and trauma-related symptomatology: a narrative review

👤 Maria Cotelli
📅 2026-10-20 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Current evidence suggests that fNIRS represents a promising and clinically feasible approach for monitoring treatment-related neurofunctional changes in PTSD and trauma-related symptomatology, particularly during non-pharmacological interventions targeting prefrontal regulatory processe...

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Transcutaneous vagus nerve stimulation during semantic verbal fluency tasks: a high-density diffuse optical tomography study

👤 Li Su 📅 2026-10-20 📖 1 min 📝 55 words FNIRS

Summary: INTRODUCTION: Transcutaneous auricular vagus nerve stimulation (taVNS) is a promising adjunct for post-stroke language recovery. There are no established biomarkers for the neuromodulatory effects of taVNS. High-density diffuse optical tomography (HD-DOT) offers a non-invasive marker of task-related...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

👤 Xuqun You
📅 2026-10-22 📖 1 min 📝 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi
📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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FroHyNet: frozen temporal priors and trainable hypergraph fusion for brain time-series classification

Jing Zhang 2026-10-23 1 min 51 words FNIRS

Summary: BACKGROUND: Functional brain time-series classification is constrained by small clinical cohorts. Handcrafted features are sample-efficient but discard temporal detail; end-to-end networks must estimate many temporal parameters from limited labels; and frozen time-series foundation models reduce tha...

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An attention-enhanced hybrid CNN-BiLSTM framework for multiclass epileptic seizure recognition using EEG signals

Ahmed H Almutairi 2026-10-22 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Electroencephalogram (EEG) signal classification plays a pivotal role in brain-computer interfaces (BCIs) and the diagnosis of neurological disorders. However, the non-stationary, non-linear, and low signal-to-noise ratio characteristics of EEG signals make accurate feature extraction ...

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Realizing Ultrahigh Polarization and Excellent Thermal Stability in Piezoceramics via Defect-Stress Engineering

Yuhui Huang 2026-10-22 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Large polarization is critical for improving the piezoelectric and actuation performance of ferroelectric materials. As a promising lead-free candidate, BiFeO(3)-BaTiO(3) inherits the excellent polarization capability of BiFeO(3). However, the actual polarization achieved in its ceramics remains far...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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Optimizing learning for control of human-machine interfaces

Preeya Khanna 2026-10-22 1 min 42 words BRAIN COMPUTER INTERFACE

Summary: Human-machine interfaces, such as augmented reality devices, hand prostheses, and brain-computer interfaces, are increasingly being used in neurorehabilitation. In this issue of Neuron, Vassiliadis et al.[1] examine the interaction between closed-loop continuous reinforcement and sensory feedback in l...

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A GABAergic preoptic-raphe circuit mediates paradoxical coupling of anxiety and social preference after comparative social deficit

Hong Lian 2026-10-22 1 min 66 words BRAIN COMPUTER INTERFACE

Summary: Social disconnection is a major risk factor for anxiety disorders, the most prevalent mental illnesses worldwide, yet its neural substrates remain unclear. Here, we establish a comparative social deficit (CSD) model in adult male mice that induces a paradoxical phenotype of increased anxiety-like be...

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ACF-Net: Design of an adaptive cross-modal fusion network for cross-subject wearable EEG-EMG gesture recognition

TongLiang Yang 2026-10-22 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: Cross-subject gesture recognition in wearable brain-computer interfaces is constrained by four structural deficiencies: symmetric cross-modal attention discarding cortical precedence over muscular execution, priors confined to post hoc analysis, invariance anchored on unimodal features, and adaptati...

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Towards semantic brain-computer interfacing: The graded inventory of semantic triggers (GIST) paradigm

Ian Daly 2026-10-22 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Semantic Brain-Computer Interfaces (sBCIs) aim to decode conceptual information directly from neural activity, enabling more efficient and natural communication for individuals with severe motor impairments. However, existing approaches are often confounded by stimulus-driven perceptual processes an...

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Preclinical models of Wilson's disease: From pathogenesis to clinical translation

Zhi-Ying Wu 2026-10-22 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Wilson's disease (WD) is an autosomal recessive disorder of copper homeostasis caused by pathogenic ATP7B variants, leading predominantly to hepatic and neurological injury. Extensive heterogeneity in genotype, age at onset, clinical presentation, and treatment response complicates both diagnosis an...

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Discussion on the closed-loop regulation strategy for acupuncture "Deqi" based on brain-computer interface and its peripheral-central mechanisms

👤 Rong Luo 📅 2026-10-22 📖 1 min 📝 72 words BRAIN COMPUTER INTERFACE

Summary: It is held that "Deqi" is the key to the therapeutic effect of acupuncture, but its evaluation has long relied on subjective feelings and lacks objective biological indicators. In the present article, we systematically review the current research status of brain-computer interface (BCI) technology b...

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RLAgent: retrieval-augmented LLM agents for code-free RNA region-ligand interaction modeling

👤 Zhe Liu 📅 2026-10-23 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: Accurately modeling RNA-ligand interactions is essential for understanding RNA function and for designing RNA-targeted therapeutics. Existing computational approaches mainly focus on predicting either full-length RNA structures or discrete binding sites, lacking explicit modeling of RNA regions that...

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Advancing neuropsychology training: Insights and advice from a qualitative analysis in 102 brain-behavior experts.

📅 2026-07-30 📖 1 min 📝 266 words NEUROPSYCHOLOGY

Summary: Objective: Neuropsychologists possess knowledge that could benefit trainees as they navigate important career decisions within their unique training pathways. This wisdom is typically passed down in an individual mentorship model wherein supervisors and advisors relay advice directly. While this mod...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001116

Sex-specific differences in amygdala resting-state functional connectivity with trait anxiety.

📅 2026-06-04 📖 1 min 📝 267 words NEUROPSYCHOLOGY

Summary: Objective: Anxiety disorders are the most common mental health conditions in the world, disproportionately affecting females and associated with significant morbidity. Neuroimaging can help us to understand the underpinnings of anxiety behaviors. The amygdala and its subnuclei have been strongly imp...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001100

Effects of simulated hearing loss on cognitive task performance: A scoping review.

📅 2026-07-30 📖 1 min 📝 235 words NEUROPSYCHOLOGY

Summary: Objective: Hearing loss (HL) and cognitive decline often co-occur in older adults. Many cognitive tests are administered using spoken instructions or stimuli, potentially biasing test results. The current scoping review aimed to provide an overview of studies using simulated HL in cognitively health...

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http://doi.org/10.1037/neu0001111

Normative cognitive test performance and typical error patterns among American Indian older adults: The Strong Heart Study.

📅 2026-06-15 📖 1 min 📝 245 words NEUROPSYCHOLOGY

Summary: Objective: The Strong Heart Study, a large cohort of American Indians from three U.S. regions, collected standardized neuropsychological assessments from participants aged 65–95 years at two visits (2010–2019), with prior documentation of lower overall score distributions and higher prevalence of co...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001093

The unity and diversity of executive functions across adulthood in a diverse Brazilian sample with varying educational attainment.

📅 2026-04-16 📖 1 min 📝 261 words NEUROPSYCHOLOGY

Summary: Objective: Executive functions (EFs) are a set of cognitive abilities associated with many social and clinical outcomes that are believed to be sensitive to culture, age, and educational attainment. Yet, this was determined using EF theoretical models and tests developed for highly educated young ad...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001085

Everyday executive functions in children with neurofibromatosis type 1: Data from nine institutions.

📅 2026-04-16 📖 1 min 📝 278 words NEUROPSYCHOLOGY

Summary: Objective: Executive function (EF) difficulties are prevalent in neurogenetic syndromes such as Neurofibromatosis Type 1 (NF1), the most common autosomal dominant condition associated with nervous system tumors. However, age-related differences in EF problems in children with NF1 remain poorly under...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001087

Integrity in six key brain regions predicts numeric working memory performance in 30,000 middle-aged and older adults: A UK Biobank magnetic resonance imaging study.

📅 2026-06-18 📖 1 min 📝 262 words NEUROPSYCHOLOGY

Summary: Objective: This study investigated the neuroanatomical correlates of working memory using the UK Biobank Numeric Memory Test (NMT; a visual backward digit-span task) in a large sample of middle-aged and older adults. Volumes of brain regions previously implicated in digit-span tasks were hypothesize...

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Heads up time for package devs

👤 /u/Gullible_Pay_6686 📅 2026-10-23 📖 1 min 📝 84 words REDDIT PYTHON

Summary: <!-- SC_OFF --> Don't you think companies or developers that manage python libraries should be given extra time before a new version release to make things compatible and fix new bugs? Like a month or two where they get full info about new features but before the version is ...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xd6oxc/heads_up_time_for_package_devs/

Show HN: Making a smart TV dumb and then smart

👤 animalhot 📅 2026-10-23 📖 1 min 📝 120 words HACKER NEWS

> **Summary:** A recent (unwanted) webOS update that got installed on TV prompted me to start going down the rabbit hole of writing new TV software. After a couple of weekends and massive help from 4 different models, finally managed to get v1 working. My daughter enjoys navigating TV through speech while I g...

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https://github.com/anubhavmalhotra/r2d2-tv

Remove pgrust: the entry is cheating

👤 f311a 📅 2026-10-23 📖 1 min 📝 13 words HACKER NEWS

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The Rise and Fall of Jev

👤 flail 📅 2026-10-23 📖 1 min 📝 13 words HACKER NEWS

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Job Loss and Cortisone Mania

👤 speckx 📅 2026-10-23 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://tilschuenemann.de/job-loss-cortisone-mania Comments URL: https://news.ycombinator.com/item?id=50206840 Points: 6 # Comments: 0

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Dutch government had hacked into Jared Kushner's phone

👤 teekert 📅 2026-10-23 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://twitter.com/jaredkushner/status/2113376217415958767 Comments URL: https://news.ycombinator.com/item?id=50206914 Points: 53 <...

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Is this the new cyber normal?

👤 drewsberry 📅 2026-10-23 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://drew.silcock.dev/blog/is-this-the-new-cyber-normal/ Comments URL: https://news.ycombinator.com/item?id=50207115 Points: 14 <...

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https://drew.silcock.dev/blog/is-this-the-new-cyber-normal/

Dogmatic Bayesianism Disorder

👤 noreply@blogger.com (Daniel Lakens)
📅 2025-12-06 📖 4 min 📝 910 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Note: This humorously intended sarcastic blog post directly mimics classifications of psychological disorders in the Diagnostic and Statistical Manual of the American Psychological Association, but Dogmatic Bayesianism is not in the DSM-5 as an actual psychologic...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/12/dogmatic-bayesianism-disorder.html

On the reliability and reproducibility of qualitative research

👤 noreply@blogger.com (Daniel Lakens)
📅 2026-02-08 📖 14 min 📝 2928 words

TWENTY PERCENT STATISTICIAN

Summary: <p class="MsoNormal">With my collaborators, I am increasingly performing qualitative research. I find qualitative research projects a useful way to improve my understanding of behaviors that I want to explore with other methods in the future. For example, some years ago I performe...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/02/on-reliability-and-reproducibility-of.html

New Papers: Optimal Filter Settings for ERP Research

👤 Steve Luck 📅 2024-02-04 📖 2 min 📝 569 words ERP BOOT CAMP

> **Summary:** <p class="">Zhang, G., Garrett, D. R., & Luck, S. J. (in press). Optimal filters for ERP research I: A general approach for selecting filter settings. Psychophysiology. https://doi.org/10.1111/psyp.14531 [<a href="https://www...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/2/4/optimal-filters

Q&A with Dr. Richard Carson, Professor of Biomedical Engineering and Radiology & Biomedical Imaging, Yale University and Yale School of Medicine

👤 Adriel Carridice 📅 2025-11-04 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/podcasts/qa-with-dr-richard-carson-professor-of-biomedical-engineering-and-radiology-biomedical-imaging-yale-university-and-yale-school-of-medicine/

Working Through Representation: Malleable Workspaces for Human-Agent Knowledge Work

👤 Nan Chen, Long Ling, Luna K. Qiu, Zilong Wang, Yuqing Yang

📅 2026-10-23 📖 1 min

📝 153 words

ARXIV CS HC

Summary: arXiv:2610.30893v1 Announce Type: new Abstract: Generative AI supports knowledge work, yet conversational histories may not capture the evolving organization that guides subsequent action. We introduce the Malleable Workspace, a conceptual model of how humans and agents form and revise a Work Repre...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30893

ABRA: Agent-Bound UI for Human-Agent Communication beyond Chat with Reusable Views

👤 Seon Gyeom Kim, Hannah Kim

📅 2026-10-23 📖 1 min 📝 123 words

ARXIV CS HC

Summary: arXiv:2610.30780v1 Announce Type: new Abstract: Agents can create rich representations, yet these representations are rarely used as a medium for expressing user intent. We present Agent-Bound UI, an interaction model that separates reusable interaction structures from task-specific meanings. Agent...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30780

Abstract Priorities, Concrete Rewards: How Stated Preferences Relate to What People Reward in AI Responses

Zihan Chen, Lei Nico Zheng, Di Zhu

2026-10-23 1 min 277 words ARXIV CS HC

Summary: arXiv:2610.30763v1 Announce Type: new Abstract: AI assistants built on large language models (LLMs) are trained on human feedback and increasingly personalized to users' stated preferences. Yet how closely these preferences reflect what people reward in responses remains unclear. We compare stated ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30763

Do We Need to "Be There" to Say Goodbye? Comparing Participatory Agency and Spatial Presence Across Digital Farewell Formats

Chan In Sio, Alex Mann, Qian Ming Jia, Lik-Hang Lee

2026-10-23 1 min 219 words ARXIV CS HC

Summary: arXiv:2610.30734v1 Announce Type: new Abstract: Digitally mediated attachment is increasingly common, yet the mechanisms of symbolic closure in hybrid physical--digital relationships remain underexplored. We address an XR design question: how different mediated farewell formats relate to perceived ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30734

Big World, Little Me: Comparing a First-Person VR Walkthrough Video and Static Slides for Communicating Height-Related Accessibility Barriers

👤 Chan In Sio, Alex Mann, Qian Ming Jia, Xuecheng Guo, Yan Zhuo, Xinyu Ning, Kaikuo Liao, Lik-Hang Lee

📅 2026-10-23 📖 1 min 📝 204 words ARXIV CS HC

Summary: arXiv:2610.30731v1 Announce Type: new Abstract: People with dwarfism and other forms of short stature may encounter barriers created by fixed-height environments and discriminatory social assumptions. This paper evaluates Big World, Little Me, a 10 min 52 s first-person walkthrough recorded from a ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30731

Impacts of generative AI skill development on academic adaptation and learning outcomes among university students

👤 Lei Fan, Fangxue Liu

📅 2026-10-23 📖 1 min 📝 225 words ARXIV CS HC

Summary: arXiv:2610.30599v1 Announce Type: new Abstract: Understanding how university students perceive the development of their generative artificial intelligence (GenAI) skills over time can clarify how GenAI use relates to learning and academic outcomes. This study examined perceived skill-development tr...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30599

Beyond Better Ideas: How Human-AI Collaboration Shapes Scientific Ideas and Human Perceptions

👤 Qian Chen, Saatvik Pradhan, Janice Ahn, Berk Atil, Angkai Li, Shijie Lu, Nobin Sarwar, Michael Ulis, Ruyuan Wan, Wenpeng Yin

📅 2026-10-23 📖 1 min 📝 214 words ARXIV CS HC

Summary: arXiv:2610.30568v1 Announce Type: new Abstract: As AI increasingly becomes a collaborator in scientific research, its impact may extend beyond improving ideas to reshaping how they are evaluated, attributed, and perceived by humans. Yet existing research has largely examined these dimensions in iso...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30568

Atlas: Making Sense of Long Horizon Agent Trajectories

👤 Sangho Suh, Joseph Chee Chang, Pao Siangliulue, Amy X. Zhang, Matt Latzke, Jonathan Bragg, Doug Downey, Daniel S. Weld

📅 2026-10-23 📖 1 min 📝 155 words ARXIV CS HC

Summary: arXiv:2610.30548v1 Announce Type: new Abstract: AI agents can now search literature, generate hypotheses, analyze datasets, and run experiments at a scale difficult for researchers to match. Yet scaling scientific execution does not automatically scale researchers' ability to make sense of what age...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30548

Exploring spatial configuration and group familiarity when co-watching 360-degree stories in social virtual reality

Portia Wang, Monique Santoso, Yujie Tao, Jeremy Bailenson

2026-10-23 1 min 188 words

ARXIV CS HC

Summary: arXiv:2610.30353v1 Announce Type: new Abstract: While traditional cinematic viewing is often characterized as a social experience, engaging with immersive stories is predominantly researched as an individual activity. This mismatch becomes increasingly problematic in social virtual reality (VR), wh...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30353

Measuring undergraduate students' reliance on generative AI during problem-solving: Scale development and validation

Chenyu Hou, Gaoxia Zhu, Vidya Sudarshan, Fun Siong Lim, Yew Soon Ong

2026-10-23 1 min 209 words

ARXIV CS HC

Summary: arXiv:2610.30195v1 Announce Type: new Abstract: Reliance on AI describes the behavioral patterns of when and how individuals depend on AI suggestions, and appropriate reliance patterns are necessary to achieve effective human-AI collaboration. Traditional measures often link reliance to decision-ma...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30195

A Formalizable Proof of the No-Supervenience Theorem: A Diagonal Limitation on the Viability of Physicalist Theories of Consciousness

Cathy M Reason
2026-10-23 1 min 90 words ARXIV QBIO NC

Summary: arXiv:2307.10178v3 Announce Type: replace Abstract: The no-supervenience theorem limits the capacity of physicalist theories to provide a comprehensive account of human consciousness. The proof of the theorem is difficult to formalize because it relies on both alethic and epistemic notions of possi...

🔗 **Read full article:**
https://arxiv.org/abs/2307.10178

Dynamics of random recurrent neural networks: colored noise and sublinear activations

German Mato, Facundo Rigatuso, Gonzalo Torroba
2026-10-23 1 min 188 words

ARXIV QBIO NC

Summary: arXiv:2610.31575v1 Announce Type: new Abstract: We study the dynamics of recurrent neural networks with random couplings, sublinear odd activation functions, and temporally correlated noise. Using large N methods, we determine stationary correlations and distinguish the loss of local linear stabili...

🔗 **Read full article:**
https://arxiv.org/abs/2610.31575

Music Between Two Minds: Individual Signatures, Shared States, and the Emergence of Creative Interaction

👤 Tatsuya Daikoku, Risa Kobayashi, Rina Tabuchi, Yoshia Abe, Yasuo Kuniyoshi

📅 2026-10-23

📖 1 min 📝 223 words ARXIV QBIO NC

Summary: arXiv:2610.31131v1 Announce Type: new Abstract: Some of the most valued forms of human creativity arise between people-in conversation, collaboration, and improvised music-yet how two minds jointly regulate the flow of novelty in real time remains unknown. Here we examined the information dynamics ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.31131

Gate-Based Feature Importance in Simulation-Based Inference for EEG: A Cautionary Tale

👤 Shrivatsa Deshmukh, Sasha Brenner, Thomas R. Kn\"osche, Nico Scherf

📅 2026-10-23 📖 1 min

📝 251 words ARXIV QBIO NC

Summary: arXiv:2610.30364v1 Announce Type: new Abstract: Simulation-based inference (SBI) provides a powerful way to fit mechanistic models to data, but offers little insight into which information in the data shapes the inferred parameter values. It could therefore benefit from robust and inexpensive inter...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30364

The medium is the message: How neurometabolic entanglement generates conscious experience

Terrence W. Deacon, Miguel Garcia-Valdecasas, Michael Jacob

2026-10-23 1 min

246 words

ARXIV QBIO NC

Summary: arXiv:2610.30217v1 Announce Type: new Abstract: Most neuroscientific accounts of consciousness assume that conscious experience is constituted by the information conveyed between neurons. The physical-chemical details of the metabolic and physiological processes that convey this information are cor...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30217

Single-cell RNA sequencing reveals vascular heterogeneity and cell–cell communication remodeling in the glioma blood–tumor barrier

1 min

31 words

BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Zhipeng Zhao, Jiachen Lang, Haiyan Gu, Luodan Wang, Jianchang Luo, Jiawang Lang, Qiongying Yao, Xinyu Qiao, Zida Wang, Ling Yin, Caidi He

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004579?dgcid=rss_sd_all

Intracerebral stimulation inducing emotions in SEEG: a systematic review

📖 1 min 📝 14 words NEUROIMAGE

Summary: Publication date: 1 November 2026 Source: NeuroImage, Volume 341 Author(s): Lucas Arrighi, Fabrice Bartolomei, Francesca Pizzo

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006038?dgcid=rss_sd_all

Localising the Seizure Onset Zone from Routine Interictal and Ictal Scalp EEG Via a Factorised Spatiotemporal Ensemble Deep Learning Framework

👤 Shi, S. 📅 2026-10-22 📖 1 min 📝 369 words BIORXIV NEUROSCIENCE

Summary: Background: Drug-resistant focal epilepsy requires precise localisation of the epileptogenic zone (EZ) for surgical resection. Current practice depends on invasive recordings or spontaneous seizure capture, a costly and risky bottleneck. Interictal scalp EEG can exhibit localising features like spik...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.16.759501v1?rss=1

Frequency-Resolved Connectivity Graphs for EEG-Based Classification of Alzheimer and Frontotemporal Dementia

👤 Radwan, M., yazidi, A., lind, P. G.
📅 2026-10-22 📖 1 min 📝 202 words
BIORXIV NEUROSCIENCE

Summary: In this study, we build EEG based graph model that uses frequency-resolved connectivity features, with Coherence, phase locking value (PLV) and phase lag index (PLI) in five frequency bands as multi-dimensional edge features. The model is used to classify Alzheimer disease (AD) and frontotemporal de...

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https://www.biorxiv.org/content/10.64898/2026.10.16.759776v1?rss=1

Learning to Pursue Goals that Conflict with Prior Experience Relies on Fast Revaluation and Slow Value Updating

👤 Pan, T.-F., Molinaro, G., Collins, A. G. E.
📅 2026-10-22 📖 1 min 📝 238 words
BIORXIV NEUROSCIENCE

Summary: Humans can flexibly pursue novel goals, learning to reach them from goal-dependent teaching signals. However, the breadth of this flexibility and its underlying mechanisms remain unclear. For example, can humans flexibly learn to reach a typically avoided outcome (e.g., +0) while avoiding a typicall...

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MR-Guided Focused Ultrasound Neuromodulation of Nucleus Accumbens to Attenuate Oral Fentanyl Motivation in A Self-Administration Rat Model

Martin, D. A., Eldirdiri, A., Ernst, T., Cunningham, E., Bernard, S., Kamdi, S., Calu, D. J., Chang, L.

2026-10-22 1 min 249 words BIORXIV NEUROSCIENCE

Summary: Introduction: Transcranial low-intensity focused ultrasound (LIFU) is being explored as a non-invasive approach for neuromodulation to treat opioid use disorder (OUD). Methods: We evaluated MR-guided focused ultrasound, with LIFU, targeting nucleus accumbens (MRgLIFU-NAc) bilaterally, using MR-acous...

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Stress, Heartbeats, and Biases: Cardiac phase and acute stress modulate affective and spatial response bias.

von Haugwitz, L., Sadr, R., Wascher, E., Larra, M. F.

2026-10-22 1 min 261 words BIORXIV NEUROSCIENCE

Summary: Threatening situations often require rapid decisions which are particularly susceptible to response bias. The perception of threat also elicits an acute stress reaction, assumed to promote automatic action tendencies via several mechanisms. Cardioafferent signals during systole, for example, increas...

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https://www.biorxiv.org/content/10.64898/2026.10.18.759627v1?rss=1

An amygdala-mediated simulation-to-decision cascade aligns interbrain dynamics to solve social dilemmas

Yina Ma 2026-10-19 1 min 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 19 October 2026; doi:10.1038/s41593-026-02473-z Using simultaneous intracranial recordings from interacting human brains, Ma et al. show that mutual cooperation emerges when amygdala activit...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02473-z

Spatiotemporal remodeling in habenula marks the onset of depression-related phenotypes

Manuel Mameli 2026-10-19 1 min 36 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 19 October 2026; doi:10.1038/s41593-026-02488-6 Molina et al. show in chronically stressed mice that neuronal and transcriptional adaptations in the habenula coincide with the onset of depre...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02488-6

From pose estimation accuracy to motor biomarker reliability: a reliability-centred evaluation framework for markerless movement analysis

Christine Fernandez-Maloigne
2026-10-01 1 min 246 words

FRONTIERS NEUROINFORMATICS

Summary: Markerless pose estimation has become an essential tool for quantitative movement analysis in neuroscience. However, current evaluation frameworks primarily rely on frame-wise spatial accuracy metrics, which provide only limited information about the reliability of the downstream kinematic biomarker...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1790241

Altered interictal alpha-band organization in childhood absence epilepsy

Justyna Paprocka
2026-10-23 1 min 318 words

FRONTIERS NEUROINFORMATICS

Summary: ObjectiveInterictal background EEG in childhood absence epilepsy (CAE) may appear normal. We asked whether drug-naive children with CAE differ from controls in alpha frequency and the regional distribution of phase-lagged synchronization, and how these differences depend on participant-specific surr...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1875377

Short-term synaptic plasticity modulates the outcome of neurodegenerative diseases

👤 Chi Chung Alan Fung 📅 2026-10-23 📖 1 min 📝 264 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAlzheimer's disease (AD) is characterized by progressive neuronal loss and synaptic dysfunction that disrupt memory-encoding neural circuits. Short-term synaptic plasticity (STP), which governs transient activity-dependent changes in synaptic efficacy on millisecond-to-second timescales,...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1923602

Orientation detection for self-calibrating neuromorphic systems

👤 Elisabetta Chicca 📅 2026-10-23 📖 1 min 📝 183 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Bioinspired sensory architectures help build adaptive touch systems with efficient hardware implementations. Orientation detection is a fundamental capability for manipulating objects in real-world scenarios, with diverse applications in robotic manipulation and manufacturing assessment processes. I...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1878557

A thalamus-centered network model of restless legs syndrome: an integrative hypothesis illustrated by three clinical phenotypes

Szabolcs Várbíró 2026-10-23 1 min 231 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Restless legs syndrome (RLS) is a clinically heterogeneous sensorimotor disorder whose pathophysiology reflects interacting biological mechanisms rather than a single causal pathway. Established influences—including brain iron biology, dopaminergic regulation, glutamatergic and GABAergic balance, ad...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1977941

Cortical responses to lipread and text-based audiovisual stimuli in adults with and without dyslexia

Milene Bonte 2026-10-23 1 min 235 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Phonetic recalibration enables listeners to adjust their perception of ambiguous speech sounds based on visual contextual information. In line with a specific difficulty in forming letter-speech sound associations, adults with dyslexia have been found to show no text-based, but intact lipread-based,...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1806964

A systems-level energetic framework for conscious processing: energetic organization as a constraint on large-scale neural integration

Holly Pollard-Wright 2026-10-23 1 min 259 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Conscious processing is associated with coordinated interactions among distributed neural systems, yet the biological constraints that enable and sustain large-scale neural integration remain incompletely understood. Contemporary theories emphasize computational, informational, and network mechanism...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1966072

Editorial: New horizons in stroke management, volume II

Nasser Kashou 2026-10-23 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2023091

Immediate effect of gait training using the hybrid assistive limb in patients in the acute phase of stroke: a prospective pilot study

Yasushi Takagi 2026-10-23 1 min 360 words FRONTIERS NEUROROBOTICS

Summary: IntroductionThe effectiveness of Hybrid Assistive Limb (HAL) gait training in patients in the acute phase of stroke remains unclear. We prospectively assessed the immediate effects of HAL gait training in this patient population.MethodsIn this prospective pilot study, four patients with acute-phase ...

🔗 **Read full article:**
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Comparison of short-term peripheral nerve stimulation, single-target supraorbital nerve pulsed radiofrequency, and dual-target supraorbital nerve–stellate ganglion pulsed radiofrequency for herpes zoster ophthalmicus–associated pain: a retrospective cohort study

Wei Wang 2026-10-23 1 min 326 words FRONTIERS NEUROSCIENCE

Summary: ObjectiveTo compare the clinical outcomes of short-term peripheral nerve stimulation (st-PNS), supraorbital nerve pulsed radiofrequency (SON-PRF), and combined supraorbital nerve and stellate ganglion pulsed radiofrequency (SON + SG-PRF) for herpes zoster ophthalmicus (HZO)-associated pain.MethodsTh...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1976324

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261023022103&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Cost-effective long-range measurement of structural displacement using a two-stage method integrating super-resolution reconstruction and sub-pixel registration

Xiaosong Li 2026-10-20 1 min 61 words LOW VISION

Summary: Displacement is an imperative metric for assessing structural performance. However, the accuracy of structural displacement measurement is constrained by the physical resolution of the camera, particularly in long-range monitoring scenarios. To address these constraints, a cost-effective long-range ...

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Physics-aware restoration and calibrated soft demodulation for turbulence-degraded OAM mode recognition

Jun Zhu 2026-10-20 1 min 58 words LOW VISION

Summary: Turbulence-degraded orbital angular momentum (OAM) mode recognition is critical for free-space optical (FSO) communication but is limited by atmospheric distortion. We propose a cascaded receiver integrating a vision transformer-based Uformer-OAM intensity-restoration network, an SE-ResNet-18 neural...

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Image-free ghost edge detection at ultra-low sampling rates via a Vision Mamba network

Rui Wang 2026-10-20 1 min 66 words LOW VISION

Summary: Edge detection is essential for extracting structural boundaries in visual scenes, but conventional deep-learning-based methods remain sensitive to degradations in the input images and involve redundant full-image acquisition. Image-free ghost edge detection provides an efficient alternative by dire...

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Ex vivo evaluation of corneal permeation and irritation potential associated with proparacaine hydrochloride misuse

Yolima Baena 2026-10-20 1 min 35 words LOW VISION

Summary: CONCLUSION: Proparacaine demonstrates dose-dependent permeation and histopathological changes, emphasizing the risks associated with misuse. The ex vivo model provides relevant and ethically aligned evidence to support ocular hazard assessment through the characterizing of histopathological changes.

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Shared variance across FrACT visual tasks: Robust positive associations but inconclusive latent structure

👤 Luca Battaglini 📅 2026-10-21 📖 1 min 📝 68 words LOW VISION

Summary: A long-standing question in vision science is whether individual differences in visual performance reflect a general visual ability or a set of largely task-specific processes. Previous studies have often reported weak associations across heterogeneous perceptual tasks, challenging the idea of a com...

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New challenges in horticultural IPM: artificial intelligence can enhance the use of biocontrol agents

👤 Andrea Lovazzano 📅 2026-10-22 📖 1 min 📝 59 words LOW VISION

Summary: Modern horticulture faces the dual imperatives of increasing global food production while mitigating the severe environmental and health impacts of conventional chemical pesticides. Biological control agents (BCAs), which utilize beneficial microorganisms, macroorganisms, semiochemical and botanical...

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Bridging the performance-efficiency gap: a resource-efficient hybrid DL-ML framework for jackfruit disease detection

👤 Hezerul Bin Abdul Karim
📅 2026-10-22 📖 1 min 📝 62 words LOW VISION

Summary: Complex environment variability and limited computational resources in agricultural settings create challenges in automated detection in tropical regions, where jackfruit diseases pose major threats to harvest productivity and food security. Conventional machine learning models often fail to general...

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Atomic Defect-Mediated Charge Trapping Enables Multimodal Plasticity in van der Waals Heterostructures for In-Sensor Neuromorphic Vision

👤 Zhiqun Lin
📅 2026-10-22 📖 1 min 📝 60 words LOW VISION

Summary: Neuromorphic hardware that unifies sensing, memory, and learning at the device level remains challenging due to limited optoelectronic comodulation and rigid architectures. Here, we report a flexible, van der Waals (vdW)-integrated MoS2/hexagonal boron nitride (h-BN)/graphene memtransistor in which ...

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Low-Dimensional Chiral Perovskites for Polarization-Sensitive Photodetection

Zhiyong Fan 2026-10-22 1 min 57 words LOW VISION

Summary: Emerging low-dimensional chiral perovskites have recently demonstrated significant application potential in circularly polarized light detection, spintronics, and quantum information technology due to their unique chiroptical activity and spin-selective properties. This review systematically summari...

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Visual context influences lateral balance while walking on winding paths

Jonathan B Dingwell 2026-10-22 1 min 28 words LOW VISION

Summary: BACKGROUND: Navigating daily environments necessitates adapting to ever-changing conditions while maintaining stability. This requires integrating sensory feedback, primarily from vision, to maintain balance and maneuverability as we walk.

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux
2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

👤 Béatrice Berçot 📅 2026-09-04 📖 1 min 📝 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan 📅 2026-09-09 📖 1 min 📝 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Clinical outcomes and safety of Five Element Acupuncture for preventive treatment of primary headache disorders: a multi-center real-world longitudinal observational study

Shengyuan Zhou 2026-10-19 1 min 49 words TDCS TACS TRNS

Summary: CONCLUSIONS: This real-world exploratory observational study suggests that FEA is associated with significant improvements in headache days, headache frequency, headache intensity, analgesic use, and quality of life, with sustained effects noted during follow-up. These findings provide preliminary o...

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Home Transcranial Direct Current Stimulation With Gamified Cognitive Control Training for Depression: A Randomized Clinical Trial

👤 Mor Nahum
📅 2026-10-19 📖 1 min 📝 45 words

TDCS TACS TRNS

Summary: CONCLUSIONS AND RELEVANCE: This RCT of home-based tDCS combined with CCT in adults with MDD was feasible and safe but did not demonstrate superior efficacy over the control condition. Future studies should refine the mechanistic alignment between stimulation and cognitive interventions to optimize t...

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Comparative chloroplast genomics of Ajania sericea and sympatric Asteraceae species identifies shared candidate molecular features associated with alpine adaptation

Xi Chen 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study showed that the chloroplast genomes of four sympatric Asteraceae species were highly conserved in overall structure, but still exhibited certain differences in gene loss, IR boundary variation, repeat sequence composition, and local hypervariable regions. These features provi...

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Brain Stimulation as a Treatment for Apathy: A Scoping Review of Effects and Safety

Lisa M McTeague 2026-10-20 1 min 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: Investigations into brain stimulation as a treatment for apathy show some promise. Conclusions were limited by small sample sizes, heterogeneous study designs, and variable outcome measures. Randomized trials with common validated apathy instruments, standardized stimulation parameters,...

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Effect of Different Cortical Targets of Transcranial Magnetic Stimulation on Freezing of Gait in Parkinsonism: A Systematic Review

Aliaa Elabd
2026-10-20 1 min 68 words TDCS TACS TRNS

Summary: BACKGROUND AND PURPOSE: Freezing of gait (FOG) is a disabling and episodic gait disturbance in patients with Parkinson's, with varying responses to noninvasive brain stimulation. Transcranial magnetic stimulation (TMS) has been explored as a therapeutic option; however, the optimal cortical target f...

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Sleep Spindles and Slow Oscillations as Potential Biomarkers and Therapeutic Targets in Mild-to-Moderate Alzheimer's Disease

👤 Thien Thanh Dang-Vu 📅 2026-10-20 📖 1 min 📝 35 words TDCS TACS TRNS

Summary: PURPOSE OF REVIEW: This review critically evaluates evidence on sleep spindles (SP), slow oscillations (SOs), and their coordination in mild-to-moderate Alzheimer's disease (AD), focusing on their relationships with cognition, AD biomarkers, and their therapeutic potential.

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

👤 Xin Zhou 📅 2026-10-20 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiple Sessions of Transcranial Direct Current Stimulation Combined With Squat Training Improve Lower-Limb Maximum Strength in Healthy Adult Men

👤 Fengxue Qi
📅 2026-10-22 📖 1 min 📝 79 words

TDCS TACS TRNS

Summary: Yan, X, Zuo, Z, Nitsche, MA, Zhang, X, and Qi, F. Multiple sessions of transcranial direct current stimulation combined with squat training improve lower-limb maximum strength in healthy adult men. J Strength Cond Res XX(X): 000-000, 2026-This study aimed to investigate the effects of transcranial d...

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Time-aware frame-aligned diffusion-inspired model for late dynamic PET frame prediction from early [¹¹C] raclopride

👤 Jae Sung Lee 📅 2026-10-22 📖 1 min 📝 42 words TDCS TACS TRNS

Summary: OBJECTIVE: To reduce the burden of dynamic PET acquisition while preserving quantitative analysis, we propose a time-aware frame-aligned diffusion-inspired residual denoising framework for predicting late dynamic PET frames from a short early acquisition window (10 min in this study) of [¹¹C] raclop...

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Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

👤 Felipe Balistieri Santinelli 📅 2026-10-18 📖 1 min 📝 70 words FNIRS

Summary: People with multiple sclerosis (pwMS) often exhibit motor fatigability and reduced foot-tapping ability. However, the coupling between performance and cortical activity during fast, repetitive and prolonged ankle movements remains unexplored. We aimed to investigate the neural correlates of a prolon...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang 📅 2026-10-19 📖 1 min 📝 64 words FNIRS

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Neural dysfunction underlying divergent thinking and its shared mechanisms with protracted withdrawal symptoms in heroin use disorder

👤 Haijun Duan
📅 2026-10-19 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSIONS: These findings reveal a sustained compensatory pathological mechanism and an imbalance in neural resource allocation during complex cognition in HUD individuals. The identified signatures of cortical activity by fNIRS offer reliable biomarkers for the rapid detection of both higher-orde...

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Functional near-infrared spectroscopy as a tool for monitoring treatment-related neurofunctional changes in posttraumatic stress disorder and trauma-related symptomatology: a narrative review

👤 Maria Cotelli
📅 2026-10-20 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Current evidence suggests that fNIRS represents a promising and clinically feasible approach for monitoring treatment-related neurofunctional changes in PTSD and trauma-related symptomatology, particularly during non-pharmacological interventions targeting prefrontal regulatory processe...

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Transcutaneous vagus nerve stimulation during semantic verbal fluency tasks: a high-density diffuse optical tomography study

👤 Li Su 📅 2026-10-20 📖 1 min 📝 55 words FNIRS

Summary: INTRODUCTION: Transcutaneous auricular vagus nerve stimulation (taVNS) is a promising adjunct for post-stroke language recovery. There are no established biomarkers for the neuromodulatory effects of taVNS. High-density diffuse optical tomography (HD-DOT) offers a non-invasive marker of task-related...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

Hua Hu
2026-10-21 1 min 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

👤 Xuqun You
📅 2026-10-22 📖 1 min 📝 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi 📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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Subjective emotional ratings dissociate brain activation and network patterns during counterfactual outcome processing

Ling Li 2026-10-19 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Human decision-making integrates outcome evaluation, counterfactual comparison, and individual traits. We investigated the neural processing of outcome-related emotions during economic decision-making by classifying feedback according to participants' subjective emotional ratings rather than objecti...

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Delineating neural contributions to electroencephalogram-based speech decoding

Shuntaro Sasai 2026-10-20 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Objective. Speech Brain-computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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An attention-enhanced hybrid CNN-BiLSTM framework for multiclass epileptic seizure recognition using EEG signals

Ahmed H Almutairi 2026-10-22 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Electroencephalogram (EEG) signal classification plays a pivotal role in brain-computer interfaces (BCIs) and the diagnosis of neurological disorders. However, the non-stationary, non-linear, and low signal-to-noise ratio characteristics of EEG signals make accurate feature extraction ...

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Realizing Ultrahigh Polarization and Excellent Thermal Stability in Piezoceramics via Defect-Stress Engineering

Yuhui Huang 2026-10-22 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Large polarization is critical for improving the piezoelectric and actuation performance of ferroelectric materials. As a promising lead-free candidate, BiFeO(3)-BaTiO(3) inherits the excellent polarization capability of BiFeO(3). However, the actual polarization achieved in its ceramics remains far...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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Optimizing learning for control of human-machine interfaces

Preeya Khanna 2026-10-22 1 min 42 words BRAIN COMPUTER INTERFACE

Summary: Human-machine interfaces, such as augmented reality devices, hand prostheses, and brain-computer interfaces, are increasingly being used in neurorehabilitation. In this issue of Neuron, Vassiliadis et al.[1] examine the interaction between closed-loop continuous reinforcement and sensory feedback in l...

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A GABAergic preoptic-raphe circuit mediates paradoxical coupling of anxiety and social preference after comparative social deficit

Hong Lian 2026-10-22 1 min 66 words BRAIN COMPUTER INTERFACE

Summary: Social disconnection is a major risk factor for anxiety disorders, the most prevalent mental illnesses worldwide, yet its neural substrates remain unclear. Here, we establish a comparative social deficit (CSD) model in adult male mice that induces a paradoxical phenotype of increased anxiety-like be...

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ACF-Net: Design of an adaptive cross-modal fusion network for cross-subject wearable EEG-EMG gesture recognition

TongLiang Yang 2026-10-22 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: Cross-subject gesture recognition in wearable brain-computer interfaces is constrained by four structural deficiencies: symmetric cross-modal attention discarding cortical precedence over muscular execution, priors confined to post hoc analysis, invariance anchored on unimodal features, and adaptati...

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Towards semantic brain-computer interfacing: The graded inventory of semantic triggers (GIST) paradigm

Ian Daly 2026-10-22 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Semantic Brain-Computer Interfaces (sBCIs) aim to decode conceptual information directly from neural activity, enabling more efficient and natural communication for individuals with severe motor impairments. However, existing approaches are often confounded by stimulus-driven perceptual processes an...

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Preclinical models of Wilson's disease: From pathogenesis to clinical translation

Zhi-Ying Wu 2026-10-22 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Wilson's disease (WD) is an autosomal recessive disorder of copper homeostasis caused by pathogenic ATP7B variants, leading predominantly to hepatic and neurological injury. Extensive heterogeneity in genotype, age at onset, clinical presentation, and treatment response complicates both diagnosis an...

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Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

2026-09-10 1 min 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000663

Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

Summary: Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000658

Distal pup cues attenuate defensive reactivity and fear-potentiated startle magnitude in postpartum rats without altering sensorimotor gating.

📅 2026-07-09 📖 1 min 📝 241 words BEHAVIORAL NEUROSCIENCE

Summary: Using a novel behavioral paradigm that integrates fear-potentiated startle (FPS) and prepulse inhibition testing, this study examined how motherhood and infant cues impact innate defensive reactivity (as measured by the magnitude of basal startle reflex), attentional processing (as measured by prepu...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000661

A novel rat spatial inference task reveals rapid deduction of transitive relationships using schemas and deliberation.

📅 2026-07-13 📖 1 min 📝 264 words BEHAVIORAL NEUROSCIENCE

Summary: Inferential reasoning is a vital cognitive ability that enables animals to navigate novel situations by leveraging existing relational knowledge of memory schema, with hypothesized roles of prefrontal cortical—hippocampal circuits. Transitive inference (TI) tasks test the ability of subjects to infe...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000660

Keep Android Open 🔓

📅 2026-02-26 📖 3 min 📝 625 words FMHY

> **Summary:** <h3 id="android-will-become-a-locked-down-platform-in-less-than-200-days" tabindex="-1">Android will become a locked-down platform in less than 200 days. </h3> In August 2025, Google announced th...

🔗 **Read full article:**
https://fmhy.net/posts/KeepAndroidOpen

Monthly Updates [April]

📅 2026-04-01 📖 4 min 📝 885 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/april-2026

Monthly Updates [October]

📅 2026-10-01 📖 5 min 📝 1097 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   Re-ordered Live Sports based on the <a href="https://challonge.com/ajqvt0wo/fullscreen" rel=...

🔗 **Read full article:**
https://fmhy.net/posts/oct-2026

I predicted data with machine learning

👤 /u/5A61Z4349 📅 2026-10-23 📖 1 min 📝 95 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I am pleased to be a part machine learning project completed by Group 6, Department of Petroleum and Mining Engineering, MIST. We predicted Plastic Viscosity and Mud Weight of drilling fluids using Decision Tree and Random Forest regressors, tuned with GridSe...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xd5wus/i_predicted_data_with_machine_learning/

Ask HN: If the models are so good, why doesn't OpenAI just start 100 companies?

👤 ex-aws-dude 📅 2026-10-22 📖 1 min 📝 93 words HACKER NEWS

> **Summary:** Title is a bit of an exaggeration but the question is more: if the frontier models are so good, why even sell them as your primary money maker? Couldn't OpenAI or Anthropic make a lot more money by just starting companies themselves that leverage the models and building their own products? Is...

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https://news.ycombinator.com/item?id=50200997

Jank – The native Clojure dialect with seamless C++ interop

👤 giancarlostoro 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

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This I Believe (1952)

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Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-05-25 📖 30 min 📝 6008 words

TWENTY PERCENT STATISTICIAN

> **Summary:** In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

👤 Adriel Carridice 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

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Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

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Multi-Channel Electromagnetic Interference Elimination for Shielding-Free MRI Using Null Operations

📅 2026-01-23 📖 1 min 📝 184 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Emerging technologies for electromagnetic interference (EMI) elimination have enabled radio frequency (RF) shielding-free magnetic resonance imaging (MRI), significantly reduced costs and increased accessibility. Existing methods often rely on multiple external sensors for EMI elimination...

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Double Banking on Knowledge: A Unified All-in-One Framework for Unpaired Multi-Modality Semi-Supervised Medical Image Segmentation

📅 2026-01-21 📖 1 min 📝 231 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Multi-modality (MM) semi-supervised learning (SSL) based medical image segmentation has recently gained increasing attention for its ability to utilize MM data and reduce reliance on labeled images. However, current methods face several challenges: (1) Multiple networks assignment hinder scalability...

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Virtual Imaging-Guided Harmonization of Chest X-Rays Using a Generative Adversarial Neural Network

📅 2026-01-26 📖 1 min 📝 247 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: To reduce variability in chest radiography (CXR) from acquisition and post-processing, and assess whether harmonization improves image quality and downstream diagnostic performance. Methods: A generative adversarial network (GAN) was trained exclusively on virtual images produced by a Vir...

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Self-Supervised Speech Representations for Sleep Apnea Severity Prediction

📅 2026-01-20 📖 1 min 📝 253 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Obstructive sleep apnea (OSA) is a prevalent yet underdiagnosed condition characterized by repetitive upper airway obstruction during sleep. Current gold standard diagnostic standards rely on polysomnography (PSG), which is resource-intensive. Since upper airway characteristics impact bot...

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Rectification of Cornea Induced Distortions in Microscopic Images for Assisted Ophthalmic Surgery

📅 2026-01-20 📖 1 min 📝 240 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Image distortions induced by the high refractive power of the eye's optical components challenge the accuracy of geometric information derived from intra-operative sensor data in ophthalmic surgery. Correcting these distortions is vital for advancing surgical assistance systems that rely ...

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Distinct Patterns of Syntactic Errors Doubly Dissociate in Chronic Poststroke Aphasia

📅 2026-10-20 📖 1 min 📝 195 words COGNITIVE NEUROSCIENCE

Summary: The lesion correlates of syntactic deficits in aphasia remain poorly understood. Previous studies have suggested that distinct error types in expressive syntax may be associated with damage to different brain regions, but the specific lesion correlates of these errors have not been fully delineated....

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Transformer Mechanisms Mimic Frontostriatal Gating Operations When Trained on Human Working Memory Tasks

📅 2026-10-20 📖 1 min 📝 237 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM) is thought to rely on sophisticated frontostriatal mechanisms for selective gating, supporting updating and readout of information to and from distinct memory “addresses” (neural populations). According to computational models, capacity limitations in WM arise due to challenges i...

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Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

Deidre Artis 2026-07-17 1 min 34 words EMBS

Summary: Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

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State-specific dynamic functional connectivity disturbances in spinal cord injury-related neuropathic pain

1 min 29 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Chuchu Sun, Haotian Xin, Ling Wang, Qian Chen, Yu Wang, Beining Yang, Xin Chen, Fang Li, Jie Lu, Nan Chen

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More is not necessarily better: A comparison of 2-mA and 4-mA prefrontal cortex anodal tDCS on endurance performance, head oxygenation, cognition, and psychophysiological responses in cyclists

📖 1 min 📝 33 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Everton Borges da Silva, Mércia Jaynne Leandro de Melo, Matheus Alves Mendonça, Daniel de Medeiros Leiros, Ehsan Amiri, Heloiana Faro, Daniel Gomes da Silva Machado

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Soft tactile stimulation engages CGRP neurons in the parabrachial nucleus

📖 1 min 📝 28 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): F. Anesten, S. Simfors, K. Ioneskou, M. Hezsö, B. Gündogdu, A. Tran, A. Stjernvall, V. Ratiglia, A. Almasri, L.S. Löken

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Humanin attenuates neuronal cellular senescence and preserves neurotransmitter release in differentiated PC12 cells

📖 1 min 📝 16 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Miki Tanahashi, Ayari Kozuka, Nobuyuki Sasakawa, Takako Niikura

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Characterizing age-dependent diamagnetic and paramagnetic susceptibility properties in 5xFAD Alzheimer's mouse brain using high-resolution MRI

📖 1 min 📝 28 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Juan Liu, Season K. Wyatt-Johnson, Jingjia Chen, Jie Chen, Zhuoheng Liu, Hongbo Wu, Li Feng, Randy R. Brutkiewicz, Nian Wang

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

Robert Oostenveld 2026-02-12 1 min 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Knowledge gaps and practices for glaucoma among adults in Palestine: a cross-sectional study

Farid K Abu Sham'a 2026-10-20 1 min 51 words LOW VISION

Summary: CONCLUSIONS: Although awareness of glaucoma is moderate in Palestine, comprehensive knowledge remains low, particularly among high-risk populations. Public health interventions should target education beyond name recognition, emphasizing asymptomatic progression, irreversible vision loss, and risk f...

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Sn/Cl Co-Doped ZnO Cathode Interfacial Layer for High-Performance Organic Photodetectors

Kai Zhang 2026-10-20 1 min 66 words LOW VISION

Summary: Organic photodetectors (OPDs) hold significant promise for applications spanning medical monitoring, bioimaging and night vision due to their unique advantages, such as flexibility, light weight, and potential for large-area fabrication. Nevertheless, the dark current density (Jd) of OPDs exhibits a...

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A thermopile array and stereo vision fusion system for small-tube temperature sensing

👤 Dongfang Hu 📅 2026-10-20 📖 1 min 📝 71 words LOW VISION

Summary: This paper presents a non-contact temperature measurement scheme for small-diameter tubes by fusing a low-resolution thermopile array (TPA) with visible-light stereo vision. Stereo vision locates the tube in three dimensions. Then, this geometry is projected onto the TPA imaging plane to constrain t...

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Cost-effective long-range measurement of structural displacement using a two-stage method integrating super-resolution reconstruction and sub-pixel registration

👤 Xiaosong Li 📅 2026-10-20 📖 1 min 📝 61 words LOW VISION

Summary: Displacement is an imperative metric for assessing structural performance. However, the accuracy of structural displacement measurement is constrained by the physical resolution of the camera, particularly in long-range monitoring scenarios. To address these constraints, a cost-effective long-range ...

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Physics-aware restoration and calibrated soft demodulation for turbulence-degraded OAM mode recognition

👤 Jun Zhu 📅 2026-10-20 📖 1 min 📝 58 words LOW VISION

Summary: Turbulence-degraded orbital angular momentum (OAM) mode recognition is critical for free-space optical (FSO) communication but is limited by atmospheric distortion. We propose a cascaded receiver integrating a vision transformer-based Uformer-OAM intensity-restoration network, an SE-ResNet-18 neural...

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Image-free ghost edge detection at ultra-low sampling rates via a Vision Mamba network

👤 Rui Wang 📅 2026-10-20 📖 1 min 📝 66 words LOW VISION

Summary: Edge detection is essential for extracting structural boundaries in visual scenes, but conventional deep-learning-based methods remain sensitive to degradations in the input images and involve redundant full-image acquisition. Image-free ghost edge detection provides an efficient alternative by dire...

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Ex vivo evaluation of corneal permeation and irritation potential associated with proparacaine hydrochloride misuse

Yolima Baena 2026-10-20 1 min 35 words LOW VISION

Summary: CONCLUSION: Proparacaine demonstrates dose-dependent permeation and histopathological changes, emphasizing the risks associated with misuse. The ex vivo model provides relevant and ethically aligned evidence to support ocular hazard assessment through the characterizing of histopathological changes.

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Shared variance across FrACT visual tasks: Robust positive associations but inconclusive latent structure

Luca Battaglini 2026-10-21 1 min 68 words LOW VISION

Summary: A long-standing question in vision science is whether individual differences in visual performance reflect a general visual ability or a set of largely task-specific processes. Previous studies have often reported weak associations across heterogeneous perceptual tasks, challenging the idea of a com...

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New challenges in horticultural IPM: artificial intelligence can enhance the use of biocontrol agents

Andrea Lovazzano 2026-10-22 1 min 59 words LOW VISION

Summary: Modern horticulture faces the dual imperatives of increasing global food production while mitigating the severe environmental and health impacts of conventional chemical pesticides. Biological control agents (BCAs), which utilize beneficial microorganisms, macroorganisms, semiochemical and botanical...

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Bridging the performance-efficiency gap: a resource-efficient hybrid DL-ML framework for jackfruit disease detection

Hezerul Bin Abdul Karim 2026-10-22 1 min 62 words LOW VISION

Summary: Complex environment variability and limited computational resources in agricultural settings create challenges in automated detection in tropical regions, where jackfruit diseases pose major threats to harvest productivity and food security. Conventional machine learning models often fail to general...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

👤 Béatrice Berçot 📅 2026-09-04 📖 1 min 📝 60 words BRAILLE

> **Summary:** CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan 📅 2026-09-09 📖 1 min 📝 57 words BRAILLE

> **Summary:** Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Written exposure therapy augmented with transcranial direct current stimulation for posttraumatic stress disorder: A pilot randomized clinical trial

STRONG STAR Consortium 2026-10-18 1 min 64 words TDCS TACS TRNS

Summary: Written exposure therapy (WET) is an evidence-based, trauma-focused psychotherapy for posttraumatic stress disorder (PTSD) that has been tested with several trauma-exposed samples. Although many individuals who receive WET and other trauma-focused psychotherapies experience a significant reduction i...

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Transcranial Direct Current Stimulation and Cranial Electrotherapy Stimulation in Athletes: Optimizing Athletic Performance and Sleep Quality Through Neuromodulation

👤 Paulina Gajniak
📅 2026-10-19 📖 1 min 📝 74 words

TDCS TACS TRNS

Summary: Neuromodulation is a procedure relying on the delivery of specific stimuli to selected sites of the nervous system and hereby influencing its activity. Even though the underlying neurobiological mechanisms are not yet fully elucidated and the research is still ongoing, neuromodulation is already bei...

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Clinical outcomes and safety of Five Element Acupuncture for preventive treatment of primary headache disorders: a multi-center real-world longitudinal observational study

Shengyuan Zhou
2026-10-19 1 min 49 words TDCS TACS TRNS

Summary: CONCLUSIONS: This real-world exploratory observational study suggests that FEA is associated with significant improvements in headache days, headache frequency, headache intensity, analgesic use, and quality of life, with sustained effects noted during follow-up. These findings provide preliminary o...

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Home Transcranial Direct Current Stimulation With Gamified Cognitive Control Training for Depression: A Randomized Clinical Trial

👤 Mor Nahum 📅 2026-10-19 📖 1 min 📝 45 words TDCS TACS TRNS

Summary: CONCLUSIONS AND RELEVANCE: This RCT of home-based tDCS combined with CCT in adults with MDD was feasible and safe but did not demonstrate superior efficacy over the control condition. Future studies should refine the mechanistic alignment between stimulation and cognitive interventions to optimize t...

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Comparative chloroplast genomics of Ajania sericea and sympatric Asteraceae species identifies shared candidate molecular features associated with alpine adaptation

Xi Chen 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study showed that the chloroplast genomes of four sympatric Asteraceae species were highly conserved in overall structure, but still exhibited certain differences in gene loss, IR boundary variation, repeat sequence composition, and local hypervariable regions. These features provi...

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Brain Stimulation as a Treatment for Apathy: A Scoping Review of Effects and Safety

Lisa M McTeague 2026-10-20 1 min 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: Investigations into brain stimulation as a treatment for apathy show some promise. Conclusions were limited by small sample sizes, heterogeneous study designs, and variable outcome measures. Randomized trials with common validated apathy instruments, standardized stimulation parameters,...

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Effect of Different Cortical Targets of Transcranial Magnetic Stimulation on Freezing of Gait in Parkinsonism: A Systematic Review

Aliaa Elabd
2026-10-20 1 min 68 words TDCS TACS TRNS

Summary: BACKGROUND AND PURPOSE: Freezing of gait (FOG) is a disabling and episodic gait disturbance in patients with Parkinson's, with varying responses to noninvasive brain stimulation. Transcranial magnetic stimulation (TMS) has been explored as a therapeutic option; however, the optimal cortical target f...

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Sleep Spindles and Slow Oscillations as Potential Biomarkers and Therapeutic Targets in Mild-to-Moderate Alzheimer's Disease

Thien Thanh Dang-Vu 2026-10-20 1 min 35 words TDCS TACS TRNS

Summary: PURPOSE OF REVIEW: This review critically evaluates evidence on sleep spindles (SP), slow oscillations (SOs), and their coordination in mild-to-moderate Alzheimer's disease (AD), focusing on their relationships with cognition, AD biomarkers, and their therapeutic potential.

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

Xin Zhou 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng 📅 2026-10-21 📖 1 min 📝 53 words TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiscale neural signatures of long-lag repetition learning: a simultaneous fNIRS-EEG study

👤 Ken Pugh 📅 2026-10-17 📖 1 min 📝 38 words FNIRS

Summary: CONCLUSIONS: Combining fNIRS and EEG with multimodal machine learning provides a richer characterization of repetition-based learning than either modality alone, establishing a framework for phenotyping language learning capacity and identifying neural markers of reading difficulties in pediatric po...

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Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

Felipe Balistieri Santinelli 2026-10-18 1 min 70 words FNIRS

Summary: People with multiple sclerosis (pwMS) often exhibit motor fatigability and reduced foot-tapping ability. However, the coupling between performance and cortical activity during fast, repetitive and prolonged ankle movements remains unexplored. We aimed to investigate the neural correlates of a prolon...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

Ming Zhang 2026-10-19 1 min 64 words FNIRS

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Neural dysfunction underlying divergent thinking and its shared mechanisms with protracted withdrawal symptoms in heroin use disorder

👤 Haijun Duan
📅 2026-10-19 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSIONS: These findings reveal a sustained compensatory pathological mechanism and an imbalance in neural resource allocation during complex cognition in HUD individuals. The identified signatures of cortical activity by fNIRS offer reliable biomarkers for the rapid detection of both higher-orde...

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Functional near-infrared spectroscopy as a tool for monitoring treatment-related neurofunctional changes in posttraumatic stress disorder and trauma-related symptomatology: a narrative review

👤 Maria Cotelli
📅 2026-10-20 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Current evidence suggests that fNIRS represents a promising and clinically feasible approach for monitoring treatment-related neurofunctional changes in PTSD and trauma-related symptomatology, particularly during non-pharmacological interventions targeting prefrontal regulatory processe...

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Transcutaneous vagus nerve stimulation during semantic verbal fluency tasks: a high-density diffuse optical tomography study

👤 Li Su 📅 2026-10-20 📖 1 min 📝 55 words FNIRS

Summary: INTRODUCTION: Transcutaneous auricular vagus nerve stimulation (taVNS) is a promising adjunct for post-stroke language recovery. There are no established biomarkers for the neuromodulatory effects of taVNS. High-density diffuse optical tomography (HD-DOT) offers a non-invasive marker of task-related...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

👤 Xuqun You
📅 2026-10-22 📖 1 min 📝 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

Qiuxia Shi 2026-10-22 1 min 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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Electroencephalography Changes in Emotional Recognition: A Signal Processing Approach Using Wavelet Features

Lakshmi Narasimhan 2026-10-17 1 min 43 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Wavelet-based EEG feature analysis provides meaningful and discriminative information for emotion recognition and may provide a robust feature set for future machine learning-based emotion recognition systems. Further studies incorporating classification algorithms and larger data sets a...

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A Compressive Neural Telemetry with Address-Event-μPacket Encoding and Duty-Cycling IR-UWB Transmitter

👤 Yao-Hong Liu 📅 2026-10-18 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: This article presents a networked wireless telemetry for 384-sensor-pixel neural data acquisition, designed to support high throughput, low power consumption, and meter-scale communication distance. The proposed event-based Address-Event-μPacket (AEμP) encoder with differential binary phase shift ke...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang 📅 2026-10-19 📖 1 min 📝 64 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Bridging the Gap Between Design and Access: Toward Visual Scene Displays for Brain-Computer Interface Use in Augmentative and Alternative Communication by Children

Ahad Behboodi 2026-10-19 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: These findings indicate that asymmetry and contextual embedding within scenes may enhance neural engagement and user experience, providing a preliminary foundation for considering photographic VSDs into pediatric BCI research.

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A spatio-temporal attention convolutional network with topology-preserving channel embedding and transformer for EEG decoding

Gang Liu 2026-10-19 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: EEG-based BCI has shown promising applications in motor rehabilitation and emotion recognition. Given the non-stationarity and variability of EEG signals, it is essential to develop classification models that can effectively extract spatio-temporal features while maintaining strong generalization ca...

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https://pubmed.ncbi.nlm.nih.gov/42896470/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261022211705&v=2.20.1

The calibration-to-accuracy mapping in P300 spellers does not transport across cohorts and was not restored by tested alignment or local recalibration

Oved Daniel 2026-10-19 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Objective
A calibration-derived score has been related to online P300-speller accuracy in the same session, always within the cohort measured. We evaluated whether a mapping from that score to expected accuracy transports to withheld cohorts.
Approach
Retrospective secondary analysis of ...

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Subjective emotional ratings dissociate brain activation and network patterns during counterfactual outcome processing

Ling Li 2026-10-19 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Human decision-making integrates outcome evaluation, counterfactual comparison, and individual traits. We investigated the neural processing of outcome-related emotions during economic decision-making by classifying feedback according to participants' subjective emotional ratings rather than objecti...

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Delineating neural contributions to electroencephalogram-based speech decoding

Shuntaro Sasai 2026-10-20 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Objective. Speech Brain-computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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An attention-enhanced hybrid CNN-BiLSTM framework for multiclass epileptic seizure recognition using EEG signals

Ahmed H Almutairi 2026-10-22 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Electroencephalogram (EEG) signal classification plays a pivotal role in brain-computer interfaces (BCIs) and the diagnosis of neurological disorders. However, the non-stationary, non-linear, and low signal-to-noise ratio characteristics of EEG signals make accurate feature extraction ...

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Realizing Ultrahigh Polarization and Excellent Thermal Stability in Piezoceramics via Defect-Stress Engineering

👤 Yuhui Huang 📅 2026-10-22 📖 1 min 📝 62 words BRAIN COMPUTER INTERFACE

Summary: Large polarization is critical for improving the piezoelectric and actuation performance of ferroelectric materials. As a promising lead-free candidate, BiFeO(3)-BaTiO(3) inherits the excellent polarization capability of BiFeO(3). However, the actual polarization achieved in its ceramics remains far...

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Psychomusicology: A resounding closing cadence.

📅 2024-01-22 📖 1 min 📝 256 words PSYCHOMUSICOLOGY

Summary: From 2012 to 2023, the American Psychological Association served as publisher of Psychomusicology: Music, Mind, and Brain. Annabel Cohen and Mark Schmuckler were the successive editors-in-chiefs during this time. As the journal is ceasing publication, the two editors reflect on the developm...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000305

Suggestibility and psychedelics: From therapeutics to social context.

📅 2024-12-12 📖 1 min 📝 175 words CLINICAL NEUROSCIENCE

Summary: Suggestibility is an important aspect of the psychedelic experience that merits greater attention in the expanding field of psychedelic research. This article provides a comprehensive road map for exploring the critical questions and issues at the intersection of suggestion, suggestibility, and psyc...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000412

Evaluating dispositional emotion-related factors in the effect of acceptance on affect reactivity: A secondary analysis of an experimental study.

📅 2024-12-30 📖 1 min 📝 257 words CLINICAL NEUROSCIENCE

Summary: Acceptance as a self-regulation strategy has beneficial effects on affect reactivity in response to stressors via reducing negative affect and state anxiety as well as improving positive affect. However, dispositional emotion-related factors, namely, experiential avoidance, emotional dysregulation, ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000416

Open monitoring meditation defined as the nonreactive meta-awareness of mind wandering.

📅 2024-12-12 📖 1 min 📝 128 words CLINICAL NEUROSCIENCE

Summary: Open monitoring meditation is generally defined as the nonreactive, meta-aware observation of moment-by-moment phenomenal experiences associated with mental contents without focusing on predetermined mental or physical objects. However, in the current literature, neither general definitions nor more...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000411

Multisensory interplay and theories of consciousness.

📅 2024-11-25 📖 1 min 📝 174 words CLINICAL NEUROSCIENCE

Summary: Sensory interplay typically refers to cases where the content of one sensory modality affects the content of another sensory modality. Although there is much evidence today of such interplay, it is unknown how to interpret them. Is it a one-way causal function between two separate states (e.g., a vi...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000407

How ghosting and stress impact vulnerable narcissism and maladaptive daydreaming: The role of mindfulness and rumination.

📅 2024-09-12 📖 1 min 📝 204 words CLINICAL NEUROSCIENCE

Summary: The present study aims to explore the influence of ghosting and stress on maladaptive daydreaming and vulnerable narcissism within a multivariate framework, using mindfulness and rumination as mediators. We employed a cross-sectional design and conducted it among 301 university students (M<...

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http://doi.org/10.1037/cns0000403

The conscious experience of thinking aloud: Are positive feelings associated with better think-aloud performance?

📅 2024-12-30 📖 1 min 📝 216 words CLINICAL NEUROSCIENCE

Summary: The think-aloud interview is a psychological procedure used to measure an individual’s cognition in solving a task. However, very little is known about how an individual’s conscious experience of the interview might be associated with their performance during the interview. This is consequential as ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000410

Mystical experiences in virtual reality.

📅 2024-04-04 📖 1 min 📝 176 words CLINICAL NEUROSCIENCE

Summary: Virtual reality (VR) has shown promise as a psychological means of inducing mystical experiences. Here, we compared the subjective responses of 48 undergraduate volunteers to script-guided multiuser perceptual experiences in VR (VR perception [VR-P]) with their responses to a control condit...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000396

Effect of a peer-led meditation club on adolescents' well-being, empathy, and compassion.

📅 2024-09-09 📖 1 min 📝 197 words CLINICAL NEUROSCIENCE

Summary: Many studies have examined the effect of meditation sessions led by experienced teachers on personal benefits for adults. The present study investigated the effect of broad-based mindfulness practices with an adolescent peer facilitator on prosocial outcomes in addition to individual gains. The 20 s...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000400

Monthly Updates [May]

📅 2026-05-01 📖 4 min 📝 892 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/may-2026

Friday Daily Thread: r/Python Meta and Free-Talk Fridays

👤 /u/AutoModerator 📅 2026-10-23 📖 1 min 📝 212 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Thread: Meta Discussions and Free Talk Friday 🎙️</h1> Welcome to Free Talk Friday on /r/Python! This is the place to discuss the r/Python community (meta discussions), Python news, projects, or anything else...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xcyvqj/friday_daily_thread_rpython_meta_and_freetalk/

LLMs are making formal software verification feasible at a much larger scale

👤 johnbender 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.anthropic.com/research/formal-verification Comments URL: https://news.ycombinator.com/item?id=50202217 Points: 81 # Commen...

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https://www.anthropic.com/research/formal-verification

AI compute is the new oil–and needs a new market

👤 ijidak 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.economist.com/finance-and-economics/2026/10/22/ai-compute-is-the-new-oil-and-needs-a-new-market Comments URL: <a href="https://news.ycombinator...

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https://www.economist.com/finance-and-economics/2026/10/22/ai-compute-is-the-new-oil-and-needs-a-new-market

Docker Is Weird

👤 psxuaw 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://codeandcake.dev/posts/2026-10-22-docker-is-weird Comments URL: https://news.ycombinator.com/item?id=50202681 Points: 58 # Co...

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https://codeandcake.dev/posts/2026-10-22-docker-is-weird

Mapping the Northern Sky

👤 tmseidman 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.maps.com/star-map-charts-stars-and-constellations-of-northern-hemisphere/ Comments URL: <a href="https://news.ycombinator.com/item?id=50202766">https://news.ycombina...

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The foolishness of "natural language programming" (1978)

👤 xiaoyu2006 📅 2026-10-23 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.cs.utexas.edu/~EWD/transcriptions/EWD06xx/EWD667.html Comments URL: https://news.ycombinator.com/item?id=50203161 Poin...

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https://www.cs.utexas.edu/~EWD/transcriptions/EWD06xx/EWD667.html

There are 13 ways to analyse a replication study, but only one of them is coherent.

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-05-03 📖 6 min 📝 1361 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Paul Simon’s says there are 50 ways to leave your lover, and a similar abundance characterizes the contemporary literature on how to analyse a replication study. The recent SCORE replication project in Nature makes this explicit by reporting “13 replication success metrics along with the number o...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/there-are-13-ways-to-analyse.html

Which Data Repository Should You Use?

noreply@blogger.com (Daniel Lakens)
2026-08-15 17 min 3483 words

TWENTY PERCENT STATISTICIAN

Summary: <p>The Center for Open Science has announced that from November 16, 2026, no new projects can be created on the Open Science Framework. After February 19 all projects will become read-only, and eventually, they anticipate that all private projects will be deleted. The OSF has raised a generation of ...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/08/which-data-repository-should-you-use.html

Neuroethical Perspectives on Neurotechnology in Latin America

Adriel Carridice
2026-09-23 1 min 0 words

BRAIN

🔗 **Read full article:**
https://brain.ieee.org/publications/research-and-white-papers/neuroethical-perspectives-on-neurotechnology-in-latin-america/

T-MRB Call for Nominations for Associate Editors

👤 Deidre Artis 📅 2026-04-22 📖 1 min 📝 16 words EMBS

> **Summary:** The post T-MRB Call for Nominations for Associate Editors appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/t-mrb-call-for-nominations-ae/#new_tab

2026 Adcom Petition is Open

👤 Deidre Artis 📅 2026-09-11 📖 1 min 📝 14 words EMBS

> **Summary:** The post 2026 Adcom Petition is Open appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/about/embs-officers/2026-embs-administrative-committee-adcom-election-2/#new_tab

Brain-inspired recurrent spiking neural networks for multiscale dynamic representation with manifolds

📖 1 min 📝 18 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Xiaojun Wang, Zaidong Lin, Bin Deng, Xiaodong Liu, Haitao Yu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006871?dgcid=rss_sd_all

Acute scalp EEG signatures of anterior thalamic DBS in epilepsy:reproducible gamma power enhancement within an aperiodic spectral-state shift

👤 Sabbaghi, E., Mattila, H., Siitama, E., Hyttinen, J., Peltola, J., Puthanmadam Subramaniyam, N.

📅 2026-10-22 📖 1 min 📝 250 words BIORXIV NEUROSCIENCE

Summary: Background: Deep brain stimulation (DBS) of the anterior nucleus of the thalamus (ANT) is an established treatment for drug-resistant epilepsy, but acute scalp-EEG signatures of stimulation remain incompletely characterized. Objective: To identify acute scalp-EEG signatures of ANT-DBS and assess the...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.16.759433v1?rss=1

Semaglutide Increases Striatal Leptin Sensitivity, Impacting Dopaminergic Response to Cocaine

Muldowney, J. J., Umlauf, M. C., Bocarsly, M. E.

2026-10-22 1 min 234 words

BIORXIV NEUROSCIENCE

Summary: There is accumulating evidence that the GLP-1 analog, semaglutide, might effectively curb intake of cocaine via brain dopaminergic functioning, despite limited brain penetrance. However, there is currently no established mechanism elucidating how chronic semaglutide treatment regulates the dopaminer...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.16.759697v1?rss=1

Multiparametric MRI and imaging transcriptomics reveal brain-wide changes in GD3 synthase deficiency

Ilic, K., Kim, E., Mlinac-Jerkovic, K., Simmons, C., Puljko, B., Balog, M., Heffer, M., Cash, D., Rosenzweig, I., Kalanj-Bognar, S.

2026-10-22 1 min 264 words

BIORXIV NEUROSCIENCE

Summary: Using an integrative imaging-transcriptomic approach that highlights the molecular complexity underlying neuroimaging phenotypes, we investigated a mouse model with disrupted ganglioside synthesis: the GD3 synthase-deficient mouse (St8sia1-null; GD3S KO). Gangliosides are membrane glycosphingolipids...

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https://www.biorxiv.org/content/10.64898/2026.10.16.759717v1?rss=1

Progression of visual-stimulus-evoked blood velocity response timing across the vascular network measured with individual-vessel phase-contrast fMRA

👤 Hu, Z., Hartung, G. A., Proulx, S., Varadarajan, D., Bollmann, S., Rosen, B. R., Polimeni, J. R.

📅 2026-10-22 📖 1 min 📝 364 words BIORXIV NEUROSCIENCE

Summary: Most functional MRI (fMRI) studies of brain function aim to measure hemodynamic responses arising from the microvasculature within the tissue parenchyma, where spatial and temporal coupling with neuronal activity is thought to be tightest. However, the measured hemodynamic responses emerge from an i...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.16.759420v1?rss=1

Cue-dependent fear learning drives nucleus accumbens spine plasticity

📅 2026-10-22 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76692-6

CLSP-REQA: a real-time quality-aware closed-loop seizure prediction framework with Mamba-BiLSTM and confidence-gated intervention

Quansheng Ren 2026-09-30 1 min 304 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionReliable seizure prediction is a prerequisite for closed-loop neurostimulation therapy, yet existing methods rarely account for the variability in EEG signal quality encountered in real-world deployment, and the overwhelming majority adopt non-strict evaluation protocols that overestimat...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1888743

Digital health meets Parkinson's disease: a bibliometric exploration of wearable sensors research and emerging frontiers (2005–2026)

Lan Xiong 2026-10-08 1 min 228 words FRONTIERS NEUROINFORMATICS

Summary: BackgroundWearable and sensor-based monitoring technologies are increasingly used to support objective, longitudinal assessment in Parkinson's disease (PD). However, the global development, knowledge structure, and emerging research frontiers of this interdisciplinary field remain incompletely mappe...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1937585

HQML-ICU: hybrid quantum-classical machine learning for ICU mortality prediction using MIMIC-IV electronic health records

Mohammed Farsi 2026-10-09 1 min 277 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Accurate prediction of mortality in the intensive care unit (ICU) is crucial for informing clinical decision-making and optimizing resource allocation. Although machine learning methodologies have enhanced traditional severity-of-illness scoring systems, they encounter intrinsic limitations in model...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1900553

Adaptive neuro-fuzzy risk index for safe and interpretable lane-change decisions in Level-3 autonomous driving

Jianwei Li 2026-10-09 1 min 250 words FRONTIERS NEUROROBOTICS

Summary: IntroductionHighway lane-change maneuvers pose a safety challenge for SAE Level 3 autonomous driving because conventional threshold-based decisions and isolated surrogate safety measures may inadequately capture the coupling between longitudinal braking demand and evolving inter-vehicle gaps. This s...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1918469

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261022183849&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261022183849&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261022183849&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261022183849&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261022183849&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Knowledge gaps and practices for glaucoma among adults in Palestine: a cross-sectional study

Farid K Abu Sham'a 2026-10-20 1 min 51 words LOW VISION

Summary: CONCLUSIONS: Although awareness of glaucoma is moderate in Palestine, comprehensive knowledge remains low, particularly among high-risk populations. Public health interventions should target education beyond name recognition, emphasizing asymptomatic progression, irreversible vision loss, and risk f...

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Sn/Cl Co-Doped ZnO Cathode Interfacial Layer for High-Performance Organic Photodetectors

Kai Zhang 2026-10-20 1 min 66 words LOW VISION

Summary: Organic photodetectors (OPDs) hold significant promise for applications spanning medical monitoring, bioimaging and night vision due to their unique advantages, such as flexibility, light weight, and potential for large-area fabrication. Nevertheless, the dark current density (Jd) of OPDs exhibits a...

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A thermopile array and stereo vision fusion system for small-tube temperature sensing

👤 Dongfang Hu 📅 2026-10-20 📖 1 min 📝 71 words LOW VISION

Summary: This paper presents a non-contact temperature measurement scheme for small-diameter tubes by fusing a low-resolution thermopile array (TPA) with visible-light stereo vision. Stereo vision locates the tube in three dimensions. Then, this geometry is projected onto the TPA imaging plane to constrain t...

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Cost-effective long-range measurement of structural displacement using a two-stage method integrating super-resolution reconstruction and sub-pixel registration

👤 Xiaosong Li 📅 2026-10-20 📖 1 min 📝 61 words LOW VISION

Summary: Displacement is an imperative metric for assessing structural performance. However, the accuracy of structural displacement measurement is constrained by the physical resolution of the camera, particularly in long-range monitoring scenarios. To address these constraints, a cost-effective long-range ...

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Physics-aware restoration and calibrated soft demodulation for turbulence-degraded OAM mode recognition

👤 Jun Zhu 📅 2026-10-20 📖 1 min 📝 58 words LOW VISION

Summary: Turbulence-degraded orbital angular momentum (OAM) mode recognition is critical for free-space optical (FSO) communication but is limited by atmospheric distortion. We propose a cascaded receiver integrating a vision transformer-based Uformer-OAM intensity-restoration network, an SE-ResNet-18 neural...

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Image-free ghost edge detection at ultra-low sampling rates via a Vision Mamba network

👤 Rui Wang 📅 2026-10-20 📖 1 min 📝 66 words LOW VISION

Summary: Edge detection is essential for extracting structural boundaries in visual scenes, but conventional deep-learning-based methods remain sensitive to degradations in the input images and involve redundant full-image acquisition. Image-free ghost edge detection provides an efficient alternative by dire...

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Ex vivo evaluation of corneal permeation and irritation potential associated with proparacaine hydrochloride misuse

Yolima Baena 2026-10-20 1 min 35 words LOW VISION

Summary: CONCLUSION: Proparacaine demonstrates dose-dependent permeation and histopathological changes, emphasizing the risks associated with misuse. The ex vivo model provides relevant and ethically aligned evidence to support ocular hazard assessment through the characterizing of histopathological changes.

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Shared variance across FrACT visual tasks: Robust positive associations but inconclusive latent structure

Luca Battaglini 2026-10-21 1 min 68 words LOW VISION

Summary: A long-standing question in vision science is whether individual differences in visual performance reflect a general visual ability or a set of largely task-specific processes. Previous studies have often reported weak associations across heterogeneous perceptual tasks, challenging the idea of a com...

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New challenges in horticultural IPM: artificial intelligence can enhance the use of biocontrol agents

Andrea Lovazzano 2026-10-22 1 min 59 words LOW VISION

Summary: Modern horticulture faces the dual imperatives of increasing global food production while mitigating the severe environmental and health impacts of conventional chemical pesticides. Biological control agents (BCAs), which utilize beneficial microorganisms, macroorganisms, semiochemical and botanical...

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Bridging the performance-efficiency gap: a resource-efficient hybrid DL-ML framework for jackfruit disease detection

Hezerul Bin Abdul Karim 2026-10-22 1 min 62 words LOW VISION

Summary: Complex environment variability and limited computational resources in agricultural settings create challenges in automated detection in tropical regions, where jackfruit diseases pose major threats to harvest productivity and food security. Conventional machine learning models often fail to general...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

👤 Béatrice Berçot 📅 2026-09-04 📖 1 min 📝 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan 📅 2026-09-09 📖 1 min 📝 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Written exposure therapy augmented with transcranial direct current stimulation for posttraumatic stress disorder: A pilot randomized clinical trial

STRONG STAR Consortium 2026-10-18 1 min 64 words TDCS TACS TRNS

Summary: Written exposure therapy (WET) is an evidence-based, trauma-focused psychotherapy for posttraumatic stress disorder (PTSD) that has been tested with several trauma-exposed samples. Although many individuals who receive WET and other trauma-focused psychotherapies experience a significant reduction i...

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Transcranial Direct Current Stimulation and Cranial Electrotherapy Stimulation in Athletes: Optimizing Athletic Performance and Sleep Quality Through Neuromodulation

👤 Paulina Gajniak
📅 2026-10-19 📖 1 min 📝 74 words

TDCS TACS TRNS

Summary: Neuromodulation is a procedure relying on the delivery of specific stimuli to selected sites of the nervous system and hereby influencing its activity. Even though the underlying neurobiological mechanisms are not yet fully elucidated and the research is still ongoing, neuromodulation is already bei...

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Clinical outcomes and safety of Five Element Acupuncture for preventive treatment of primary headache disorders: a multi-center real-world longitudinal observational study

Shengyuan Zhou
2026-10-19 1 min 49 words TDCS TACS TRNS

Summary: CONCLUSIONS: This real-world exploratory observational study suggests that FEA is associated with significant improvements in headache days, headache frequency, headache intensity, analgesic use, and quality of life, with sustained effects noted during follow-up. These findings provide preliminary o...

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Home Transcranial Direct Current Stimulation With Gamified Cognitive Control Training for Depression: A Randomized Clinical Trial

👤 Mor Nahum
📅 2026-10-19 📖 1 min 📝 45 words

TDCS TACS TRNS

Summary: CONCLUSIONS AND RELEVANCE: This RCT of home-based tDCS combined with CCT in adults with MDD was feasible and safe but did not demonstrate superior efficacy over the control condition. Future studies should refine the mechanistic alignment between stimulation and cognitive interventions to optimize t...

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Comparative chloroplast genomics of Ajania sericea and sympatric Asteraceae species identifies shared candidate molecular features associated with alpine adaptation

Xi Chen 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study showed that the chloroplast genomes of four sympatric Asteraceae species were highly conserved in overall structure, but still exhibited certain differences in gene loss, IR boundary variation, repeat sequence composition, and local hypervariable regions. These features provi...

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Brain Stimulation as a Treatment for Apathy: A Scoping Review of Effects and Safety

Lisa M McTeague 2026-10-20 1 min 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: Investigations into brain stimulation as a treatment for apathy show some promise. Conclusions were limited by small sample sizes, heterogeneous study designs, and variable outcome measures. Randomized trials with common validated apathy instruments, standardized stimulation parameters,...

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Effect of Different Cortical Targets of Transcranial Magnetic Stimulation on Freezing of Gait in Parkinsonism: A Systematic Review

Aliaa Elabd
2026-10-20 1 min 68 words TDCS TACS TRNS

Summary: BACKGROUND AND PURPOSE: Freezing of gait (FOG) is a disabling and episodic gait disturbance in patients with Parkinson's, with varying responses to noninvasive brain stimulation. Transcranial magnetic stimulation (TMS) has been explored as a therapeutic option; however, the optimal cortical target f...

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Sleep Spindles and Slow Oscillations as Potential Biomarkers and Therapeutic Targets in Mild-to-Moderate Alzheimer's Disease

👤 Thien Thanh Dang-Vu 📅 2026-10-20 📖 1 min 📝 35 words TDCS TACS TRNS

Summary: PURPOSE OF REVIEW: This review critically evaluates evidence on sleep spindles (SP), slow oscillations (SOs), and their coordination in mild-to-moderate Alzheimer's disease (AD), focusing on their relationships with cognition, AD biomarkers, and their therapeutic potential.

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

👤 Xin Zhou 📅 2026-10-20 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng 📅 2026-10-21 📖 1 min 📝 53 words TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiscale neural signatures of long-lag repetition learning: a simultaneous fNIRS-EEG study

👤 Ken Pugh 📅 2026-10-17 📖 1 min 📝 38 words FNIRS

Summary: CONCLUSIONS: Combining fNIRS and EEG with multimodal machine learning provides a richer characterization of repetition-based learning than either modality alone, establishing a framework for phenotyping language learning capacity and identifying neural markers of reading difficulties in pediatric po...

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Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

Felipe Balistieri Santinelli 2026-10-18 1 min 70 words FNIRS

Summary: People with multiple sclerosis (pwMS) often exhibit motor fatigability and reduced foot-tapping ability. However, the coupling between performance and cortical activity during fast, repetitive and prolonged ankle movements remains unexplored. We aimed to investigate the neural correlates of a prolon...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

Ming Zhang 2026-10-19 1 min 64 words FNIRS

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Neural dysfunction underlying divergent thinking and its shared mechanisms with protracted withdrawal symptoms in heroin use disorder

👤 Haijun Duan
📅 2026-10-19 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSIONS: These findings reveal a sustained compensatory pathological mechanism and an imbalance in neural resource allocation during complex cognition in HUD individuals. The identified signatures of cortical activity by fNIRS offer reliable biomarkers for the rapid detection of both higher-orde...

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Functional near-infrared spectroscopy as a tool for monitoring treatment-related neurofunctional changes in posttraumatic stress disorder and trauma-related symptomatology: a narrative review

👤 Maria Cotelli
📅 2026-10-20 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Current evidence suggests that fNIRS represents a promising and clinically feasible approach for monitoring treatment-related neurofunctional changes in PTSD and trauma-related symptomatology, particularly during non-pharmacological interventions targeting prefrontal regulatory processe...

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Transcutaneous vagus nerve stimulation during semantic verbal fluency tasks: a high-density diffuse optical tomography study

👤 Li Su 📅 2026-10-20 📖 1 min 📝 55 words FNIRS

Summary: INTRODUCTION: Transcutaneous auricular vagus nerve stimulation (taVNS) is a promising adjunct for post-stroke language recovery. There are no established biomarkers for the neuromodulatory effects of taVNS. High-density diffuse optical tomography (HD-DOT) offers a non-invasive marker of task-related...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

👤 Xuqun You
📅 2026-10-22 📖 1 min 📝 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

Qiuxia Shi 2026-10-22 1 min 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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Electroencephalography Changes in Emotional Recognition: A Signal Processing Approach Using Wavelet Features

Lakshmi Narasimhan 2026-10-17 1 min 43 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Wavelet-based EEG feature analysis provides meaningful and discriminative information for emotion recognition and may provide a robust feature set for future machine learning-based emotion recognition systems. Further studies incorporating classification algorithms and larger data sets a...

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A Compressive Neural Telemetry with Address-Event-μPacket Encoding and Duty-Cycling IR-UWB Transmitter

👤 Yao-Hong Liu 📅 2026-10-18 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: This article presents a networked wireless telemetry for 384-sensor-pixel neural data acquisition, designed to support high throughput, low power consumption, and meter-scale communication distance. The proposed event-based Address-Event-μPacket (AEμP) encoder with differential binary phase shift ke...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang 📅 2026-10-19 📖 1 min 📝 64 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Bridging the Gap Between Design and Access: Toward Visual Scene Displays for Brain-Computer Interface Use in Augmentative and Alternative Communication by Children

Ahad Behboodi 2026-10-19 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: These findings indicate that asymmetry and contextual embedding within scenes may enhance neural engagement and user experience, providing a preliminary foundation for considering photographic VSDs into pediatric BCI research.

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A spatio-temporal attention convolutional network with topology-preserving channel embedding and transformer for EEG decoding

Gang Liu 2026-10-19 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: EEG-based BCI has shown promising applications in motor rehabilitation and emotion recognition. Given the non-stationarity and variability of EEG signals, it is essential to develop classification models that can effectively extract spatio-temporal features while maintaining strong generalization ca...

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The calibration-to-accuracy mapping in P300 spellers does not transport across cohorts and was not restored by tested alignment or local recalibration

Oved Daniel 2026-10-19 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Objective
A calibration-derived score has been related to online P300-speller accuracy in the same session, always within the cohort measured. We evaluated whether a mapping from that score to expected accuracy transports to withheld cohorts.
Approach
Retrospective secondary analysis of ...

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Subjective emotional ratings dissociate brain activation and network patterns during counterfactual outcome processing

Ling Li 2026-10-19 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Human decision-making integrates outcome evaluation, counterfactual comparison, and individual traits. We investigated the neural processing of outcome-related emotions during economic decision-making by classifying feedback according to participants' subjective emotional ratings rather than objecti...

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Delineating neural contributions to electroencephalogram-based speech decoding

Shuntaro Sasai 2026-10-20 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Objective. Speech Brain-computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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An attention-enhanced hybrid CNN-BiLSTM framework for multiclass epileptic seizure recognition using EEG signals

Ahmed H Almutairi 2026-10-22 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Electroencephalogram (EEG) signal classification plays a pivotal role in brain-computer interfaces (BCIs) and the diagnosis of neurological disorders. However, the non-stationary, non-linear, and low signal-to-noise ratio characteristics of EEG signals make accurate feature extraction ...

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Realizing Ultrahigh Polarization and Excellent Thermal Stability in Piezoceramics via Defect-Stress Engineering

👤 Yuhui Huang 📅 2026-10-22 📖 1 min 📝 62 words BRAIN COMPUTER INTERFACE

> **Summary:** Large polarization is critical for improving the piezoelectric and actuation performance of ferroelectric materials. As a promising lead-free candidate, BiFeO(3)-BaTiO(3) inherits the excellent polarization capability of BiFeO(3). However, the actual polarization achieved in its ceramics remains far...

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Jared Kushner Hacked Phone Reveals Secret Negotiations, Millions in Transactions

👤 ck2 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.vanityfair.com/story/jared-kushner-hacked-phone-secret-negotiations Comments URL: <a href="https://news.ycombinator.com/item?id=50201259">https://news.ycombinator.com/item...

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Railpass

👤 CharlesW 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

Summary: Article URL: https://train.samismith.com/ Comments URL: https://news.ycombinator.com/item?id=50201414 Points: 8 # Comments: 3

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Are meta-scientists ignoring philosophy of science?

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-07-04 📖 8 min 📝 1681 words

TWENTY PERCENT STATISTICIAN

Summary: Are meta-scientists ignoring philosophy of science (PoS)? Are they re-inventing the wheel? A recent panel at the Metascience conference engaged with this question, and the first sentence of the abstract states “Critics argue t...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/are-meta-scientists-ignoring-philosophy.html

Type S and M errors as a “rhetorical tool”

👤 noreply@blogger.com (Daniel Lakens)
📅 2025-09-28 📖 17 min 📝 3572 words

TWENTY PERCENT STATISTICIAN

Summary: Update 30/09/2025: I have added a reply by Andrew Gelman below my original blog post. We recently posted a preprint criticizing the idea of Type S and M errors (https://osf.io/2phzb_v1). From our abstract: “While these concepts have been pr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/09/type-s-and-m-errors-as-rhetorical-tool.html

Why we should stop using statistical techniques that have not been adequately vetted by experts in psychology

👤 noreply@blogger.com (Daniel Lakens)
📅 2025-10-29 📖 27 min 📝 5516 words

TWENTY PERCENT STATISTICIAN

Summary: In a recent post on Bluesky, where Richard Morey reflects on a paper he published with Clintin Davis-Stober that points out concerns with the p-curve method (Morey & Davis-Stober, 2025), he writes: <p cla...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/10/why-we-should-stop-using-statistical.html

ERP Decoding for Everyone: Software and Webinar

👤 Steve Luck 📅 2023-06-23 📖 2 min 📝 420 words ERP BOOT CAMP

Summary: You can access the recording h strong>.
You can access the final PDF of the slides <a href="https://ucdavis.box.com/s/f...

🔗 **Read full article:**
https://erpinfo.org/blog/2023/6/23/decoding-webinar

Important Changes to the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-05 📖 2 min 📝 444 words ERP BOOT CAMP

Summary: We are disappointed to announce that we will not be holding a regular 10-day ERP Boot Camp this summer. We have held Boot Camps nearly every summer since 2007, supported by a series of generous grants from NIMH that allowed us to provide scholarships for all attendees. Unf...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/5/changes-to-the-2024-erp-boot-camp

Standardized Measurement Error now available in BrainVision Analyzer and MNE-Python

👤 Steve Luck 📅 2025-12-19 📖 1 min 📝 253 words ERP BOOT CAMP

Summary: <p class="">A few years ago, we developed a universal metric of data quality for averaged ERPs called the standardized measurement error (click here for more information or watch our <a href="https://www.youtube.com/watch?v=tEKsx0p53rs" t...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/12/19/sme-now-available-in-bva-and-mne-python

Transcranial Focused Ultrasound: A Transformative Tool for Intracranial Ablation, Drug Delivery, and Neuromodulation

📅 2025-11-11 📖 1 min 📝 136 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Transcranial focused ultrasound (tFUS) is an emerging neuromodulation and therapeutic technology offering noninvasive, submillimeter precision for targeting deep brain structures. Unlike transcranial magnetic stimulation (TMS) and transcranial electric stimulation (tES), which are limited by depth-f...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11241049

Join the Mailing List

👤 dziura 📅 2025-12-10 📖 1 min 📝 13 words EMBS

> **Summary:** The post Join the Mailing List appeared first on IEEE EMBS.

🔗 **Read full article:**
https://embc.embs.org/2026/#new_tab

EMBS MentorLink: Fostering Careers Together

👤 Deidre Artis 📅 2026-08-18 📖 1 min 📝 14 words EMBS

> **Summary:** The post EMBS MentorLink: Fostering Careers Together appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/embs-mentorlink-fostering-careers-together/

Alterations in the populations of peripheral immune cells in the blood of a mouse model of progressive myoclonus epilepsy of Lafora type

📖 1 min 📝 17 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): L. Baños-Carrión, A. Campos-Rodriguez, M.A. Garcia-Gimeno, P. Sanz

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004580?dgcid=rss_sd_all

Gait-combined transcranial alternating current stimulation around gait cycle frequency over cerebral motor areas modulates spatiotemporal gait variabilities

📖 1 min 📝 20 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Ryosuke Kitatani, Runa Sorimachi, Akane Amano, Rima Watanabe, Naofumi Otsuru, Hideaki Onishi

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Effort perception and endurance running performance following prefrontal transcranial direct current stimulation (tDCS): a randomized crossover-controlled trial

1 min | 27 words | NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Géraldine Martens, Stéphanie Hody, Stephen Bornheim, Luca Angius, Louis De Beaumont, Jean-François Kaux, Aurore Thibaut, Thierry Bury, Florence Schleich

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VO659, a CAG repeat-targeting antisense oligonucleotide, lowers multiple neurotoxic mutant huntingtin proteins in Huntington's disease

Röttgering, B., Weij, R., Bijl, S., Verheul, R., Testerink, J., Toonen, L., Bragg, R., Carroll, J. B., Beekman, C., Datson, N.

2026-10-22 | 1 min | 200 words | BIORXIV NEUROSCIENCE

Summary: Huntington's disease (HD) is caused by a CAG repeat expansion in the huntingtin (HTT) gene, encoding an elongated polyglutamine stretch in the corresponding mutant HTT (mHTT) protein. HD pathogenesis is driven by multiple neurotoxic mHTT protein species, including HTT1a and repeat-associated non-AUG...

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A brain-behavior generative model links individual brain dynamics to perceptual decisions

👤 Wang, C., Yang, G., Wang, S., Luo, H.

📅 2026-10-22 📖 1 min 📝 191 words

BIORXIV NEUROSCIENCE

Summary: Perceptual decision speed varies across individuals and trials. Explaining this variability requires linking individual brain anatomy, neural dynamics, and behavior, yet whole-brain models typically reproduce brain signals without behavior, whereas behavioral models lack anatomical constraints. Here...

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Aberrant cortical reorganization and astrogliosis underlie memory deficit post-traumatic brain injury: a longitudinal multiscale multimodal study

👤 Tan, J., Patel, S., Xu, T., Dorr, A., Almanza, D. L. V., Osmann, J., Rabelo, T., Vecchio, L., McLaurin, J., Wang, C., Lipsman, N., Burke, M., Aubert, I., Hamani, C., Stefanovic, B., Goubran, M.

📅 2026-10-22 📖 1 min 📝 135 words

BIORXIV NEUROSCIENCE

Summary: Integrating high-dimensional neuroimaging data across spatial scales remains a fundamental bottleneck in quantitative neuroscience. Here, we apply emerging analytical techniques including functional connectome gradient mapping and 3D segmentation of whole-brain cellular resolution microscopy data in...

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Multiquantal transmission supports dynamic synaptic strength in vivo

Patel, P., Higley, M.
2026-10-22 1 min 136 words BIORXIV NEUROSCIENCE

Summary: In the central nervous system, synaptic transmission occurs through the probabilistic release of neurotransmitter vesicles following presynaptic action potentials (AP). Traditional models posit a single vesicle release event per AP, though experimental evidence indicates multivesicular release (MVR)...

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Early Insulin Signaling Defects and Oxidative Stress Kinase Activation Precede Tau Pathology in PS19 Tau Transgenic Mice

Elahi, M., Burd, I., Yang, P., Shen, W.-B.
2026-10-22 1 min 152 words BIORXIV NEUROSCIENCE

Summary: Pathological tau accumulation is a hallmark of Alzheimers disease (AD), yet its relationship with impaired insulin signaling remains unclear. Previously, we reported altered IRS1 phosphorylation reduced p IR beta, and GLUT3 expression in young AD transgenic mice prior to overt pathology. Here, we in...

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Dopaminergic Medication Improves Force Control and Modifies Motor Unit Activity During the Steadiest and Least Steady Periods in Parkinsons Disease

Alaei, P., Holobar, A., Wile, D. J., Jakobi, J. M.

2026-10-22 1 min 248 words

BIORXIV NEUROSCIENCE

Summary: Parkinsons disease (PD) impairs steady force production, but how dopaminergic medication alters force control and the motor unit (MU) activity is unclear. This study examined force steadiness and biceps brachii MU behavior in 16 people with PD ON and OFF dopaminergic medication. Participants perform...

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Structured recurrence of latent neural processes predicts memory recognition performance

Christensen, K. F., Klarlund, M., Fernandez-Rubio, G., Rosso, M., Streton, S., Schiassi, M., Kringelbach, M. L., Vuust, P., Bonetti, L.

2026-10-22 1 min 202 words BIORXIV NEUROSCIENCE

Summary: Understanding cognition requires explaining not only which neural processes are engaged, but how their relationships evolve through time. We tested whether the recurrent organization of low-dimensional whole-brain dynamics relates to auditory recognition behaviour beyond activation magnitude. After ...

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Contribution of the Late Positive Potential to Emotional Memory: A Joint Neuro-Cognitive Approach to Memory Prediction

Hellerstedt, R., Gunn, J., Talmi, D.
2026-10-22 1 min 297 words

BIORXIV NEUROSCIENCE

Summary: Emotional events are better remembered than neutral events. The emotional Context Maintenance and Retrieval model (eCMR) explains this effect by simulating preferential encoding of emotional items, implemented by a free parameter that scales the learning rate. Incorporating a neural signal measured ...

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Withdrawal latency in a human hot plate analog tracks inter-individual differences in pain sensitivity

Keuter, L., Goltermann, O., Jessberger, J., Asan, L., Büchel, C.
2026-10-22 1 min 245 words

BIORXIV NEUROSCIENCE

Summary: Pain research relies mainly on self-report in humans and nocifensive behavior in animals. This asymmetry limits cross-species translation and neglects movement as integral to human pain responses. We therefore developed a human analog of the rodent hot plate test and tested whether temporal and kine...

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FOXG1 spatiotemporally orchestrates subtype fate specification and patch/matrix identity of striatal spiny projection neurons

Shuangshuang DongLijuan SunFeng XiongFeier HuangBaoshen ZhangYonghui WuDa MiWei HuangRu BaHanchuan PengChunjie ZhaoaJiangsu Provincial Key Laboratory of Critical Care Medicine, Department of Histology and Embryology, School of Medicine, Southeast University, Nanjing, Jiangsu 210009, ChinabNew Cornerstone Science Laboratory, Institute for Brain and Intelligence, Southeast University, Nanjing 210096, ChinacSchool of Biological Science and Medical Engineering, Southeast University, Nanjing, Jiangsu 210096, Chinadhttps://ror.org/03cve4549State Key Laboratory of Membrane Biology, International Data Group/McGovern Institute for Brain Research, Tsinghua-Peking Joint Centre for Life Sciences, School of Life Sciences, Tsinghua University, Beijing 100084, Chinaehttps://ror.org/034t30j35Institute of Neuroscience, Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai 200031, Chinafhttps://ror.org/013q1eq08New Cornerstone Science Laboratory, Institute for Brain and Intelligence, Fudan University, Shanghai 200433, Chinaghttps://ror.org/013q1eq08Shanghai Academy of Natural Sciences, Fudan University, Shanghai 200433, ChinahDepartment of Anesthesiology, Zhongda Hospital, Southeast University, Nanjing 210009, China

📅 2026-10-15 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 42, October 2026.
SignificanceThe striatum functions as a gateway to the basal ganglia and plays crucial roles in regulating movement, emotion, reward processing, and cognitive functions. The striatal projection neurons consist...

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https://www.pnas.org/doi/abs/10.1073/pnas.2603031123?af=R

Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu
2026-10-16 1 min 222 words FRONTIERS NEUROROBOTICS

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Mapping unintended trigeminal nerve activation during transcranial alternating current stimulation (tACS): a computational study

Sulagna Sahu, Munish Chauhan, Saurav Z K Sajib, Enock Boakye and Rosalind J Sadleir

2026-10-15 1 min 211 words JOURNAL NEURAL ENGINEERING

Summary: Objective. To systematically characterize activation thresholds for peripheral trigeminal nerve branches under clinical transcranial alternating current stimulation (tACS) parameters using a computational modeling approach. Approach. An MRI-derived finite element head model was coupled with multicom...

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Involuntary facial muscle activity during imagined vocalisation contaminates EEG and enables emotion decoding

Yichen Tang, Paul M Corballis and Luke E Hallum

2026-10-15 1 min 281 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Decoding imagined speech from electroencephalography (EEG) recordings is potentially useful for brain-computer interfaces. Previous studies have focused on decoding semantic information from EEG, leaving the decoding of emotion — an important component of human communication — largely une...

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Assessing consciousness by near-nerve electrostimulation in intensive rehabilitation

Piergiuseppe Liuzzi, Sara Ballanti, Andrea Mannini, Bahia Hakiki, Rachele Burali, Tanita Toci, Maria Chiara Carrozza, Calogero Maria Oddo and Antonello Grippo

2026-10-15 1 min 172 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Consciousness diagnosis in prolonged disorders of consciousness (pDoC) relies on the Coma Recovery Scale–Revised, yet consciousness or its potential for recovery may escape behavioural detection. We tested whether near-nerve stimulation (NNS) of the median nerve, compared to standard surf...

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Prioritized nonlinear modeling of shared dynamics across neural populations

Trisha Jha, Dong Song and Maryam M Shanechi

2026-10-15 1 min 284 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Advances in neural recording technology enable simultaneous measurements across multiple brain regions, providing new opportunities to study inter-regional interactions. However, several challenges arise when developing nonlinear dynamical models of cross-region interactions. In particula...

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https://iopscience.iop.org/article/10.1088/1741-2552/ae9bee

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt

2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

> **Summary:** The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

> **Summary:** The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Birth of the Nigerian Sleep Society: A Milestone for Sleep Medicine in Nigeria and Africa

👤 G E Erhabor 📅 2026-10-20 📖 1 min 📝 72 words LOW VISION

Summary: Sleep disorders are significant and are of major public health concern with high socioeconomic impact. However, they are under-recognised and under-treated in Nigeria and most of Africa, because of poor access to sleep specialists, sleep studies and limited treatment options. Prior to the inaugurati...

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Knowledge gaps and practices for glaucoma among adults in Palestine: a cross-sectional study

Farid K Abu Sham'a 2026-10-20 1 min 51 words LOW VISION

Summary: CONCLUSIONS: Although awareness of glaucoma is moderate in Palestine, comprehensive knowledge remains low, particularly among high-risk populations. Public health interventions should target education beyond name recognition, emphasizing asymptomatic progression, irreversible vision loss, and risk f...

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Sn/Cl Co-Doped ZnO Cathode Interfacial Layer for High-Performance Organic Photodetectors

Kai Zhang 2026-10-20 1 min 66 words LOW VISION

Summary: Organic photodetectors (OPDs) hold significant promise for applications spanning medical monitoring, bioimaging and night vision due to their unique advantages, such as flexibility, light weight, and potential for large-area fabrication. Nevertheless, the dark current density (Jd) of OPDs exhibits a...

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A thermopile array and stereo vision fusion system for small-tube temperature sensing

👤 Dongfang Hu 📅 2026-10-20 📖 1 min 📝 71 words LOW VISION

Summary: This paper presents a non-contact temperature measurement scheme for small-diameter tubes by fusing a low-resolution thermopile array (TPA) with visible-light stereo vision. Stereo vision locates the tube in three dimensions. Then, this geometry is projected onto the TPA imaging plane to constrain t...

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Cost-effective long-range measurement of structural displacement using a two-stage method integrating super-resolution reconstruction and sub-pixel registration

👤 Xiaosong Li 📅 2026-10-20 📖 1 min 📝 61 words LOW VISION

Summary: Displacement is an imperative metric for assessing structural performance. However, the accuracy of structural displacement measurement is constrained by the physical resolution of the camera, particularly in long-range monitoring scenarios. To address these constraints, a cost-effective long-range ...

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Physics-aware restoration and calibrated soft demodulation for turbulence-degraded OAM mode recognition

👤 Jun Zhu 📅 2026-10-20 📖 1 min 📝 58 words LOW VISION

Summary: Turbulence-degraded orbital angular momentum (OAM) mode recognition is critical for free-space optical (FSO) communication but is limited by atmospheric distortion. We propose a cascaded receiver integrating a vision transformer-based Uformer-OAM intensity-restoration network, an SE-ResNet-18 neural...

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Image-free ghost edge detection at ultra-low sampling rates via a Vision Mamba network

👤 Rui Wang 📅 2026-10-20 📖 1 min 📝 66 words LOW VISION

Summary: Edge detection is essential for extracting structural boundaries in visual scenes, but conventional deep-learning-based methods remain sensitive to degradations in the input images and involve redundant full-image acquisition. Image-free ghost edge detection provides an efficient alternative by dire...

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Ex vivo evaluation of corneal permeation and irritation potential associated with proparacaine hydrochloride misuse

Yolima Baena 2026-10-20 1 min 35 words LOW VISION

Summary: CONCLUSION: Proparacaine demonstrates dose-dependent permeation and histopathological changes, emphasizing the risks associated with misuse. The ex vivo model provides relevant and ethically aligned evidence to support ocular hazard assessment through the characterizing of histopathological changes.

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Shared variance across FrACT visual tasks: Robust positive associations but inconclusive latent structure

Luca Battaglini 2026-10-21 1 min 68 words LOW VISION

Summary: A long-standing question in vision science is whether individual differences in visual performance reflect a general visual ability or a set of largely task-specific processes. Previous studies have often reported weak associations across heterogeneous perceptual tasks, challenging the idea of a com...

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New challenges in horticultural IPM: artificial intelligence can enhance the use of biocontrol agents

Andrea Lovazzano 2026-10-22 1 min 59 words LOW VISION

Summary: Modern horticulture faces the dual imperatives of increasing global food production while mitigating the severe environmental and health impacts of conventional chemical pesticides. Biological control agents (BCAs), which utilize beneficial microorganisms, macroorganisms, semiochemical and botanical...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

👤 Béatrice Berçot 📅 2026-09-04 📖 1 min 📝 60 words BRAILLE

> **Summary:** CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan 📅 2026-09-09 📖 1 min 📝 57 words BRAILLE

> **Summary:** Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Written exposure therapy augmented with transcranial direct current stimulation for posttraumatic stress disorder: A pilot randomized clinical trial

👤 STRONG STAR Consortium 📅 2026-10-18 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: Written exposure therapy (WET) is an evidence-based, trauma-focused psychotherapy for posttraumatic stress disorder (PTSD) that has been tested with several trauma-exposed samples. Although many individuals who receive WET and other trauma-focused psychotherapies experience a significant reduction i...

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Transcranial Direct Current Stimulation and Cranial Electrotherapy Stimulation in Athletes: Optimizing Athletic Performance and Sleep Quality Through Neuromodulation

👤 Paulina Gajniak
📅 2026-10-19 📖 1 min 📝 74 words

TDCS TACS TRNS

Summary: Neuromodulation is a procedure relying on the delivery of specific stimuli to selected sites of the nervous system and hereby influencing its activity. Even though the underlying neurobiological mechanisms are not yet fully elucidated and the research is still ongoing, neuromodulation is already bei...

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Clinical outcomes and safety of Five Element Acupuncture for preventive treatment of primary headache disorders: a multi-center real-world longitudinal observational study

Shengyuan Zhou
2026-10-19 1 min 49 words TDCS TACS TRNS

Summary: CONCLUSIONS: This real-world exploratory observational study suggests that FEA is associated with significant improvements in headache days, headache frequency, headache intensity, analgesic use, and quality of life, with sustained effects noted during follow-up. These findings provide preliminary o...

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Home Transcranial Direct Current Stimulation With Gamified Cognitive Control Training for Depression: A Randomized Clinical Trial

Mor Nahum 2026-10-19 1 min 45 words TDCS TACS TRNS

Summary: CONCLUSIONS AND RELEVANCE: This RCT of home-based tDCS combined with CCT in adults with MDD was feasible and safe but did not demonstrate superior efficacy over the control condition. Future studies should refine the mechanistic alignment between stimulation and cognitive interventions to optimize t...

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Comparative chloroplast genomics of Ajania sericea and sympatric Asteraceae species identifies shared candidate molecular features associated with alpine adaptation

Xi Chen 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study showed that the chloroplast genomes of four sympatric Asteraceae species were highly conserved in overall structure, but still exhibited certain differences in gene loss, IR boundary variation, repeat sequence composition, and local hypervariable regions. These features provi...

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Brain Stimulation as a Treatment for Apathy: A Scoping Review of Effects and Safety

Lisa M McTeague 2026-10-20 1 min 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: Investigations into brain stimulation as a treatment for apathy show some promise. Conclusions were limited by small sample sizes, heterogeneous study designs, and variable outcome measures. Randomized trials with common validated apathy instruments, standardized stimulation parameters,...

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Effect of Different Cortical Targets of Transcranial Magnetic Stimulation on Freezing of Gait in Parkinsonism: A Systematic Review

Aliaa Elabd
2026-10-20 1 min 68 words TDCS TACS TRNS

Summary: BACKGROUND AND PURPOSE: Freezing of gait (FOG) is a disabling and episodic gait disturbance in patients with Parkinson's, with varying responses to noninvasive brain stimulation. Transcranial magnetic stimulation (TMS) has been explored as a therapeutic option; however, the optimal cortical target f...

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Sleep Spindles and Slow Oscillations as Potential Biomarkers and Therapeutic Targets in Mild-to-Moderate Alzheimer's Disease

👤 Thien Thanh Dang-Vu 📅 2026-10-20 📖 1 min 📝 35 words TDCS TACS TRNS

Summary: PURPOSE OF REVIEW: This review critically evaluates evidence on sleep spindles (SP), slow oscillations (SOs), and their coordination in mild-to-moderate Alzheimer's disease (AD), focusing on their relationships with cognition, AD biomarkers, and their therapeutic potential.

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

👤 Xin Zhou 📅 2026-10-20 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

Fanshuo Zeng 2026-10-21 1 min 53 words TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Neckline-free open-source cap design and improved motion resistance for perioperative fNIRS procedures

Audrey K Bowden 2026-10-17 1 min 60 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) caps are limited by motion artifacts and unstable optode-scalp coupling, which degrade signal quality during extended and naturalistic monitoring sessions. We present a fabric-based, chinstrap-free headcap designed to improve mechanical stability and opt...

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Multiscale neural signatures of long-lag repetition learning: a simultaneous fNIRS-EEG study

Ken Pugh 2026-10-17 1 min 38 words FNIRS

Summary: CONCLUSIONS: Combining fNIRS and EEG with multimodal machine learning provides a richer characterization of repetition-based learning than either modality alone, establishing a framework for phenotyping language learning capacity and identifying neural markers of reading difficulties in pediatric po...

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Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

Felipe Balistieri Santinelli 2026-10-18 1 min 70 words FNIRS

Summary: People with multiple sclerosis (pwMS) often exhibit motor fatigability and reduced foot-tapping ability. However, the coupling between performance and cortical activity during fast, repetitive and prolonged ankle movements remains unexplored. We aimed to investigate the neural correlates of a prolon...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang
📅 2026-10-19 📖 1 min 📝 64 words FNIRS

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Neural dysfunction underlying divergent thinking and its shared mechanisms with protracted withdrawal symptoms in heroin use disorder

👤 Haijun Duan
📅 2026-10-19 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSIONS: These findings reveal a sustained compensatory pathological mechanism and an imbalance in neural resource allocation during complex cognition in HUD individuals. The identified signatures of cortical activity by fNIRS offer reliable biomarkers for the rapid detection of both higher-orde...

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Functional near-infrared spectroscopy as a tool for monitoring treatment-related neurofunctional changes in posttraumatic stress disorder and trauma-related symptomatology: a narrative review

👤 Maria Cotelli
📅 2026-10-20 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Current evidence suggests that fNIRS represents a promising and clinically feasible approach for monitoring treatment-related neurofunctional changes in PTSD and trauma-related symptomatology, particularly during non-pharmacological interventions targeting prefrontal regulatory processe...

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Transcutaneous vagus nerve stimulation during semantic verbal fluency tasks: a high-density diffuse optical tomography study

👤 Li Su 📅 2026-10-20 📖 1 min 📝 55 words FNIRS

Summary: INTRODUCTION: Transcutaneous auricular vagus nerve stimulation (taVNS) is a promising adjunct for post-stroke language recovery. There are no established biomarkers for the neuromodulatory effects of taVNS. High-density diffuse optical tomography (HD-DOT) offers a non-invasive marker of task-related...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

Xuqun You 2026-10-22 1 min 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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An exploration in mental health care: The launch of China's first psychiatric noninvasive brain-computer interface (BCI) and neuromodulation clinic

Tianhong Zhang 2026-10-17 1 min 2 words BRAIN COMPUTER INTERFACE

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A sliding window and Fourier decomposition method based 1D-CNN framework for motor imagery EEG classification

M Ramakrishnan 2026-10-17 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) signal classification is fundamental requirement for effective motor imagery (MI) brain-computer interface (BCI) applications. However, current MI classification methods often depend on handcrafted spatial filtering or complex time-frequency transformations that increase...

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Electroencephalography Changes in Emotional Recognition: A Signal Processing Approach Using Wavelet Features

Lakshmi Narasimhan 2026-10-17 1 min 43 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Wavelet-based EEG feature analysis provides meaningful and discriminative information for emotion recognition and may provide a robust feature set for future machine learning-based emotion recognition systems. Further studies incorporating classification algorithms and larger data sets a...

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https://pubmed.ncbi.nlm.nih.gov/42890958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261022152458&v=2.20.1

A Compressive Neural Telemetry with Address-Event-μPacket Encoding and Duty-Cycling IR-UWB Transmitter

👤 Yao-Hong Liu 📅 2026-10-18 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: This article presents a networked wireless telemetry for 384-sensor-pixel neural data acquisition, designed to support high throughput, low power consumption, and meter-scale communication distance. The proposed event-based Address-Event-μPacket (AEμP) encoder with differential binary phase shift ke...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang 📅 2026-10-19 📖 1 min 📝 64 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Bridging the Gap Between Design and Access: Toward Visual Scene Displays for Brain-Computer Interface Use in Augmentative and Alternative Communication by Children

Ahad Behboodi 2026-10-19 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: These findings indicate that asymmetry and contextual embedding within scenes may enhance neural engagement and user experience, providing a preliminary foundation for considering photographic VSDs into pediatric BCI research.

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A spatio-temporal attention convolutional network with topology-preserving channel embedding and transformer for EEG decoding

Gang Liu 2026-10-19 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: EEG-based BCI has shown promising applications in motor rehabilitation and emotion recognition. Given the non-stationarity and variability of EEG signals, it is essential to develop classification models that can effectively extract spatio-temporal features while maintaining strong generalization ca...

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https://pubmed.ncbi.nlm.nih.gov/42896470/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261022152458&v=2.20.1

The calibration-to-accuracy mapping in P300 spellers does not transport across cohorts and was not restored by tested alignment or local recalibration

Oved Daniel 2026-10-19 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Objective
A calibration-derived score has been related to online P300-speller accuracy in the same session, always within the cohort measured. We evaluated whether a mapping from that score to expected accuracy transports to withheld cohorts.
Approach
Retrospective secondary analysis of ...

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Subjective emotional ratings dissociate brain activation and network patterns during counterfactual outcome processing

Ling Li 2026-10-19 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Human decision-making integrates outcome evaluation, counterfactual comparison, and individual traits. We investigated the neural processing of outcome-related emotions during economic decision-making by classifying feedback according to participants' subjective emotional ratings rather than objecti...

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Delineating neural contributions to electroencephalogram-based speech decoding

Shuntaro Sasai 2026-10-20 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Objective. Speech Brain-computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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https://pubmed.ncbi.nlm.nih.gov/42899673/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261022152458&v=2.20.1

Monthly Updates [March]

2026-03-01 2 min 571 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/mar-2026

Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 954 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/june-2026

Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 700 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/aug-2026

Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1079 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/sept-2026

Timed every test in my pytest suite: 3% of the tests ate 77% of the time

👤 /u/lxl2008 📅 2026-10-22 📖 2 min 📝 402 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Some context first: I'm one person building an open-source research tool, and most of the code is written by AI coding agents (Claude Code and Codex) running in parallel. They write code quickly. What slowed everything down was the test suite: on one recent change t...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xcpai5/timed_every_test_in_my_pytest_suite_3_of_the/

Mini brain organoids 'reading' Braille key step towards biohybrid computing

👤 CharlesW 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.frontiersin.org/news/2026/10/16/mini-brain-organoids-reading-braille-biohybrid-computing-frontiers-artificial-intelligence<...

🔗 **Read full article:**
https://www.frontiersin.org/news/2026/10/16/mini-brain-organoids-reading-braille-biohybrid-computing-frontiers-artificial-intelligence

A farewell to Arc browser

👤 suhdonghwi 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://arc.net/ Comments URL: https://news.ycombinator.com/item?id=50197601 Points: 134 # Comments: 100

🔗 **Read full article:**
https://arc.net/

Nvidia OpenShell

👤 niemandhier 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/NVIDIA/OpenShell Comments URL: https://news.ycombinator.com/item?id=50198675 Points: 82 # Comments: 39

🔗 **Read full article:**
https://github.com/NVIDIA/OpenShell

Show HN: agent-usage-guard – a Claude Code mod to protect your usage window

👤 zzyyxx332211 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/rbartoli/agent-usage-guard Comments URL: https://news.ycombinator.com/item?id=50198936 Points: 9 # Comments: 6

🔗 **Read full article:**
https://github.com/rbartoli/agent-usage-guard

AI analysis of CT scans finds 42.9% of patients have coronary inflammation

👤 brandonb 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: <a href="https://www.biograph.com/newsroom/biograph-completes-first-u.s.-study-with-caristo-diagnostics%E2%80%99-ai-powered-coronary-inflammation-detection-finding-risk-in-over-42-of-asymptomatic-patients">https://www.biograph.com/newsroom/biograph-completes-first-u.s.-study-with-car...

🔗 **Read full article:**
https://www.biograph.com/newsroom/biograph-completes-first-u.s.-study-with-caristo-diagnostics%E2%80%99-ai-powered-coronary-inflammation-detection-finding-risk-in-over-42-of-asymptomatic-patients

AI and the Enclosure of the Scientific Commons

👤 jrepinc 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://arxiv.org/abs/2610.27251 Comments URL: https://news.ycombinator.com/item?id=50199230 Points: 14 # Comments: 1

🔗 **Read full article:**
https://arxiv.org/abs/2610.27251

Award-Winning AI Literature

👤 jamesjyu 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://jamesyu.org/award-winning-ai-literature/ Comments URL: https://news.ycombinator.com/item?id=50199232 Points: 13 # Comments: 3

🔗 **Read full article:**
https://jamesyu.org/award-winning-ai-literature/

Show HN: UI micro interactions prompts for AI

👤 elwingo1 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.typeui.sh/micro-interactions Comments URL: https://news.ycombinator.com/item?id=50199290 Points: 12 # Comments: 2

🔗 **Read full article:**
https://www.typeui.sh/micro-interactions

Show HN: Build a world on your phone, explore it in XR on Vision Pro and Quest

👤 suejunghuh0 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://3d.camilleworld.com/play/ Comments URL: https://news.ycombinator.com/item?id=50199349 Points: 3 # Comments: 0

🔗 **Read full article:**
https://3d.camilleworld.com/play/

Hedra is a 3D open world game I wrote from scratch while in highschool

👤 maxilevi 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/maxilevi/project-hedra/tree/pre-ai Comments URL: https://news.ycombinator.com/item?id=50199389 Points: 13 # Comments...

🔗 **Read full article:**
https://github.com/maxilevi/project-hedra/tree/pre-ai

Registration is now full for the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-16 📖 1 min 📝 106 words ERP BOOT CAMP

> **Summary:** The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Call for Student/YP Deputy Editor

👤 Deidre Artis 📅 2026-09-29 📖 1 min 📝 14 words EMBS

Summary: The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Vitamin B6 augments antidepressant activity of venlafaxine by restoring cortical EEG oscillations, behavior and oxidative biomarkers in chronic unpredictable mild stress (CUMS)-induced model of depression-like behavior

📖 1 min 📝 30 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Fashwa Khan Tareen, Sana Javaid, Waseem Ashraf, Syed Muhammad Muneeb Anjum, Rana Muhammad Zahid Mushtaq, Asad Abrar, Faleh Alqahtani, Imran Imran

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004506?dgcid=rss_sd_all

Stage-specific exposure to an activity-permissive medium enhances neuronal activity in oligodendrocyte-enriched cortical organoids

📖 1 min 📝 34 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Clara Chung, Michelle J. Kim, Grace Field, Konstantina Pilarinos, Elizabeth K. Kharitonova, Natalie Baker Campbell, Marie Shi, Christopher V. Gabel, Joseph L. Orofino, Ella Zeldich

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004634?dgcid=rss_sd_all

Cue-responsive mPFC neuronal populations are associated with response execution and response withholding in a Go/No-Go task

📖 1 min 📝 20 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Shunqi Wang, Yiting Li, Yize Cai, Xinyu Yang, Chong Sun, Chaolin Ma

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006640?dgcid=rss_sd_all

Relationships between speech-in-noise comprehension, lip reading ability and cortical speech tracking in talking avatars as well as natural talking heads

📖 1 min 📝 22 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Jasmin Riegel, Alina Schüller, Constantin Jehn, Alexander Wißmann, Steffen Zeiler, Dorothea Kolossa, Tobias Reichenbach

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926005975?dgcid=rss_sd_all

Interplay of action-based prediction and top-down attention: EEG evidence for joint modulation of late perceptual processing

📖 1 min 📝 16 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Thomas Holstein, Jean-Christophe Sarrazin, Bruno Berberian, Andrea Desantis

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006087?dgcid=rss_sd_all

Multivariate prediction of white matter BOLD activity and its relationship to marginal connectivity

📖 1 min 📝 15 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Muwei Li, Zhaohua Ding, John C Gore

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006129?dgcid=rss_sd_all

Amplitude-conditioned brain dynamics reveal widespread negative modulation and targeted positive modulation of functional networks

📖 1 min 📝 15 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Mohammad Reza Salehi, Ali BashirGonbadi, Hamid Soltanian-Zadeh

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006191?dgcid=rss_sd_all

Precise modeling of task-related sensorimotor activation based on simultaneous surface electromyography

📖 1 min 📝 22 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Monika Jasenská, Pavel Hok, Martin Kojan, Ondřej Burkot, Barbora Kolářová, Aleš Holobar, Petr Hluštík

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006154?dgcid=rss_sd_all

Neural dynamics of auditory attention: Interactions between top-down and bottom-up processes

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): S. Graceffo, M. Elhilali

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006002?dgcid=rss_sd_all

Neurovascular coupling in simultaneous EEG-fNIRS studies: A scoping review of analytical frameworks and clinical applications

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Yu Wu, Rajamanickam Yuvaraj, Stéphane Perrey, Hasan Ayaz, Wei-Peng Teo

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006130?dgcid=rss_sd_all

A sparse neural code for acoustic targets enables speech-like vocal flexibility in parrots

👤 Zhao, Z., Plump, A. N., Jones, C., Teoh, H. K., Kardon, B., Goldberg, J. H. 📅 2026-10-22 📖 1 min

📝 137 words BIORXIV NEUROSCIENCE

Summary: English speech flexibly recombines 44 phonemes into >20,000 words and limitless phrases. Here we identify a novel neural code supporting combinatorial vocal control in budgerigar parrots. Deep learning based acoustic analyses show that both English imitated and natural warble songs are composed of s...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.21.760739v1?rss=1

Activation of basal forebrain cholinergic neurons promotes the attribution of incentive salience to reward cues in male rats

👤 Rakowski, E. A., King, C. P., Mukherjee, A., Holmes, E., Hannan, L., Loney, G. C., Meyer, P. J.

📅 2026-10-22 📖 1 min 📝 257 words BIORXIV NEUROSCIENCE

Summary: Background: Reward-associated environmental stimuli (cues) can gain incentive salience in some individuals, thereby becoming attractive, reinforcing, and instigators of motivational states. In rodents, the tendency to attribute incentive salience can be measured using a Pavlovian Conditioned Approac...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.20.760494v1?rss=1

Constricted pupil states index neural replay during wakeful rest in humans

Dash, D., Claudino, L., Sugata, H., Iwane, F., Cohen, L., Buch, E. R.

2026-10-22 1 min

160 words

BIORXIV NEUROSCIENCE

Summary: Neural replay has been linked to skill consolidation during wakeful rest, yet its physiological context in humans remains poorly understood. In animals, hippocampal ripples and associated replay occur more frequently within intrinsically oriented brain states associated with constricted pupils, sugg...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.21.760673v1?rss=1

Dopaminergic reward signals shape foraging decisions through learned value expectations

Kingsbury, L. E., May, H., Uchida, N.

2026-10-22 1 min 261 words

BIORXIV NEUROSCIENCE

Summary: A fundamental function of the nervous system is to direct animals towards life-sustaining resources and enable adaptive strategies for acquiring them. Foraging, the challenge of securing these resources from the environment, requires making decisions that balance local gains against costs, opening a...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.21.760765v1?rss=1

Cross-cohort exosomal miRNA profiling identifies a robust Parkinson’s disease signature with multi-omics evidence for mitochondrial and immune pathology

📅 2026-10-22 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41531-026-01566-7

In vivo replication of a fully functional nerve net in a clonally reproducing animal

📅 2026-10-22 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11184-x

Occupational sex segregation is strongly predicted by psychological sex differences

📅 2026-10-22 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-75691-x

Correction for Liu et al., A spinoreticular pathway mediates nocifensive responses to noxious mechanical stimuli

📅 2026-10-21 📖 1 min 📝 15 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2635736123?af=R

Compensation of the effects of temperature on a motor system in the crab, Cancer borealis

👤 Kathleen JacquerieJames M. DiMartinoAnanya DalalJi ZengEve Marderahttps://ror.org/05abbep66Volen Center and Biology Department, Brandeis University, Waltham, MA 02454

📅 2026-10-21 📖 1 min 📝 50 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceCold-blooded animals that live in climates with significant seasonal changes in ambient temperature must have myriad mechanisms to function over a wide range of environmental conditions. We explore...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2625185123?af=R

Translational control of microglial metabolism, inflammatory responses, and survival by the eIF4E-binding proteins

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📅 2026-10-21 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceThe link between Alzheimer’s disease (AD) risk and microglial function is well-known, yet the mechanisms governing microglial responses remain unclear. Postmortem and experimental studies demonstra...

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Naturalistic spatial education engages distinct neural processes for mental rotation, cognitive load, and linguistic demands

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📅 2026-10-21 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceSpatial visualization skill is a foundational component of human intelligence and a powerful lever for success in science, technology, engineering, and mathematics (STEM). Yet little is known about...

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C1ql2 gates stress vulnerability by regulating microglia-mediated synaptic pruning in the BLA-vHPC circuit

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📅 2026-10-21 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceStress-related emotional disorders involve maladaptive synaptic remodeling in the basolateral amygdala (BLA), which is highly heterogeneous across neurons projecting to distinct downstream targets....

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https://www.pnas.org/doi/abs/10.1073/pnas.2600854123?af=R

Window leakage in clinical EEG machine learning: matched subject-independent evaluation across ADHD and Parkinson cohorts

Burak Aggul 2026-09-29 1 min 255 words FRONTIERS NEUROINFORMATICS

Summary: Windowed electroencephalography (EEG) pipelines can produce misleadingly high performance when segments from the same participant occur in both training and testing. We define this contamination as window leakage and test it in two clinically and technically distinct public cohorts: 121 children (61...

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BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

Tingwei Quan 2026-10-07 1 min 139 words FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

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Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

Bocheng Wang 2026-10-07 1 min 217 words FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

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Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

Ansel Y. Rodríguez-González 2026-10-07 1 min 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

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Constrained Hebbian learning supports efficient representational allocation under structural constraints

Andreas Knoblauch
2026-10-07 1 min 253 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionBiological systems are constrained by anatomy and metabolism, with synaptic maintenance consuming substantial energy and spatial limitations restricting connectivity. These constraints favor neural codes that compress behaviorally relevant information into low-redundancy patterns. Here, ...

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Mechanonociceptive tactile feedback enhances prosthetic grasp control during repeated human-in-the-loop interaction

Elif Hocaoglu
2026-10-15 1 min 215 words

FRONTIERS NEUROROBOTICS

Summary: Upper-limb prosthetic systems continue to face a fundamental challenge in providing sensory feedback that remains functionally effective during repeated interaction. This study investigates innocuous painful feedback (IPF), a controlled mechanonociceptive tactile modality intended to provide salient...

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Brain-inspired predictive visual attention for spatiotemporal semantic understanding of complex film and television videos

Peng Han
2026-10-22 1 min 275 words FRONTIERS NEUROROBOTICS

Summary: Due to the unique characteristics of film and television, such as montage editing and narrative structure jumps, semantically salient targets frequently experience temporal feature drift, making it difficult for existing models to maintain stable semantic understanding output when keyframes are lost...

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Single-channel EEG denoising and artifact removal: a systematic scoping review of methods, benchmarking, and reproducibility

Usman Qamar Shaikh, Anubha Manju Kalra, Andrew Lowe and Imran Khan Niazi
2026-10-14

1 min 239 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Single-channel and wearable electroencephalography (EEG) reduce acquisition burden but limit the spatial information available to conventional multichannel artifact-removal methods. We conducted a systematic scoping review of single-channel-compatible EEG artifact removal, linking method ...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb0ce

Phase-locked closed-loop ultrasound stimulation of hippocampal slow gamma modulates CA3-CA1 coherence and improves spatial memory in APP/PS1 mice

👤 Zhenyu Xie, Hui Ji, Mürsel Karadas, Yiyao Zhang and Yi Yuan

📅 2026-10-14 📖 1 min 📝 235 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Impaired hippocampal slow gamma oscillations and CA3-CA1 dysfunction may contribute to spatial memory deficits in early Alzheimer's disease (AD). This study examined whether phase-locked closed-loop transcranial ultrasound stimulation (TUS) could modulate CA3-CA1 activity and improve memo...

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Ordered onset of regional desynchronization and a transition in movement-class discriminability during cue-paced motor imagery

👤 Seho Lee

📅 2026-10-21 📖 1 min 📝 298 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Motor imagery recruits a distributed network whose components and large-scale organization are well characterized. The temporal order in which those components are engaged is not: the latency at which each region begins to desynchronize within a trial has not been compared statistically i...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb161

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

Robert Oostenveld 2026-02-12 1 min 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Behaviorally-relevant features of observed actions dominate cortical representational geometry in natural vision

Samuel A Nastase 2026-10-19 1 min 67 words LOW VISION

Summary: We effortlessly extract behaviorally relevant information from dynamic visual input to understand the actions of others. Here, we compare behavioral judgments of action meaning against other models capturing semantic and visual features. We measured brain activity using fMRI while participants viewe...

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Birth of the Nigerian Sleep Society: A Milestone for Sleep Medicine in Nigeria and Africa

G E Erhabor 2026-10-20 1 min 72 words LOW VISION

Summary: Sleep disorders are significant and are of major public health concern with high socioeconomic impact. However, they are under-recognised and under-treated in Nigeria and most of Africa, because of poor access to sleep specialists, sleep studies and limited treatment options. Prior to the inaugurati...

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Knowledge gaps and practices for glaucoma among adults in Palestine: a cross-sectional study

Farid K Abu Sham'a
2026-10-20 1 min 51 words LOW VISION

Summary: CONCLUSIONS: Although awareness of glaucoma is moderate in Palestine, comprehensive knowledge remains low, particularly among high-risk populations. Public health interventions should target education beyond name recognition, emphasizing asymptomatic progression, irreversible vision loss, and risk f...

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Sn/Cl Co-Doped ZnO Cathode Interfacial Layer for High-Performance Organic Photodetectors

Kai Zhang
2026-10-20 1 min 66 words LOW VISION

Summary: Organic photodetectors (OPDs) hold significant promise for applications spanning medical monitoring, bioimaging and night vision due to their unique advantages, such as flexibility, light weight, and potential for large-area fabrication. Nevertheless, the dark current density (Jd) of OPDs exhibits a...

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A thermopile array and stereo vision fusion system for small-tube temperature sensing

👤 Dongfang Hu 📅 2026-10-20 📖 1 min 📝 71 words LOW VISION

Summary: This paper presents a non-contact temperature measurement scheme for small-diameter tubes by fusing a low-resolution thermopile array (TPA) with visible-light stereo vision. Stereo vision locates the tube in three dimensions. Then, this geometry is projected onto the TPA imaging plane to constrain t...

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Cost-effective long-range measurement of structural displacement using a two-stage method integrating super-resolution reconstruction and sub-pixel registration

👤 Xiaosong Li 📅 2026-10-20 📖 1 min 📝 61 words LOW VISION

Summary: Displacement is an imperative metric for assessing structural performance. However, the accuracy of structural displacement measurement is constrained by the physical resolution of the camera, particularly in long-range monitoring scenarios. To address these constraints, a cost-effective long-range ...

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Physics-aware restoration and calibrated soft demodulation for turbulence-degraded OAM mode recognition

👤 Jun Zhu 📅 2026-10-20 📖 1 min 📝 58 words LOW VISION

Summary: Turbulence-degraded orbital angular momentum (OAM) mode recognition is critical for free-space optical (FSO) communication but is limited by atmospheric distortion. We propose a cascaded receiver integrating a vision transformer-based Uformer-OAM intensity-restoration network, an SE-ResNet-18 neural...

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Image-free ghost edge detection at ultra-low sampling rates via a Vision Mamba network

👤 Rui Wang 📅 2026-10-20 📖 1 min 📝 66 words LOW VISION

Summary: Edge detection is essential for extracting structural boundaries in visual scenes, but conventional deep-learning-based methods remain sensitive to degradations in the input images and involve redundant full-image acquisition. Image-free ghost edge detection provides an efficient alternative by dire...

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Ex vivo evaluation of corneal permeation and irritation potential associated with proparacaine hydrochloride misuse

Yolima Baena 2026-10-20 1 min 35 words LOW VISION

Summary: CONCLUSION: Proparacaine demonstrates dose-dependent permeation and histopathological changes, emphasizing the risks associated with misuse. The ex vivo model provides relevant and ethically aligned evidence to support ocular hazard assessment through the characterizing of histopathological changes.

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Shared variance across FrACT visual tasks: Robust positive associations but inconclusive latent structure

Luca Battaglini 2026-10-21 1 min 68 words LOW VISION

Summary: A long-standing question in vision science is whether individual differences in visual performance reflect a general visual ability or a set of largely task-specific processes. Previous studies have often reported weak associations across heterogeneous perceptual tasks, challenging the idea of a com...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

👤 Béatrice Berçot 📅 2026-09-04 📖 1 min 📝 60 words BRAILLE

> **Summary:** CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan 📅 2026-09-09 📖 1 min 📝 57 words BRAILLE

> **Summary:** Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Written exposure therapy augmented with transcranial direct current stimulation for posttraumatic stress disorder: A pilot randomized clinical trial

STRONG STAR Consortium 2026-10-18 1 min 64 words TDCS TACS TRNS

Summary: Written exposure therapy (WET) is an evidence-based, trauma-focused psychotherapy for posttraumatic stress disorder (PTSD) that has been tested with several trauma-exposed samples. Although many individuals who receive WET and other trauma-focused psychotherapies experience a significant reduction i...

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Transcranial Direct Current Stimulation and Cranial Electrotherapy Stimulation in Athletes: Optimizing Athletic Performance and Sleep Quality Through Neuromodulation

👤 Paulina Gajniak
📅 2026-10-19 📖 1 min 📝 74 words

TDCS TACS TRNS

Summary: Neuromodulation is a procedure relying on the delivery of specific stimuli to selected sites of the nervous system and hereby influencing its activity. Even though the underlying neurobiological mechanisms are not yet fully elucidated and the research is still ongoing, neuromodulation is already bei...

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Clinical outcomes and safety of Five Element Acupuncture for preventive treatment of primary headache disorders: a multi-center real-world longitudinal observational study

Shengyuan Zhou
2026-10-19 1 min 49 words TDCS TACS TRNS

Summary: CONCLUSIONS: This real-world exploratory observational study suggests that FEA is associated with significant improvements in headache days, headache frequency, headache intensity, analgesic use, and quality of life, with sustained effects noted during follow-up. These findings provide preliminary o...

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Home Transcranial Direct Current Stimulation With Gamified Cognitive Control Training for Depression: A Randomized Clinical Trial

👤 Mor Nahum 📅 2026-10-19 📖 1 min 📝 45 words TDCS TACS TRNS

Summary: CONCLUSIONS AND RELEVANCE: This RCT of home-based tDCS combined with CCT in adults with MDD was feasible and safe but did not demonstrate superior efficacy over the control condition. Future studies should refine the mechanistic alignment between stimulation and cognitive interventions to optimize t...

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Comparative chloroplast genomics of Ajania sericea and sympatric Asteraceae species identifies shared candidate molecular features associated with alpine adaptation

Xi Chen 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study showed that the chloroplast genomes of four sympatric Asteraceae species were highly conserved in overall structure, but still exhibited certain differences in gene loss, IR boundary variation, repeat sequence composition, and local hypervariable regions. These features provi...

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Brain Stimulation as a Treatment for Apathy: A Scoping Review of Effects and Safety

Lisa M McTeague 2026-10-20 1 min 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: Investigations into brain stimulation as a treatment for apathy show some promise. Conclusions were limited by small sample sizes, heterogeneous study designs, and variable outcome measures. Randomized trials with common validated apathy instruments, standardized stimulation parameters,...

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Effect of Different Cortical Targets of Transcranial Magnetic Stimulation on Freezing of Gait in Parkinsonism: A Systematic Review

Aliaa Elabd
2026-10-20 1 min 68 words TDCS TACS TRNS

Summary: BACKGROUND AND PURPOSE: Freezing of gait (FOG) is a disabling and episodic gait disturbance in patients with Parkinson's, with varying responses to noninvasive brain stimulation. Transcranial magnetic stimulation (TMS) has been explored as a therapeutic option; however, the optimal cortical target f...

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Sleep Spindles and Slow Oscillations as Potential Biomarkers and Therapeutic Targets in Mild-to-Moderate Alzheimer's Disease

👤 Thien Thanh Dang-Vu 📅 2026-10-20 📖 1 min 📝 35 words TDCS TACS TRNS

Summary: PURPOSE OF REVIEW: This review critically evaluates evidence on sleep spindles (SP), slow oscillations (SOs), and their coordination in mild-to-moderate Alzheimer's disease (AD), focusing on their relationships with cognition, AD biomarkers, and their therapeutic potential.

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

👤 Xin Zhou 📅 2026-10-20 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Prior photon measurement density functions constructed in standard channel space for functional near-infrared spectroscopy

👤 Chao-Zhe Zhu 📅 2026-10-17 📖 1 min 📝 65 words FNIRS

Summary: Photon measurement density function (PMDF) determines the measurement locations and their associated sensitivity to the hemodynamic changes for an fNIRS channel. It is therefore essential for pre-experimental optode arrangement and post-experimental signal analysis and interpretation in fNIRS studie...

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Neckline-free open-source cap design and improved motion resistance for perioperative fNIRS procedures

👤 Audrey K Bowden 📅 2026-10-17 📖 1 min 📝 60 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) caps are limited by motion artifacts and unstable optode-scalp coupling, which degrade signal quality during extended and naturalistic monitoring sessions. We present a fabric-based, chinstrap-free headcap designed to improve mechanical stability and opt...

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Multiscale neural signatures of long-lag repetition learning: a simultaneous fNIRS-EEG study

Ken Pugh 2026-10-17 1 min 38 words FNIRS

Summary: CONCLUSIONS: Combining fNIRS and EEG with multimodal machine learning provides a richer characterization of repetition-based learning than either modality alone, establishing a framework for phenotyping language learning capacity and identifying neural markers of reading difficulties in pediatric po...

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Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

Felipe Balistieri Santinelli 2026-10-18 1 min 70 words FNIRS

Summary: People with multiple sclerosis (pwMS) often exhibit motor fatigability and reduced foot-tapping ability. However, the coupling between performance and cortical activity during fast, repetitive and prolonged ankle movements remains unexplored. We aimed to investigate the neural correlates of a prolon...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang
📅 2026-10-19 📖 1 min 📝 64 words FNIRS

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Neural dysfunction underlying divergent thinking and its shared mechanisms with protracted withdrawal symptoms in heroin use disorder

👤 Haijun Duan
📅 2026-10-19 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSIONS: These findings reveal a sustained compensatory pathological mechanism and an imbalance in neural resource allocation during complex cognition in HUD individuals. The identified signatures of cortical activity by fNIRS offer reliable biomarkers for the rapid detection of both higher-orde...

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Functional near-infrared spectroscopy as a tool for monitoring treatment-related neurofunctional changes in posttraumatic stress disorder and trauma-related symptomatology: a narrative review

👤 Maria Cotelli
📅 2026-10-20 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Current evidence suggests that fNIRS represents a promising and clinically feasible approach for monitoring treatment-related neurofunctional changes in PTSD and trauma-related symptomatology, particularly during non-pharmacological interventions targeting prefrontal regulatory processe...

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Transcutaneous vagus nerve stimulation during semantic verbal fluency tasks: a high-density diffuse optical tomography study

👤 Li Su 📅 2026-10-20 📖 1 min 📝 55 words FNIRS

Summary: INTRODUCTION: Transcutaneous auricular vagus nerve stimulation (taVNS) is a promising adjunct for post-stroke language recovery. There are no established biomarkers for the neuromodulatory effects of taVNS. High-density diffuse optical tomography (HD-DOT) offers a non-invasive marker of task-related...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

Hua Hu 2026-10-21 1 min 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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An exploration in mental health care: The launch of China's first psychiatric noninvasive brain-computer interface (BCI) and neuromodulation clinic

Tianhong Zhang 2026-10-17 1 min 2 words BRAIN COMPUTER INTERFACE

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A sliding window and Fourier decomposition method based 1D-CNN framework for motor imagery EEG classification

M Ramakrishnan 2026-10-17 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) signal classification is fundamental requirement for effective motor imagery (MI) brain-computer interface (BCI) applications. However, current MI classification methods often depend on handcrafted spatial filtering or complex time-frequency transformations that increase...

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Electroencephalography Changes in Emotional Recognition: A Signal Processing Approach Using Wavelet Features

Lakshmi Narasimhan 2026-10-17 1 min 43 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Wavelet-based EEG feature analysis provides meaningful and discriminative information for emotion recognition and may provide a robust feature set for future machine learning-based emotion recognition systems. Further studies incorporating classification algorithms and larger data sets a...

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A Compressive Neural Telemetry with Address-Event-μPacket Encoding and Duty-Cycling IR-UWB Transmitter

👤 Yao-Hong Liu 📅 2026-10-18 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: This article presents a networked wireless telemetry for 384-sensor-pixel neural data acquisition, designed to support high throughput, low power consumption, and meter-scale communication distance. The proposed event-based Address-Event-μPacket (AEμP) encoder with differential binary phase shift ke...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang 📅 2026-10-19 📖 1 min 📝 64 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Bridging the Gap Between Design and Access: Toward Visual Scene Displays for Brain-Computer Interface Use in Augmentative and Alternative Communication by Children

Ahad Behboodi 2026-10-19 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: These findings indicate that asymmetry and contextual embedding within scenes may enhance neural engagement and user experience, providing a preliminary foundation for considering photographic VSDs into pediatric BCI research.

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A spatio-temporal attention convolutional network with topology-preserving channel embedding and transformer for EEG decoding

Gang Liu 2026-10-19 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: EEG-based BCI has shown promising applications in motor rehabilitation and emotion recognition. Given the non-stationarity and variability of EEG signals, it is essential to develop classification models that can effectively extract spatio-temporal features while maintaining strong generalization ca...

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The calibration-to-accuracy mapping in P300 spellers does not transport across cohorts and was not restored by tested alignment or local recalibration

Oved Daniel 2026-10-19 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Objective
A calibration-derived score has been related to online P300-speller accuracy in the same session, always within the cohort measured. We evaluated whether a mapping from that score to expected accuracy transports to withheld cohorts.
Approach
Retrospective secondary analysis of ...

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Subjective emotional ratings dissociate brain activation and network patterns during counterfactual outcome processing

Ling Li 2026-10-19 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Human decision-making integrates outcome evaluation, counterfactual comparison, and individual traits. We investigated the neural processing of outcome-related emotions during economic decision-making by classifying feedback according to participants' subjective emotional ratings rather than objecti...

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Delineating neural contributions to electroencephalogram-based speech decoding

Shuntaro Sasai 2026-10-20 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Objective. Speech Brain-computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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A research on the typical images of animals in the dreams of Naxi.

2026-09-10 1 min 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

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Philosophical and medical analysis of dreams in avicenna's perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

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http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

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http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

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http://doi.org/10.1037/drm0000349

When the body suffers, the mind dreams: The lived experience of cancer.

📅 2026-09-10 📖 1 min 📝 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

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Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

Summary: This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

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http://doi.org/10.1037/drm0000350

Coping strategies and dreams—The impact of waking-life distress on negative dream emotions.

📅 2026-05-18 📖 1 min 📝 183 words DREAMING

Summary: The relationship between waking life and dreaming seems complex, and various studies show that dreams are influenced by waking-life experiences. However, only a few studies deal with the interactions between coping strategies in waking life and dream emotions. The main goal of this study is to inves...

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http://doi.org/10.1037/drm0000346

Emudhra compromise affects older versions of key Python libraries - uv, certifi, pip, sigstore, pixi

👤 /u/Ska82 📅 2026-10-22 📖 1 min 📝 23 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> https://discuss.python.org/t/information-for-python-users-regarding-emudhra-emsign-ca-certificate-security-incident/109576 </...

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Jailbreaking the Switch 2

👤 ledoge 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

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Several Linux kernel vulnerabilities accidentally posted to public mailing list

👤 mrkn1 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

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Amazon identifies Oct 17 PHP2Shell exploitation prior to public disclosure

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The Last Museum: Semantic search across 5.8M artworks from major museums

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SBD: We Have a Filesystem at Home

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eMudhra to Exit Global SSL/TLS Certificate Business

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What's Getting Pwned

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Quad9 (DNS Resolver) to Exit France

👤 farrokhi 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

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New software package: ERPLAB Studio

Steve Luck | 2024-06-12 | 2 min | 449 words | ERP BOOT CAMP

Summary: <p class="">We are excited to announce the release of a new EEG/ERP analysis package, ERPLAB Studio. We think it's a huge improvement over the classic EEGLAB user interface. See our cheesy <a href="https://www.youtube.com/watch?v=lIaKVQ9DD6E">...

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https://erpinfo.org/blog/2024/6/11/erplab-studio

SW-VEI-Net: A Physics-Informed Deep Neural Network for Shear Wave Viscoelasticity Imaging

2026-01-12 | 1 min | 226 words | TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Quantitative viscoelasticity imaging via shear wave elastography (SWE) remains challenging due to complex wave physics and limitations of conventional reconstruction methods. To address this, we present SW-VEI-Net, a physics-informed neural network (PINN) that simultaneously reconstructs the shear e...

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Feasibility of Dual Probe Pulse Wave Imaging of the Abdominal Aorta

📅 2026-01-12 📖 1 min 📝 233 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Pulse wave velocity (PWV) is an indirect measure of vessel stiffness, that has the potential to serve as a meaningful parameter for risk stratification of vascular diseases, such as abdominal aortic aneurysms (AAAs). However, assessing the PWV and pulse wave patterns in the complete abdominal aorta ...

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Dynamic Cardiac Event Detection From Single-Arm Wearable ECG via a Contrastive Multitask Framework

📅 2026-01-12 📖 1 min 📝 246 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Upper-limb electrocardiogram (ECG) acquired from a single-arm wearable device offers a practical approach for dynamic cardiac monitoring. This study addresses whether single-channel Arm-ECG, characterized by higher noise and distinct morphology compared to standard 12-lead ECG, can reliab...

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A Dual-Energy CBCT With Reduced Scatter and Cone Beam Artifacts Using an X-Ray Source Array and Interlaced Spectral Filters

📅 2026-01-12 📖 1 min 📝 249 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: To design a dual-energy cone beam computed tomography (DE-CBCT) scanner with reduced scatter and cone beam artifacts. Methods: The scanner designed for maxillofacial imaging comprises a carbon nanotube (CNT) X-ray source array with multiple focal spots (“sources”) and an energy integratin...

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Muscle Synergy-Guided Reinforcement Learning for Embodied Musculoskeletal Motion Skill Learning

📅 2026-01-12 📖 1 min 📝 247 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Background: Acquiring human-like motor skills in embodied musculoskeletal models is challenging due to the high dimensionality and redundancy of muscle actuators. Methods: Inspired by human motor control coordination patterns, we introduce a synergy-guided reinforcement learning framework integratin...

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Network Models of Neurodegeneration: Bridging Neuronal Dynamics and Disease Progression

📅 2026-01-16 📖 1 min 📝 196 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Neurodegenerative diseases are characterized by the accumulation of misfolded proteins and widespread disruptions in brain function. Computational modeling has advanced our understanding of these processes, but efforts have traditionally focused on either neuronal dynamics or the biological processe...

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Optogenetics: Pinpoint Light on Precise Neuromodulation

📅 2025-11-06 📖 1 min 📝 137 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Optogenetics has emerged as a pivotal tool in neuroscience, enabling intricate modulation of targeted neurons within the nervous system. Despite its transformative potential, achieving high spatiotemporal resolution in neuromodulation remains a significant challenge, particularly in free-behaving an...

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Emerging Noninvasive Approaches for the Suppression of Pathological Tremor

📅 2025-12-16 📖 1 min 📝 215 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Pathological tremor affects over 40 million people worldwide, significantly impairing daily activities and quality of life. Pharmacological treatments show limited efficacy, with up to 30% discontinuation rates, while surgical interventions like deep brain stimulation achieve significant tremor redu...

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Assistive Trajectory Planning for Lower Limb Exoskeletons: Strategies From Laboratory-Optimized Gait to Environmentally-Adaptive Locomotion Through Multimodal Parameter Awareness

📅 2025-12-30 📖 1 min 📝 182 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Lower limb assistive exoskeletons (LLEs) show great potential to reduce metabolic cost, improve walking performance, and correct abnormal gait patterns. Among their core control architectures, assistive trajectory planning is key to determining system responsiveness and effectiveness under varying l...

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Selective cognitive changes in a multidrug mouse model of chemotherapy-induced adverse effects

📖 1 min 📝 13 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Jeffrey Goris, Rudi D’Hooge

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A Hierarchical-Dynamic integration model for the formation of interpersonal neural synchrony during spoken communication

📖 1 min 📝 13 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yating Bai, Qingfang Zhang

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Identification of a shared blood–brain transcriptomic signature in Parkinson's disease using bioinformatic and network pharmacology approaches

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Hafsa Mohammad, Akanksha Singh, Dipika Bansal

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TEMPORARY REMOVAL: Corrigendum to “LPI alleviates Alzheimer’s disease pathology via the GPR55 receptor” [Neuroscience 611 (2026) 99–110]

📖 1 min 📝 26 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Wenjie Xu, Jiaqi Cao, Yibo Liu, Zimo Wei, Xu Zha, Shumin Xie, Xicheng Liu, Wei Wang, Chen Zhang

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Sex-based measurement invariance in Reinforcement Sensitivity Theory: An exploratory structural equation model based on biological variables

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 1 November 2026 Source: NeuroImage, Volume 341 Author(s): Elena Lacomba-Arnau, Agustín Martínez-Molina, Óscar Pérez-Díaz, Maria Picó-Pérez, Alfonso Barrós-Loscertales

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Effects of intensive sensorimotor rehabilitation in resting state functional connectivity in a rodent model of Cerebral Palsy

📖 1 min 📝 22 words NEUROIMAGE

Summary: Publication date: 1 November 2026 Source: NeuroImage, Volume 341 Author(s): Eduardo Sanches, Gabriel Schirmbeck, Jérôme Mairesse, Dini Ho, Charlotte Dennez, Olivier Baud, Stéphane Sizonenko

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A new neural code for position: generalizable spatial maps in mouse medial frontal cortex

👤 Sen, R., Posani, L., Medoff, A., Zhang, B., Abbott, L., Axel, R., Fusi, S. 📅 2026-10-22 📖 1 min

📝 158 words BIORXIV NEUROSCIENCE

Summary: Navigation requires the brain to encode spatial relationships that support extrapolation from explored to unexplored regions of space. We identified a novel neural code for space in the mouse medial frontal cortex (mFC) that is well suited for this form of spatial generalization. Using calcium imagi...

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Development of Neonatal Functional Brain Network Organisation and Its Association with Childhood Neurodevelopmental Outcomes

Chang, Q., Fenn-Moltu, S., Chew, A., Bonthrone, A. F., Gal-Er, B., Nosarti, C., Arichi, T., McAlonan, G., Counsell, S., Batalle, D.

2026-10-22 1 min 199 words BIORXIV NEUROSCIENCE

Summary: The human brain undergoes rapid dynamic changes in the neonatal period that are essential for later cognitive and behavioural performance. Using resting-state fMRI data from 394 neonates (325 term-born, 69 preterm-born) within the first month after birth, we characterised the early development of fu...

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Parrots imitate song and dance with mixed vocal-gestural neural signals

Jones, C., Teoh, H. K., Carpenter, J., Roeser, A., Chen, R., Kardon, B., Zhao, Z., Cohen, I., Goldberg, J. H.

2026-10-22 1 min 137 words BIORXIV NEUROSCIENCE

Summary: Humans shrug when saying "I don't know" and nod when saying "yes," yet it remains unclear how the brain coordinates vocal and gestural motor systems for learned, multimodal communication. Here we find that budgerigar parrots precisely imitate the fine temporal alignment between song and affiliative ...

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Hypothalamic plasticity for alloparenting in male mice

O'Neill, P. T., Yan, R., Wang, Y., Wang, Q., Yin, L., Lhamo, G., Rhodes, D. J., Little, J., Cheng, H., Chen, L., Shoham, S., Chen, X., Froemke, R. C., Lin, D.

2026-10-22 1 min 141 words BIORXIV NEUROSCIENCE

Summary: Repeated exposure to young can induce caregiving in virgin female mice, a phenomenon known as pup sensitization, but whether virgin males can similarly acquire parental behavior and the neural mechanisms underlying this transition remain unclear. We show that 7 days of continuous pup exposure reliab...

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Individual Differences in Sustained Attention Tune Semantic Abstraction in Neural Encoding of Narratives

Gao, S., Leong, Y. C., Rosenberg, M. D.
2026-10-22 1 min 309 words

BIORXIV NEUROSCIENCE

Summary: Neural encoding modeling for narrative comprehension -- using LLM embeddings to predict the temporal dynamics of neural activation -- is widely used to evaluate how well the brain represents certain semantic information. This approach, however, typically collapses semantic representation into a scal...

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Coordinated cerebellar-midbrain activity for predictive signaling during fear learning in mice

Urrutia Desmaison, J. D., Lena, C., Popa, D.
2026-10-22 1 min 133 words

BIORXIV NEUROSCIENCE

Summary: Learning from experience to assess threat is crucial for survival and mental health. This function is known to involve several midbrain areas, notably the ventrolateral periaqueductal grey (vlPAG) and dorsal raphe nuclei (DRN). Recently it has been shown that fear learning involves projections from ...

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Task-general brain state dynamics relate to trial-to-trial performance and individual response strategy

Yamashita, A., Sawayama, M., Amano, K.

2026-10-22 · 1 min · 151 words

BIORXIV NEUROSCIENCE

Summary: Behavior fluctuates from moment to moment, yet individuals also show characteristic response strategies across tasks. Whether these two levels of variation relate to common brain dynamics is unknown. Using an inside-out approach across 10 cognitive tasks during fMRI, we identified recurring brain st...

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https://www.biorxiv.org/content/10.64898/2026.10.21.760632v1?rss=1

Projection populations form a dorsoventral gradient of reward representations in the lateral septum

Isaac, J., Karkare, S. C., Balasubramanian, H., Olah, V., Rowan, M., Murugan, M.

2026-10-22

1 min · 193 words

BIORXIV NEUROSCIENCE

Summary: Several decades ago, the septal region was identified as the most dependable location of intracranial self-stimulation (ICSS) in the brain, but the neural substrates that mediate septal self-stimulation are still unknown. One part of the septal region, the lateral septum (LS), contributes to a numbe...

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An organotypic hippocampo-prefrontal co-culture model for studying long-range connectivity and circuit plasticity

Stoehr, P., Grigoryan, G., Lenz, M., Vlachos, A.

2026-10-22 1 min 172 words

BIORXIV NEUROSCIENCE

Summary: The hippocampal projection to the medial prefrontal cortex (mPFC) is a key component of brain networks involved in cognition and emotion. Here, we establish and validate an organotypic entorhino-hippocampo-prefrontal co-culture model that enables experimental investigation of long-range circuit conn...

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Inferring structural complexity from dynamical measures

Gaglioti, G., Cattani, A., Nieus, T., Momi, D., Comolatti, R., Dalla Porta, L., Sarasso, S., Massimini, M.

2026-10-22 1 min 250 words

BIORXIV NEUROSCIENCE

Summary: Complexity, commonly defined as the coexistence of integration and differentiation, is a fundamental property of thalamocortical networks and is regarded as a prerequisite for adaptive brain function and consciousness. A variety of metrics have been developed to estimate neural complexity from recor...

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Maltreatment predicts centiles of individual variation from normative brain growth in children

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A UCP1-IRES-Cre knock-in mouse enables specific brown adipocyte targeting without CNS off-target expression

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Maternal mental health remains a missing priority

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https://www.nature.com/articles/s44220-026-00752-z

Multi-scale integration identifies an ox-LDL-autophagy-inflammasome framework linking osteoporosis and neurodegeneration

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Network geometry shapes the transformation of task representations across the human cortex

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Minocycline attenuates gulf war illness-related pain sensitization and preserves descending control of nociception in mice

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Behavioral genomics identifies Creld1 as a cross-modal regulator of somatosensation

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https://www.nature.com/articles/s41593-026-02484-w

Behavioral genomics identifies Creld1 as a cross-modal regulator of somatosensation

👤 Bailong Xiao 📅 2026-10-22 📖 1 min 📝 42 words NATURE NEUROSCIENCE

> **Summary:** Nature Neuroscience, Published online: 22 October 2026; doi:10.1038/s41593-026-02484-w Jiang et al. developed a robust postnatal knockout platform for somatosensory genomics and behavioral screening, identifying Creld1 as a cross...

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https://www.nature.com/articles/s41593-026-02484-w

Network geometry shapes the transformation of task representations across the human cortex

Michael W. Cole 2026-10-22 1 min 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 22 October 2026; doi:10.1038/s41593-026-02486-8 The dimensionality of intrinsic brain connectivity systematically shapes how task representations compress and expand across the cortex, and a...

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https://www.nature.com/articles/s41593-026-02486-8

Approximate matrix exponential: a fast and accurate solver for Markov kinetic models of ion channels

David Alton McDougall 2026-10-06 1 min 148 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: A new method is presented for simulating Markov kinetic models of ion channels, which are biologically realistic but difficult to simulate. Computing the analytic solution is prohibitively expensive, necessitating the use of numerical methods. The new method creates a piecewise polynomial approximat...

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Different modes of stimulation at the Neiguan acupoint (PC6) regulate different EEG microstates under VR immersion conditions

Guangjun Wang 2026-10-22 1 min 302 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAcupuncture at the Neiguan acupoint (PC6) alleviates postoperative and chemotherapy-induced nausea and vomiting. Nevertheless, the central regulatory effects of PC6 stimulation on virtual reality motion sickness (VRMS), a form of physiological vertigo triggered by visual-vestibular senso...

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Anticipating interactions: a predictive diffusion framework for evolving bimanual manipulation

Shenggang Wei 2026-10-06 1 min 241 words

FRONTIERS NEUROROBOTICS

Summary: IntroductionBimanual robotic manipulation remains a significant challenge in embodied AI, as it requires synchronized motion planning, inter-arm collision avoidance, and dynamic role allocation over long task horizons. Although existing diffusion-based policies and affordance learning methods have d...

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Retraction: Intervention effect of rehabilitation robotic bed under machine learning combined with intensive motor training on stroke patients with hemiplegia

📅 2026-10-14 📖 1 min 📝 0 words FRONTIERS NEUROROBOTICS

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Multiparametric MRI radiomics and machine learning for outcome prediction in acute ischemic stroke: a retrospective study of 353 patients

👤 Tingting Li 📅 2026-10-22 📖 1 min 📝 285 words FRONTIERS NEUROSCIENCE

Summary: BackgroundAcute ischemic stroke (AIS) remains a leading cause of death and disability worldwide. Early prediction for clinical outcomes is essential for guiding therapy. Multiparametric Magnetic Resonance Imaging (MRI) offers detailed characterization of ischemic tissue and may predict prognosis in ...

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Striatal and orbitofrontal grey matter volumes are associated with cognitive event related potentials during associative learning task performance in male and female children and adolescents

Almira Kustubayeva 2026-10-22 1 min 303 words FRONTIERS NEUROSCIENCE

Summary: IntroductionThe individual differences in gray matter volumes and event-related potentials may underlie variability in associative learning. However, little is known about the sex differences in these structure–function relationships across development.MethodsThe current study investigates associati...

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R. Beyette 2026-10-22 1 min 267 words FRONTIERS NEUROSCIENCE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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The mammalian regulatory architecture: biobehavioral portals and the organization of physiological state

Stephen W. Porges 2026-10-22 1 min 214 words FRONTIERS NEUROSCIENCE

Summary: IntroductionNeuromodulatory interventions can produce overlapping adaptive outcomes despite engaging substantially different biological pathways. Pathway-centered approaches identify how interventions initially engage the nervous system and characterize pathway-specific mechanisms. Convergence acros...

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Continuous performance tests in the assessment of children with attention-deficit/hyperactivity disorder: a narrative review

Xiaoke Zhao 2026-10-22 1 min 231 words FRONTIERS NEUROSCIENCE

Summary: Attention-deficit/hyperactivity disorder (ADHD) is diagnosed primarily through clinical interviews, behavioral observations, and rating scales, while continuous performance tests (CPTs) provide complementary, performance-based measures of sustained attention, response inhibition, and response consis...

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Microglia in the neurogenic niche: emerging roles of immunometabolism

Yan Liu 2026-10-22 1 min 136 words FRONTIERS NEUROSCIENCE

Summary: Adult neurogenesis is dynamically regulated by the neurogenic niche, yet the mechanisms governing its remodeling are not fully elucidated. Current research has increasingly extended beyond the intrinsic properties of neural stem cells to consider the broader microenvironment in which neurogenesis oc...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Behaviorally-relevant features of observed actions dominate cortical representational geometry in natural vision

👤 Samuel A Nastase 📅 2026-10-19 📖 1 min 📝 67 words LOW VISION

Summary: We effortlessly extract behaviorally relevant information from dynamic visual input to understand the actions of others. Here, we compare behavioral judgments of action meaning against other models capturing semantic and visual features. We measured brain activity using fMRI while participants viewe...

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Birth of the Nigerian Sleep Society: A Milestone for Sleep Medicine in Nigeria and Africa

👤 G E Erhabor 📅 2026-10-20 📖 1 min 📝 72 words LOW VISION

Summary: Sleep disorders are significant and are of major public health concern with high socioeconomic impact. However, they are under-recognised and under-treated in Nigeria and most of Africa, because of poor access to sleep specialists, sleep studies and limited treatment options. Prior to the inaugurati...

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Knowledge gaps and practices for glaucoma among adults in Palestine: a cross-sectional study

👤 Farid K Abu Sham'a 📅 2026-10-20 📖 1 min 📝 51 words LOW VISION

Summary: CONCLUSIONS: Although awareness of glaucoma is moderate in Palestine, comprehensive knowledge remains low, particularly among high-risk populations. Public health interventions should target education beyond name recognition, emphasizing asymptomatic progression, irreversible vision loss, and risk f...

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Sn/Cl Co-Doped ZnO Cathode Interfacial Layer for High-Performance Organic Photodetectors

Kai Zhang 2026-10-20 1 min 66 words LOW VISION

Summary: Organic photodetectors (OPDs) hold significant promise for applications spanning medical monitoring, bioimaging and night vision due to their unique advantages, such as flexibility, light weight, and potential for large-area fabrication. Nevertheless, the dark current density (Jd) of OPDs exhibits a...

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A thermopile array and stereo vision fusion system for small-tube temperature sensing

Dongfang Hu 2026-10-20 1 min 71 words LOW VISION

Summary: This paper presents a non-contact temperature measurement scheme for small-diameter tubes by fusing a low-resolution thermopile array (TPA) with visible-light stereo vision. Stereo vision locates the tube in three dimensions. Then, this geometry is projected onto the TPA imaging plane to constrain t...

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Cost-effective long-range measurement of structural displacement using a two-stage method integrating super-resolution reconstruction and sub-pixel registration

Xiaosong Li 2026-10-20 1 min 61 words LOW VISION

Summary: Displacement is an imperative metric for assessing structural performance. However, the accuracy of structural displacement measurement is constrained by the physical resolution of the camera, particularly in long-range monitoring scenarios. To address these constraints, a cost-effective long-range ...

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Physics-aware restoration and calibrated soft demodulation for turbulence-degraded OAM mode recognition

Jun Zhu 2026-10-20 1 min 58 words LOW VISION

Summary: Turbulence-degraded orbital angular momentum (OAM) mode recognition is critical for free-space optical (FSO) communication but is limited by atmospheric distortion. We propose a cascaded receiver integrating a vision transformer-based Uformer-OAM intensity-restoration network, an SE-ResNet-18 neural...

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Image-free ghost edge detection at ultra-low sampling rates via a Vision Mamba network

Rui Wang 2026-10-20 1 min 66 words LOW VISION

Summary: Edge detection is essential for extracting structural boundaries in visual scenes, but conventional deep-learning-based methods remain sensitive to degradations in the input images and involve redundant full-image acquisition. Image-free ghost edge detection provides an efficient alternative by dire...

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Ex vivo evaluation of corneal permeation and irritation potential associated with proparacaine hydrochloride misuse

Yolima Baena 2026-10-20 1 min 35 words LOW VISION

Summary: CONCLUSION: Proparacaine demonstrates dose-dependent permeation and histopathological changes, emphasizing the risks associated with misuse. The ex vivo model provides relevant and ethically aligned evidence to support ocular hazard assessment through the characterizing of histopathological changes.

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Shared variance across FrACT visual tasks: Robust positive associations but inconclusive latent structure

👤 Luca Battaglini 📅 2026-10-21 📖 1 min 📝 68 words LOW VISION

Summary: A long-standing question in vision science is whether individual differences in visual performance reflect a general visual ability or a set of largely task-specific processes. Previous studies have often reported weak associations across heterogeneous perceptual tasks, challenging the idea of a com...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

👤 Martina M Mensi 📅 2026-06-25 📖 1 min 📝 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

> **Summary:** CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Written exposure therapy augmented with transcranial direct current stimulation for posttraumatic stress disorder: A pilot randomized clinical trial

👤 STRONG STAR Consortium 📅 2026-10-18 📖 1 min 📝 64 words TDCS TACS TRNS

> **Summary:** Written exposure therapy (WET) is an evidence-based, trauma-focused psychotherapy for posttraumatic stress disorder (PTSD) that has been tested with several trauma-exposed samples. Although many individuals who receive WET and other trauma-focused psychotherapies experience a significant reduction i...

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Transcranial Direct Current Stimulation and Cranial Electrotherapy Stimulation in Athletes: Optimizing Athletic Performance and Sleep Quality Through Neuromodulation

👤 Paulina Gajniak
📅 2026-10-19 📖 1 min 📝 74 words

TDCS TACS TRNS

Summary: Neuromodulation is a procedure relying on the delivery of specific stimuli to selected sites of the nervous system and hereby influencing its activity. Even though the underlying neurobiological mechanisms are not yet fully elucidated and the research is still ongoing, neuromodulation is already bei...

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Clinical outcomes and safety of Five Element Acupuncture for preventive treatment of primary headache disorders: a multi-center real-world longitudinal observational study

👤 Shengyuan Zhou
📅 2026-10-19 📖 1 min 📝 49 words

TDCS TACS TRNS

Summary: CONCLUSIONS: This real-world exploratory observational study suggests that FEA is associated with significant improvements in headache days, headache frequency, headache intensity, analgesic use, and quality of life, with sustained effects noted during follow-up. These findings provide preliminary o...

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Home Transcranial Direct Current Stimulation With Gamified Cognitive Control Training for Depression: A Randomized Clinical Trial

👤 Mor Nahum 📅 2026-10-19 📖 1 min 📝 45 words TDCS TACS TRNS

Summary: CONCLUSIONS AND RELEVANCE: This RCT of home-based tDCS combined with CCT in adults with MDD was feasible and safe but did not demonstrate superior efficacy over the control condition. Future studies should refine the mechanistic alignment between stimulation and cognitive interventions to optimize t...

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Comparative chloroplast genomics of Ajania sericea and sympatric Asteraceae species identifies shared candidate molecular features associated with alpine adaptation

Xi Chen 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study showed that the chloroplast genomes of four sympatric Asteraceae species were highly conserved in overall structure, but still exhibited certain differences in gene loss, IR boundary variation, repeat sequence composition, and local hypervariable regions. These features provi...

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Brain Stimulation as a Treatment for Apathy: A Scoping Review of Effects and Safety

Lisa M McTeague 2026-10-20 1 min 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: Investigations into brain stimulation as a treatment for apathy show some promise. Conclusions were limited by small sample sizes, heterogeneous study designs, and variable outcome measures. Randomized trials with common validated apathy instruments, standardized stimulation parameters,...

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Effect of Different Cortical Targets of Transcranial Magnetic Stimulation on Freezing of Gait in Parkinsonism: A Systematic Review

Aliaa Elabd
2026-10-20 1 min 68 words TDCS TACS TRNS

Summary: BACKGROUND AND PURPOSE: Freezing of gait (FOG) is a disabling and episodic gait disturbance in patients with Parkinson's, with varying responses to noninvasive brain stimulation. Transcranial magnetic stimulation (TMS) has been explored as a therapeutic option; however, the optimal cortical target f...

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Sleep Spindles and Slow Oscillations as Potential Biomarkers and Therapeutic Targets in Mild-to-Moderate Alzheimer's Disease

👤 Thien Thanh Dang-Vu 📅 2026-10-20 📖 1 min 📝 35 words TDCS TACS TRNS

Summary: PURPOSE OF REVIEW: This review critically evaluates evidence on sleep spindles (SP), slow oscillations (SOs), and their coordination in mild-to-moderate Alzheimer's disease (AD), focusing on their relationships with cognition, AD biomarkers, and their therapeutic potential.

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

👤 Xin Zhou 📅 2026-10-20 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Prior photon measurement density functions constructed in standard channel space for functional near-infrared spectroscopy

👤 Chao-Zhe Zhu 📅 2026-10-17 📖 1 min 📝 65 words FNIRS

Summary: Photon measurement density function (PMDF) determines the measurement locations and their associated sensitivity to the hemodynamic changes for an fNIRS channel. It is therefore essential for pre-experimental optode arrangement and post-experimental signal analysis and interpretation in fNIRS studie...

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Neckline-free open-source cap design and improved motion resistance for perioperative fNIRS procedures

👤 Audrey K Bowden 📅 2026-10-17 📖 1 min 📝 60 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) caps are limited by motion artifacts and unstable optode-scalp coupling, which degrade signal quality during extended and naturalistic monitoring sessions. We present a fabric-based, chinstrap-free headcap designed to improve mechanical stability and opt...

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Multiscale neural signatures of long-lag repetition learning: a simultaneous fNIRS-EEG study

👤 Ken Pugh 📅 2026-10-17 📖 1 min 📝 38 words FNIRS

Summary: CONCLUSIONS: Combining fNIRS and EEG with multimodal machine learning provides a richer characterization of repetition-based learning than either modality alone, establishing a framework for phenotyping language learning capacity and identifying neural markers of reading difficulties in pediatric po...

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Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

👤 Felipe Balistieri Santinelli 📅 2026-10-18 📖 1 min 📝 70 words FNIRS

Summary: People with multiple sclerosis (pwMS) often exhibit motor fatigability and reduced foot-tapping ability. However, the coupling between performance and cortical activity during fast, repetitive and prolonged ankle movements remains unexplored. We aimed to investigate the neural correlates of a prolon...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang
📅 2026-10-19 📖 1 min 📝 64 words FNIRS

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Neural dysfunction underlying divergent thinking and its shared mechanisms with protracted withdrawal symptoms in heroin use disorder

👤 Haijun Duan
📅 2026-10-19 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSIONS: These findings reveal a sustained compensatory pathological mechanism and an imbalance in neural resource allocation during complex cognition in HUD individuals. The identified signatures of cortical activity by fNIRS offer reliable biomarkers for the rapid detection of both higher-orde...

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Functional near-infrared spectroscopy as a tool for monitoring treatment-related neurofunctional changes in posttraumatic stress disorder and trauma-related symptomatology: a narrative review

👤 Maria Cotelli
📅 2026-10-20 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Current evidence suggests that fNIRS represents a promising and clinically feasible approach for monitoring treatment-related neurofunctional changes in PTSD and trauma-related symptomatology, particularly during non-pharmacological interventions targeting prefrontal regulatory processe...

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Transcutaneous vagus nerve stimulation during semantic verbal fluency tasks: a high-density diffuse optical tomography study

👤 Li Su 📅 2026-10-20 📖 1 min 📝 55 words FNIRS

Summary: INTRODUCTION: Transcutaneous auricular vagus nerve stimulation (taVNS) is a promising adjunct for post-stroke language recovery. There are no established biomarkers for the neuromodulatory effects of taVNS. High-density diffuse optical tomography (HD-DOT) offers a non-invasive marker of task-related...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

Hua Hu 2026-10-21 1 min 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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An exploration in mental health care: The launch of China's first psychiatric noninvasive brain-computer interface (BCI) and neuromodulation clinic

Tianhong Zhang 2026-10-17 1 min 2 words BRAIN COMPUTER INTERFACE

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A sliding window and Fourier decomposition method based 1D-CNN framework for motor imagery EEG classification

M Ramakrishnan 2026-10-17 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) signal classification is fundamental requirement for effective motor imagery (MI) brain-computer interface (BCI) applications. However, current MI classification methods often depend on handcrafted spatial filtering or complex time-frequency transformations that increase...

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Electroencephalography Changes in Emotional Recognition: A Signal Processing Approach Using Wavelet Features

Lakshmi Narasimhan 2026-10-17 1 min 43 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Wavelet-based EEG feature analysis provides meaningful and discriminative information for emotion recognition and may provide a robust feature set for future machine learning-based emotion recognition systems. Further studies incorporating classification algorithms and larger data sets a...

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A Compressive Neural Telemetry with Address-Event-μPacket Encoding and Duty-Cycling IR-UWB Transmitter

👤 Yao-Hong Liu 📅 2026-10-18 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: This article presents a networked wireless telemetry for 384-sensor-pixel neural data acquisition, designed to support high throughput, low power consumption, and meter-scale communication distance. The proposed event-based Address-Event-μPacket (AEμP) encoder with differential binary phase shift ke...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang 📅 2026-10-19 📖 1 min 📝 64 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Bridging the Gap Between Design and Access: Toward Visual Scene Displays for Brain-Computer Interface Use in Augmentative and Alternative Communication by Children

Ahad Behboodi 2026-10-19 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: These findings indicate that asymmetry and contextual embedding within scenes may enhance neural engagement and user experience, providing a preliminary foundation for considering photographic VSDs into pediatric BCI research.

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A spatio-temporal attention convolutional network with topology-preserving channel embedding and transformer for EEG decoding

Gang Liu 2026-10-19 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: EEG-based BCI has shown promising applications in motor rehabilitation and emotion recognition. Given the non-stationarity and variability of EEG signals, it is essential to develop classification models that can effectively extract spatio-temporal features while maintaining strong generalization ca...

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The calibration-to-accuracy mapping in P300 spellers does not transport across cohorts and was not restored by tested alignment or local recalibration

Oved Daniel 2026-10-19 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Objective
A calibration-derived score has been related to online P300-speller accuracy in the same session, always within the cohort measured. We evaluated whether a mapping from that score to expected accuracy transports to withheld cohorts.
Approach
Retrospective secondary analysis of ...

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Subjective emotional ratings dissociate brain activation and network patterns during counterfactual outcome processing

Ling Li 2026-10-19 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Human decision-making integrates outcome evaluation, counterfactual comparison, and individual traits. We investigated the neural processing of outcome-related emotions during economic decision-making by classifying feedback according to participants' subjective emotional ratings rather than objecti...

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Delineating neural contributions to electroencephalogram-based speech decoding

Shuntaro Sasai 2026-10-20 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Objective. Speech Brain-computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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Implicit learning of melodic structure: A role for pitch?

2024-01-22 1 min 180 words PSYCHOMUSICOLOGY

Summary: Growing evidence suggests that pitch influences musical processing, with melodic processing being enhanced in higher pitch ranges (e.g., Fujioka et al., 2005) and rhythmic processing being enhanced in lower pitches, and these effects may have a basis in elementary properties of the auditory system (...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000303

To those who feel unloved, misunderstood, or afraid

📅 2026-09-22 📖 3 min 📝 683 words FMHY

> **Summary:** I understand the world can feel devoid of love, but I swear, there is a spark within humanity that has the capability to care deeply. To love unconditionally. It doesn't always shine through us all, but it exists, it's real, it's tangible, and no one is exempt from it. We often see it shin...

🔗 **Read full article:**
https://fmhy.net/posts/love

I made a free Python ML cheat sheet repo — 9 topics

👤 /u/SK-Thaara_27 📅 2026-10-22 📖 1 min 📝 168 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Hey everyone, I put together a complete Python ML reference repo that covers everything I kept having to Google while learning: 📌 What's inside: • NumPy — arrays, broadcasting, linear algebra • Pandas — DataFrames, groupby, merge, datetim...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xcebo6/i_made_a_free_python_ml_cheat_sheet_repo_9_topics/

Parsing EDIFACT with syntax trees

👤 /u/alexprengere 📅 2026-10-22 📖 1 min 📝 94 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Edison is a Python library for parsing EDIFACT messages. Instead of treating a message as a flat list of segments, it models messages as a syntax tree: segments can define groups of child segments, recursively. You define the syntax tree once, then parse, val...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xccpkl/parsing_edifact_with_syntax_trees/

WorldTUI: I trained a 360M model to pretend to be a terminal

👤 alphax314 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://simedw.com/2026/10/21/world-model-terminal/ Comments URL: https://news.ycombinator.com/item?id=50190363 Points: 215 # Comments: 6...

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OpenAI CLI

👤 armcat 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

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🔗 **Read full article:**
https://github.com/openai/openai-cli

Achieving AI-style reasoning without an LLM

👤 edwardchalk 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://fleetingswallow.com/minotaur-ai-deep-dive-achieving-reasoning-without-a-language-model/ Comments URL: <a href="https://news.ycombinator.com/item?id=50192868">h...

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CVE-2026-111943: Compromised Intermediate Certificate Authority

👤 keks24 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://nvd.nist.gov/vuln/detail/cve-2026-111943 Comments URL: https://news.ycombinator.com/item?id=50192900 Points: 86 # Comments: 23

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Faster than earcut: triangulating 3.7M map polygons in 489 ms

👤 molind 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

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https://globus.software/blog/2026/10/16/cutting-map-polygons/

Recording and slides now available for ERPLAB Studio webinar

Steve Luck
2024-06-28 1 min 30 words ERP BOOT CAMP

Summary: We held a webinar to demonstration ERPLAB Studio on 28 June 2024. Click here to access a recording. Click here to access a PDF of the slides.<...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/28/recording-and-slides-now-available-for-erplab-studio-webinar

GenXR-Bench: Benchmarking Coding Agents for Interactive XR Development

William Huang, David Li, Xiang 'Anthony' Chen, Xun Qian, Nels Numan, Zhongyi Zhou, Hyunsung Cho, Salman Muin Kayser Chishti, Ding Xia, Xincheng Huang, Jiahao Ren, Faraz Faruqi, Tianyu Xu, Yang Zhang, David Kim, Ruofei Du

2026-10-22 1 min 181 words ARXIV CS HC

Summary: arXiv:2610.29014v1 Announce Type: cross Abstract: Coding agents are rapidly advancing from single-file generation to long-horizon repository-scale engineering, yet the evaluation of their capabilities to author interactive extended reality (XR) applications remains an open challenge. Unlike traditi...

🔗 **Read full article:**
https://arxiv.org/abs/2610.29014

Artificial Intelligence and Job Design: A Dynamic Framework

Amin Shirangi, Todd L. Pittinsky

2026-10-22 1 min 158 words ARXIV CS HC

Summary: arXiv:2610.28830v1 Announce Type: cross Abstract: This study aims to deepen the understanding of artificial intelligence's (AI) impact on work by introducing and describing the AI-Job Dynamics Framework. Drawing from sociotechnical systems theory, complexity theory, and organizational learning theo...

🔗 **Read full article:**
https://arxiv.org/abs/2610.28830

Scaling Therapy Supervision: Assessing Psychotherapy Fidelity Using Large Language Models

Arvind Pillai, Alexis Roach, Jason Holden, Varsha D Badal, Subigya Nepal, Yu Yvonne Wu, Yuliang Chen, Sudarshan Regmi, Sarah Masud Preum, Nicholas C Jacobson, Nikhil Singh, Eric Granholm, Colin Depp, Andrew Campbell

2026-10-22 1 min 159 words ARXIV CS HC

Summary: arXiv:2610.28772v1 Announce Type: cross Abstract: Psychotherapy fidelity assessment measures how faithfully a therapist adheres to a treatment protocol, a critical determinant of positive patient outcomes. However, fidelity assessment is bottlenecked by manual expert review, rendering the process l...

🔗 **Read full article:**
https://arxiv.org/abs/2610.28772

Warping the Workspace: Expanding Visual Access with Adjustable Reach Mapping for Humanoid Teleoperation

Keigo Minamida, Koya Dendo, Yuto Nakamura, Jun Rekimoto

2026-10-22 1 min 79 words

ARXIV CS HC

Summary: arXiv:2610.28712v1 Announce Type: cross Abstract: First-person humanoid teleoperation ties access to side and rear targets to operator rotation. We present coupled visual--motor remapping, pairing a panoramic azimuth warp with its inverse upper-limb yaw transform. A working prototype implements Glo...

🔗 **Read full article:**
https://arxiv.org/abs/2610.28712

Measurement-Guided Conversion of Laboratory Eye-Tracking Data for VR Authentication

Md Arif Shahriar, Mehedi Hasan Raju, Oleg V. Komogortsev

2026-10-22 1 min 153 words

ARXIV CS HC

Summary: arXiv:2610.30032v1 Announce Type: new Abstract: Eye-movement authentication in virtual reality (VR) needs longitudinal training data, yet most such datasets come from lab trackers. We convert lab recordings to a VR headset's signal quality while keeping their eye movement, using only unpaired VR re...

🔗 **Read full article:**
https://arxiv.org/abs/2610.30032

The Serendiptionnaire: a Questionnaire for Measuring Experienced Serendipity

Brett Binst, Annelien Smets
2026-10-22 1 min 175 words ARXIV CS HC

Summary: arXiv:2610.29487v1 Announce Type: new Abstract: Serendipity is important to design for, yet it remains difficult to measure and evaluate. No validated questionnaire exists, so studies either borrow ResQue's novelty scale, which does not measure serendipity, or build their own items, which makes fin...

🔗 **Read full article:**
https://arxiv.org/abs/2610.29487

VIRA-Bench: Diagnosing Pedagogical Contingency Understanding in Online Tutoring Videos

Weihang Wang, Hansu Gu, Guanming Liu, Peng Zhang, Ning Gu, Tun Lu
2026-10-22 1 min 203 words ARXIV CS HC

Summary: arXiv:2610.29428v1 Announce Type: new Abstract: Understanding a one-to-one tutoring video requires more than recognizing its events. A model must connect a visible teacher intervention to the learner response it elicits and to the follow-up adjusted to that response. Existing video and educational-...

🔗 **Read full article:**
https://arxiv.org/abs/2610.29428

A Stratified Multi-Rater Evaluation of LLM-Based Virtual Standardized Patients with a Deployed Data-Generation Platform

Yongjin Yi, Jinyong Park, Sejoong Kim
2026-10-22 1 min 244 words ARXIV CS HC

Summary: arXiv:2610.29312v1 Announce Type: new Abstract: The Objective Structured Clinical Examination (OSCE) is widely used to assess clinical communication skills in medical education, requiring trainees to take histories and reason clinically with standardized patients (SPs). Large language model (LLM)-b...

🔗 **Read full article:**
https://arxiv.org/abs/2610.29312

From AI-Free Gates to AI-First Work: How Engineering Students in India Navigate University Education and Campus Placements Without AI Guidelines

Samhitha Goutham, Ziwei Gu
2026-10-22 1 min 177 words ARXIV CS HC

Summary: arXiv:2610.29140v1 Announce Type: new Abstract: Engineering graduates in India enter a technology sector that is being reshaped by generative AI, yet the route into that sector runs through examinations and campus placement screens that largely exclude AI. Institutions in India have an approved int...

🔗 **Read full article:**
https://arxiv.org/abs/2610.29140

Scriptorium: System Design and Implementation of a Multi-Human, Multi-AI Collaborative Digital Library Search Platform

Abbas Pirmoradi, Milad Momeni, Anna Chu, Orland Hoeber

2026-10-22 1 min 157 words

ARXIV CS HC

Summary: arXiv:2610.28929v1 Announce Type: new Abstract: This technical report documents the design and implementation of Scriptorium, a digital library search interface designed to support multi-human, multi-AI collaboration: full cross-user transparency of search activity combined with a Manager--Worker a...

🔗 **Read full article:**
https://arxiv.org/abs/2610.28929

Distinct weak antisymmetric interactions shape human brain functions as probability fluxes

Yoshiaki Horiike, Shin Fujishiro

2026-10-22 1 min 204 words

ARXIV QBIO NC

Summary: arXiv:2508.20961v3 Announce Type: replace-cross Abstract: The functional computation of the human brain is hypothesized to arise from the collective behaviour of the underlying neural network. A method based on an idea from equilibrium statistical mechanics has been applied to infer the underlying ...

🔗 **Read full article:**
https://arxiv.org/abs/2508.20961

Binding-Motivated Contextuality: A Cross-Domain Cyclic Test in Perception and Judgment

Adam Y. Shavit 2026-10-22 1 min 270 words ARXIV QBIO NC

Summary: arXiv:2609.23977v3 Announce Type: replace Abstract: Psychophysics and decision research study perceptual binding and judgment contextuality separately. We argue a binding-motivated perceptual task and judgment contextuality are scored against the same cyclic noncontextuality inequalities and share ...

🔗 **Read full article:**
https://arxiv.org/abs/2609.23977

Perceived vertical and eye level as one orientation process: a closed-form account of the Li-Matin combination rules for egocentric space

A. Y. Shavit 2026-10-22 1 min 302 words ARXIV QBIO NC

Summary: arXiv:2607.19681v4 Announce Type: replace Abstract: Li and Matin (2005a, 2005b) reported three regularities of the induced visually-perceived vertical (VPV), the direction judged upright. For small tilts, a peripheral line shifts it nearly linearly with tilt, two lines combine nearly linearly, and ...

🔗 **Read full article:**
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Predictive and feedback signals differently shape the formation of language representations

Shuguang Yang, Shaoyun Yu, Xin Jiang, Suiping Wang, Gangyi Feng

2026-10-22 1 min

197 words

ARXIV QBIO NC

Summary: arXiv:2605.09409v2 Announce Type: replace Abstract: Adults vary greatly in how effectively they learn a new language, but the signals driving the learning processes and individual differences remain unclear. Over seven days, we tracked behavioral learning and collected fMRI data on day 1 from 102 a...

🔗 **Read full article:**
https://arxiv.org/abs/2605.09409

Bootstrapping Life-Inspired Machine Intelligence: The Biological Route from Chemistry to Cognition and Creativity

Giovanni Pezzulo, Michael Levin

2026-10-22 1 min

200 words

ARXIV QBIO NC

Summary: arXiv:2602.08079v3 Announce Type: replace Abstract: Achieving advanced machine intelligence remains a central challenge in AI research, often approached through scaling neural architectures and generative models. However, biological systems offer a broader repertoire of strategies for adaptive, goa...

🔗 **Read full article:**
https://arxiv.org/abs/2602.08079

Interpretable Embeddings of Speech Enhance and Explain Brain Encoding Performance of Audio Models

Riki Shimizu, Richard J. Antonello, Chandan Singh, Nima Mesgarani

2026-10-22 1 min

178 words

ARXIV QBIO NC

Summary: arXiv:2507.16080v3 Announce Type: replace Abstract: Speech foundation models (SFMs) are increasingly hailed as powerful computational models of human speech perception. However, since their representations are inherently black-box, it remains unclear what drives their alignment with brain responses...

🔗 **Read full article:**
https://arxiv.org/abs/2507.16080

Population Neuroscience 3.0: Bridging large-N and small-N paradigms in brain-wide association studies

Zi-Xuan Zhou, Xi-Nian Zuo

2026-10-22 1 min 224 words

ARXIV QBIO NC

Summary: arXiv:2212.04195v4 Announce Type: replace Abstract: Brain-wide association studies (BWAS) leveraging large datasets, have led to divergent and sometimes opposing views on research practices and priorities. The debates span multiple dimensions, including large-N vs.\ small-N sampling, cross-sectiona...

🔗 **Read full article:**
https://arxiv.org/abs/2212.04195

Bio-Inspired Auditory Processing: Coupling Parallel Resonate-and-Fire Dynamics with Ternary Recurrent Networks for Speech Recognition

Pavlos Poulos, Katelyn Sadorf, Themos Stafylakis, Costas A. Anastassiou

2026-10-22 1 min

254 words

ARXIV QBIO NC

Summary: arXiv:2610.29720v1 Announce Type: cross Abstract: Spiking Neural Networks (SNNs) promise energy-efficient inference through sparse, event-driven computation, yet remain constrained on continuous, context-dependent tasks such as automatic speech recognition (ASR). A key bottleneck is the Leaky Integ...

Read full article:
https://arxiv.org/abs/2610.29720

A Reverse-Gold Theorem: Positive-Data Identification by Substrate-Bounded Contrastive Learners

Rakesh Sengupta

2026-10-22 1 min 241 words

ARXIV QBIO NC

Summary: arXiv:2610.28747v1 Announce Type: cross Abstract: Gold's theorem shows that no learner can identify an arbitrary language in a superfinite class from positive presentations alone. We prove a positive result of a different shape. Fix an innate \emph{recursive contrast operator} R mapping any strin...

Read full article:
https://arxiv.org/abs/2610.28747

Vagus nerve stimulation strengthens beta- and gamma-rate auditory steady-state responses in rat cortex

Shinichi Kumagai, Tomoyo Isoguchi Shiramatsu, Karin Oshima, Yohei Ishishita, Kenji Ibayashi, Yoshiyuki Onuki, Naoto Kunii, Kensuke Kawai, Hirokazu Takahashi

2026-10-22 1 min 233 words ARXIV QBIO NC

Summary: arXiv:2610.29042v1 Announce Type: new Abstract: Background: Gamma entrainment using sensory stimulation (GENUS) is under study as a treatment for neurological disease, but cortical engagement is variable. Vagus nerve stimulation (VNS) modulates cortical state and sound-evoked beta-gamma activity, s...

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https://arxiv.org/abs/2610.29042

Psychological distress and sleep quality related to cognitive function in older adults with cognitive impairment

2026-10-22 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s41598-026-56229-7

The associations between muscle strength and executive function among Chinese adolescents

📅 2026-10-22 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76586-7

Prosthesis embodiment predicts reduced amputation salience as reflected in dreams

📅 2026-10-22 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-77088-2

Individualized morphometric-similarity deviations in autism linked to cortical hierarchy and microarchitecture

📅 2026-10-22 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s42003-026-11166-z

CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-10-22 1 min 231 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMotor imagery-based brain-computer interface (MI-BCI) holds great potential in neurorehabilitation and motor assistance. Data quality and motor intention decoding effectiveness are closely associated with participants' engagement. However, previous MI tasks ignored active engagement regula...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1908608

Prefrontal cortical morphometric covariance and cognitive task performance in healthy young adults: an exploratory analysis of Human Connectome Project data

Roman Chivers 2026-10-22 1 min 343 words FRONTIERS HUMAN NEUROSCIENCE

Summary: The prefrontal cortex (PFC) is implicated in many cognitive tasks, including working memory, reward-based decision-making, executive control, and metacognition. This exploratory analysis investigated whether individual expression of PFC cortical thickness and surface area covariance components is as...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1976371

Explainable fMRI decoding reveals neurobiological signatures of emotional arousal complementary to traditional analyses

👤 Daniel Agostinho, Davide Borra, Miguel Castelo-Branco and Marco Simões

📅 2026-10-13 📖 1 min

📝 283 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Understanding how emotional arousal is represented in the brain is fundamental for elucidating affective regulation and its alterations in autism spectrum disorder. Classical neuroimaging approaches, such as the general linear model (GLM), have provided important insights into emotional p...

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https://iopscience.iop.org/article/10.1088/1741-2552/aead27

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt

📅 2025-01-09 📖 1 min 📝 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Behaviorally-relevant features of observed actions dominate cortical representational geometry in natural vision

Samuel A Nastase 2026-10-19 1 min 67 words LOW VISION

Summary: We effortlessly extract behaviorally relevant information from dynamic visual input to understand the actions of others. Here, we compare behavioral judgments of action meaning against other models capturing semantic and visual features. We measured brain activity using fMRI while participants viewe...

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Birth of the Nigerian Sleep Society: A Milestone for Sleep Medicine in Nigeria and Africa

G E Erhabor 2026-10-20 1 min 72 words LOW VISION

Summary: Sleep disorders are significant and are of major public health concern with high socioeconomic impact. However, they are under-recognised and under-treated in Nigeria and most of Africa, because of poor access to sleep specialists, sleep studies and limited treatment options. Prior to the inaugurati...

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Knowledge gaps and practices for glaucoma among adults in Palestine: a cross-sectional study

Farid K Abu Sham'a 2026-10-20 1 min 51 words LOW VISION

Summary: CONCLUSIONS: Although awareness of glaucoma is moderate in Palestine, comprehensive knowledge remains low, particularly among high-risk populations. Public health interventions should target education beyond name recognition, emphasizing asymptomatic progression, irreversible vision loss, and risk f...

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Sn/Cl Co-Doped ZnO Cathode Interfacial Layer for High-Performance Organic Photodetectors

Kai Zhang 2026-10-20 1 min 66 words LOW VISION

Summary: Organic photodetectors (OPDs) hold significant promise for applications spanning medical monitoring, bioimaging and night vision due to their unique advantages, such as flexibility, light weight, and potential for large-area fabrication. Nevertheless, the dark current density (Jd) of OPDs exhibits a...

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A thermopile array and stereo vision fusion system for small-tube temperature sensing

Dongfang Hu 2026-10-20 1 min 71 words LOW VISION

Summary: This paper presents a non-contact temperature measurement scheme for small-diameter tubes by fusing a low-resolution thermopile array (TPA) with visible-light stereo vision. Stereo vision locates the tube in three dimensions. Then, this geometry is projected onto the TPA imaging plane to constrain t...

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Cost-effective long-range measurement of structural displacement using a two-stage method integrating super-resolution reconstruction and sub-pixel registration

Xiaosong Li 2026-10-20 1 min 61 words LOW VISION

Summary: Displacement is an imperative metric for assessing structural performance. However, the accuracy of structural displacement measurement is constrained by the physical resolution of the camera, particularly in long-range monitoring scenarios. To address these constraints, a cost-effective long-range ...

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Physics-aware restoration and calibrated soft demodulation for turbulence-degraded OAM mode recognition

Jun Zhu 2026-10-20 1 min 58 words LOW VISION

Summary: Turbulence-degraded orbital angular momentum (OAM) mode recognition is critical for free-space optical (FSO) communication but is limited by atmospheric distortion. We propose a cascaded receiver integrating a vision transformer-based Uformer-OAM intensity-restoration network, an SE-ResNet-18 neural...

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Image-free ghost edge detection at ultra-low sampling rates via a Vision Mamba network

Rui Wang 2026-10-20 1 min 66 words LOW VISION

Summary: Edge detection is essential for extracting structural boundaries in visual scenes, but conventional deep-learning-based methods remain sensitive to degradations in the input images and involve redundant full-image acquisition. Image-free ghost edge detection provides an efficient alternative by dire...

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Ex vivo evaluation of corneal permeation and irritation potential associated with proparacaine hydrochloride misuse

Yolima Baena 2026-10-20 1 min 35 words LOW VISION

Summary: CONCLUSION: Proparacaine demonstrates dose-dependent permeation and histopathological changes, emphasizing the risks associated with misuse. The ex vivo model provides relevant and ethically aligned evidence to support ocular hazard assessment through the characterizing of histopathological changes.

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Shared variance across FrACT visual tasks: Robust positive associations but inconclusive latent structure

Luca Battaglini 2026-10-21 1 min 68 words LOW VISION

Summary: A long-standing question in vision science is whether individual differences in visual performance reflect a general visual ability or a set of largely task-specific processes. Previous studies have often reported weak associations across heterogeneous perceptual tasks, challenging the idea of a com...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

> **Summary:** CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Written exposure therapy augmented with transcranial direct current stimulation for posttraumatic stress disorder: A pilot randomized clinical trial

👤 STRONG STAR Consortium 📅 2026-10-18 📖 1 min 📝 64 words TDCS TACS TRNS

> **Summary:** Written exposure therapy (WET) is an evidence-based, trauma-focused psychotherapy for posttraumatic stress disorder (PTSD) that has been tested with several trauma-exposed samples. Although many individuals who receive WET and other trauma-focused psychotherapies experience a significant reduction i...

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Transcranial Direct Current Stimulation and Cranial Electrotherapy Stimulation in Athletes: Optimizing Athletic Performance and Sleep Quality Through Neuromodulation

👤 Paulina Gajniak
📅 2026-10-19 📖 1 min 📝 74 words

TDCS TACS TRNS

Summary: Neuromodulation is a procedure relying on the delivery of specific stimuli to selected sites of the nervous system and hereby influencing its activity. Even though the underlying neurobiological mechanisms are not yet fully elucidated and the research is still ongoing, neuromodulation is already bei...

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Clinical outcomes and safety of Five Element Acupuncture for preventive treatment of primary headache disorders: a multi-center real-world longitudinal observational study

👤 Shengyuan Zhou
📅 2026-10-19 📖 1 min 📝 49 words

TDCS TACS TRNS

Summary: CONCLUSIONS: This real-world exploratory observational study suggests that FEA is associated with significant improvements in headache days, headache frequency, headache intensity, analgesic use, and quality of life, with sustained effects noted during follow-up. These findings provide preliminary o...

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Home Transcranial Direct Current Stimulation With Gamified Cognitive Control Training for Depression: A Randomized Clinical Trial

👤 Mor Nahum 📅 2026-10-19 📖 1 min 📝 45 words TDCS TACS TRNS

Summary: CONCLUSIONS AND RELEVANCE: This RCT of home-based tDCS combined with CCT in adults with MDD was feasible and safe but did not demonstrate superior efficacy over the control condition. Future studies should refine the mechanistic alignment between stimulation and cognitive interventions to optimize t...

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Comparative chloroplast genomics of Ajania sericea and sympatric Asteraceae species identifies shared candidate molecular features associated with alpine adaptation

Xi Chen 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study showed that the chloroplast genomes of four sympatric Asteraceae species were highly conserved in overall structure, but still exhibited certain differences in gene loss, IR boundary variation, repeat sequence composition, and local hypervariable regions. These features provi...

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Brain Stimulation as a Treatment for Apathy: A Scoping Review of Effects and Safety

Lisa M McTeague 2026-10-20 1 min 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: Investigations into brain stimulation as a treatment for apathy show some promise. Conclusions were limited by small sample sizes, heterogeneous study designs, and variable outcome measures. Randomized trials with common validated apathy instruments, standardized stimulation parameters,...

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Effect of Different Cortical Targets of Transcranial Magnetic Stimulation on Freezing of Gait in Parkinsonism: A Systematic Review

Aliaa Elabd
2026-10-20 1 min 68 words TDCS TACS TRNS

Summary: BACKGROUND AND PURPOSE: Freezing of gait (FOG) is a disabling and episodic gait disturbance in patients with Parkinson's, with varying responses to noninvasive brain stimulation. Transcranial magnetic stimulation (TMS) has been explored as a therapeutic option; however, the optimal cortical target f...

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Sleep Spindles and Slow Oscillations as Potential Biomarkers and Therapeutic Targets in Mild-to-Moderate Alzheimer's Disease

👤 Thien Thanh Dang-Vu 📅 2026-10-20 📖 1 min 📝 35 words TDCS TACS TRNS

Summary: PURPOSE OF REVIEW: This review critically evaluates evidence on sleep spindles (SP), slow oscillations (SOs), and their coordination in mild-to-moderate Alzheimer's disease (AD), focusing on their relationships with cognition, AD biomarkers, and their therapeutic potential.

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

👤 Xin Zhou 📅 2026-10-20 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

Fanshuo Zeng
2026-10-21 1 min 53 words TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Prior photon measurement density functions constructed in standard channel space for functional near-infrared spectroscopy

👤 Chao-Zhe Zhu 📅 2026-10-17 📖 1 min 📝 65 words FNIRS

Summary: Photon measurement density function (PMDF) determines the measurement locations and their associated sensitivity to the hemodynamic changes for an fNIRS channel. It is therefore essential for pre-experimental optode arrangement and post-experimental signal analysis and interpretation in fNIRS studie...

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Neckline-free open-source cap design and improved motion resistance for perioperative fNIRS procedures

👤 Audrey K Bowden 📅 2026-10-17 📖 1 min 📝 60 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) caps are limited by motion artifacts and unstable optode-scalp coupling, which degrade signal quality during extended and naturalistic monitoring sessions. We present a fabric-based, chinstrap-free headcap designed to improve mechanical stability and opt...

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Multiscale neural signatures of long-lag repetition learning: a simultaneous fNIRS-EEG study

Ken Pugh 2026-10-17 1 min 38 words FNIRS

Summary: CONCLUSIONS: Combining fNIRS and EEG with multimodal machine learning provides a richer characterization of repetition-based learning than either modality alone, establishing a framework for phenotyping language learning capacity and identifying neural markers of reading difficulties in pediatric po...

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Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

Felipe Balistieri Santinelli 2026-10-18 1 min 70 words FNIRS

Summary: People with multiple sclerosis (pwMS) often exhibit motor fatigability and reduced foot-tapping ability. However, the coupling between performance and cortical activity during fast, repetitive and prolonged ankle movements remains unexplored. We aimed to investigate the neural correlates of a prolon...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang
📅 2026-10-19 📖 1 min 📝 64 words FNIRS

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Neural dysfunction underlying divergent thinking and its shared mechanisms with protracted withdrawal symptoms in heroin use disorder

👤 Haijun Duan
📅 2026-10-19 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSIONS: These findings reveal a sustained compensatory pathological mechanism and an imbalance in neural resource allocation during complex cognition in HUD individuals. The identified signatures of cortical activity by fNIRS offer reliable biomarkers for the rapid detection of both higher-orde...

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Functional near-infrared spectroscopy as a tool for monitoring treatment-related neurofunctional changes in posttraumatic stress disorder and trauma-related symptomatology: a narrative review

👤 Maria Cotelli
📅 2026-10-20 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Current evidence suggests that fNIRS represents a promising and clinically feasible approach for monitoring treatment-related neurofunctional changes in PTSD and trauma-related symptomatology, particularly during non-pharmacological interventions targeting prefrontal regulatory processe...

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Transcutaneous vagus nerve stimulation during semantic verbal fluency tasks: a high-density diffuse optical tomography study

👤 Li Su 📅 2026-10-20 📖 1 min 📝 55 words FNIRS

Summary: INTRODUCTION: Transcutaneous auricular vagus nerve stimulation (taVNS) is a promising adjunct for post-stroke language recovery. There are no established biomarkers for the neuromodulatory effects of taVNS. High-density diffuse optical tomography (HD-DOT) offers a non-invasive marker of task-related...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

Hua Hu 2026-10-21 1 min 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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An exploration in mental health care: The launch of China's first psychiatric noninvasive brain-computer interface (BCI) and neuromodulation clinic

Tianhong Zhang 2026-10-17 1 min 2 words BRAIN COMPUTER INTERFACE

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A sliding window and Fourier decomposition method based 1D-CNN framework for motor imagery EEG classification

M Ramakrishnan 2026-10-17 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) signal classification is fundamental requirement for effective motor imagery (MI) brain-computer interface (BCI) applications. However, current MI classification methods often depend on handcrafted spatial filtering or complex time-frequency transformations that increase...

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Electroencephalography Changes in Emotional Recognition: A Signal Processing Approach Using Wavelet Features

Lakshmi Narasimhan 2026-10-17 1 min 43 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Wavelet-based EEG feature analysis provides meaningful and discriminative information for emotion recognition and may provide a robust feature set for future machine learning-based emotion recognition systems. Further studies incorporating classification algorithms and larger data sets a...

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A Compressive Neural Telemetry with Address-Event-μPacket Encoding and Duty-Cycling IR-UWB Transmitter

👤 Yao-Hong Liu 📅 2026-10-18 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: This article presents a networked wireless telemetry for 384-sensor-pixel neural data acquisition, designed to support high throughput, low power consumption, and meter-scale communication distance. The proposed event-based Address-Event-μPacket (AEμP) encoder with differential binary phase shift ke...

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Temporal synchrony promotes motor recovery and ipsilesional cortical reorganization in BCI-based stroke rehabilitation: a randomized controlled trial

👤 Ming Zhang 📅 2026-10-19 📖 1 min 📝 64 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Synchronous BCI training was associated with superior clinical improvement compared with asynchronous and sham conditions in subacute stroke. The benefit was observed only when robotic assistance was contingent on real-time task performance, despite all three groups receiving comparable...

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Bridging the Gap Between Design and Access: Toward Visual Scene Displays for Brain-Computer Interface Use in Augmentative and Alternative Communication by Children

Ahad Behboodi 2026-10-19 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: These findings indicate that asymmetry and contextual embedding within scenes may enhance neural engagement and user experience, providing a preliminary foundation for considering photographic VSDs into pediatric BCI research.

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A spatio-temporal attention convolutional network with topology-preserving channel embedding and transformer for EEG decoding

Gang Liu 2026-10-19 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: EEG-based BCI has shown promising applications in motor rehabilitation and emotion recognition. Given the non-stationarity and variability of EEG signals, it is essential to develop classification models that can effectively extract spatio-temporal features while maintaining strong generalization ca...

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The calibration-to-accuracy mapping in P300 spellers does not transport across cohorts and was not restored by tested alignment or local recalibration

Oved Daniel 2026-10-19 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Objective
A calibration-derived score has been related to online P300-speller accuracy in the same session, always within the cohort measured. We evaluated whether a mapping from that score to expected accuracy transports to withheld cohorts.
Approach
Retrospective secondary analysis of ...

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Subjective emotional ratings dissociate brain activation and network patterns during counterfactual outcome processing

Ling Li 2026-10-19 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Human decision-making integrates outcome evaluation, counterfactual comparison, and individual traits. We investigated the neural processing of outcome-related emotions during economic decision-making by classifying feedback according to participants' subjective emotional ratings rather than objecti...

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Delineating neural contributions to electroencephalogram-based speech decoding

Shuntaro Sasai 2026-10-20 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Objective. Speech Brain-computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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https://pubmed.ncbi.nlm.nih.gov/42899673/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261022011300&v=2.20.1

Whoever got bored and put random stuff in there code

/u/Randomstuff0550 2026-10-22 1 min 78 words REDDIT PYTHON

Summary: <!-- SC_OFF --> Bro, I got bored, so i put some random stuff in my code, dont worry - I used a hashtag, thats preety crazy, Python there, 120 characters <pre><code>#Discord #hi #Hello! #What are we doing? #Typing in python #Can the output see us? #No, because of the tag...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xc702c/whoever_got_bored_and_put_random_stuff_in_there/

Postgres recovery_prefetch can cause outages

👤 plaur782 📅 2026-10-21 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://ardentperf.com/2026/10/20/postgres-recovery_prefetch-considered-harmful-and-causing-outages/ Comments URL: <a href="https://news.ycombinator.com/item?id=5...

🔗 **Read full article:**
https://ardentperf.com/2026/10/20/postgres-recovery_prefetch-considered-harmful-and-causing-outages/

mitmproxy adds --emudhra option

👤 ahlCVA 📅 2026-10-21 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/mitmproxy/mitmproxy/commit/169234499088fc2c414644a67efaed3d64ae2e3b Comments URL: <a href="https://news.ycombinator.com/item?id=50189232">https://news.ycombi...

🔗 **Read full article:**
https://github.com/mitmproxy/mitmproxy/commit/169234499088fc2c414644a67efaed3d64ae2e3b

LLM agents play Pokemon showdown

👤 grabnero 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://olivergrabner.com/research/pokemon/ Comments URL: https://news.ycombinator.com/item?id=50190334 Points: 21 # Comments: 11

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https://olivergrabner.com/research/pokemon/

The recently proposed o(n log n) integer multiplication algorithm [pdf]

👤 craigcitro 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://web.maths.unsw.edu.au/~davidharvey/research/subnlogn.pdf Comments URL: https://news.ycombinator.com/item?id=50190455 Points...

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https://web.maths.unsw.edu.au/~davidharvey/research/subnlogn.pdf

I can't believe PowerShell is good

👤 corbindavenport 📅 2026-10-22 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.spacebar.news/powershell-is-actually-good/ Comments URL: https://news.ycombinator.com/item?id=50190959 Points: 22 # Commen...

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https://www.spacebar.news/powershell-is-actually-good/

New Paper: Does the P3b component reflect working memory updating?

👤 Steve Luck 📅 2025-03-21 📖 7 min 📝 1540 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Simmons, A. M., Kiat, J. E., & Luck, S. J. (in press). Enhanced working memory representations for rare events. Psychophysiology. https://doi.org/10.1111/psyp.70038 [<a href="https://doi.org/10.1101/2024.03.20...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/3/20/new-paper-oddball

10-Day ERP Boot Camp to be held June 15-24, 2026 in Davis, California

👤 Steve Luck 📅 2025-08-20 📖 1 min 📝 125 words ERP BOOT CAMP

Summary: We have received another 5 years of funding from the National Institute of Mental Health, so we plan to hold ERP Boot Camps in each of the next 5 summers. The next one will be June 15-24, 2026 in Davis, California. The application portal will open around January 1, 2026. A...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/8/20/boot-camp-summer-2026

Call for Papers: IEEE Brain Special Issue

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 36 words BRAIN

Summary: In a unique interdisciplinary collaboration with the IEEE's Society on Social Implications of Technology (SSIT) and IEEE Brain, J-FLEX is joining forces to explore both the technology of the Internet-of-Medical-Things (IoMT) solutions and medical wearables/ implantables.

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-journal-on-flexible-electronics/

IEEE Brain Joins the American Brain Coalition

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 61 words BRAIN

Summary: IEEE Brain is pleased to announce its acceptance as a nonprofit member of the American Brain Coalition (ABC), a prestigious alliance of over 150 organizations dedicated to advancing brain research, advocacy, and improving treatments for individuals affected by brain conditions. The ABC Board has ent...

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-brain-joins-the-american-brain-coalition-as-a-nonprofit-member/

Call for Papers: IEEE Transactions on Human-Machine Systems

👤 Adriel Carridice 📅 2025-06-18 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/call-for-papers-ieee-transactions-on-human-machine-systems/

Q&A with Dr. Alain Dagher, Neurologist and Professor at McGill University and the Montreal Neurological Institute

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-alain-dagher-neurologist-and-professor-at-mcgill-university-and-the-montreal-neurological-institute/

IEEE Advances Responsible Neurotechnology Standard

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/news-and-insights/ieee-brain-news/ieee-advances-responsible-neurotechnology-standard/

Q&A with Dhruv Vaish, PhD student at UC Berkeley and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/podcasts/qa-with-dhruv-vaish-phd-student-at-uc-berkeley-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Hippocampus mediates conceptual generalization of pain modulation

Sutterlin-Guindon, D., Wager, T. D., Koban, L.

2026-10-21 1 min 233 words

BIORXIV NEUROSCIENCE

Summary: Pain is strongly influenced by expectations and learning from previous experience, such as in classical conditioning. Conditioned responses and expectations can generalize to perceptually and conceptually related cues, but how generalization influences pain experience and the neurobiological process...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.16.759532v1?rss=1

Perceptual repulsion and decisional attraction in male body serial dependence: a dimensional dissociation in channel architecture

Cass, J., Talbot, D.

2026-10-21 1 min 251 words

BIORXIV NEUROSCIENCE

Summary: Our judgments of the world are shaped by what we have just seen. This serial dependence is often treated as one bias, but it can constitute opposing influences: an attractive pull toward recent input and a repulsive push away from it. These have been linked to different stages, decisional and percep...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.15.756928v1?rss=1

Polyglutamine-expanded TBP rewires neuronal and muscle proteomes beyond transcription in a Drosophila model of SCA17

Prifti, M. V., Richardson, K., Hussein, F. H., Kula, T., Stephan, A., Wang, Y.-D., Wang, W., High, A. A., Poudel, S., Demontis, F., Todi, S. V.

2026-10-21 1 min 150 words BIORXIV NEUROSCIENCE

Summary: Spinocerebellar ataxia type 17 (SCA17) is an age-related neurodegenerative disorder caused by polyglutamine expansion in the TATA-binding protein (TBP). TBP is a core transcription factor, but pathogenic polyglutamine expansion likely rewires cellular programs beyond transcription. Here, we utilized...

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Cross-modal alignment with language tokens for unified multilingual image-to-text generation

Guanjun Lin

2026-10-14 1 min 274 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Current multimodal translation systems typically employ a cascaded pipeline where image captioning and machine translation are performed sequentially, leading to error propagation and semantic drift. This paper introduces a unified neural architecture that integrates visual encoding, English caption...

Read full article:
https://www.frontiersin.org/articles/10.3389/fncom.2026.1826455

Embodied multimodal human-computer interaction for English language learning: integrating speech, gestures, emotion, and adaptive feedback

Yao Tang 2026-10-14 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: Current English language learning systems are often limited by single-dimensional learning-state perception, insufficient modeling of affective-cognitive characteristics, and delayed or inflexible feedback regulation. To address these limitations, this study proposes an affective-cognitive gated cro...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1920739

Temporal signatures of prediction error reveal external forces on an embodied robot

Christian Balkenius 2026-10-14 1 min 218 words FRONTIERS NEUROROBOTICS

Summary: Embodied robots must distinguish self-generated sensory consequences from externally caused perturbations. In predictive processing, such deviation is accounted for as prediction error, but it is commonly treated primarily as a signal to be minimised. Here, we test whether the temporal profile of lo...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1918275

Thursday Daily Thread: Python Careers, Courses, and Furthering Education!

👤 /u/AutoModerator 📅 2026-10-22 📖 1 min 📝 219 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Thread: Professional Use, Jobs, and Education 🏢</h1> Welcome to this week's discussion on Python in the professional world! This is your spot to talk about job hunting, career growth, and educational resources in Python. Please note, this thread is <stron...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xc3ijg/thursday_daily_thread_python_careers_courses_and/

CleanFrame: A quieter home screen for Samsung The Frame

👤 ingve 📅 2026-10-21 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/nthnclrk/CleanFrame Comments URL: https://news.ycombinator.com/item?id=50187659 Points: 17 # Comments: 2

🔗 **Read full article:**
https://github.com/nthnclrk/CleanFrame

PHP web runtimes (CVE-2026-109618 and CVE-2026-109619)

👤 hayd 📅 2026-10-21 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://aws.amazon.com/security/security-bulletins/2026-143-aws/ Comments URL: https://news.ycombinator.com/item?id=50187708 Points...

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https://aws.amazon.com/security/security-bulletins/2026-143-aws/

Railway Security Disclosure

👤 passenger 📅 2026-10-21 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.railway.com/p/security-disclosure-wednesday-october-21st Comments URL: https://news.ycombinator.com/item?id=50188228 ...

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https://blog.railway.com/p/security-disclosure-wednesday-october-21st

Qpack amplification attacks against HTTP/3 servers

👤 kozika 📅 2026-10-21 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://foxio.io/blog/header-accordions-qpack-amplification-attacks-against-http-3-servers Comments URL: <a href="https://news.ycombinator.com/item?id=50188612">https://new...

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https://foxio.io/blog/header-accordions-qpack-amplification-attacks-against-http-3-servers

There isn't an easy exploit for this PHP vulnerability

👤 zkldi 📅 2026-10-21 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://eliastaylor.com/noexploit/ Comments URL: https://news.ycombinator.com/item?id=50188948 Points: 10 # Comments: 1

🔗 **Read full article:**
https://eliastaylor.com/noexploit/

Decoding Spikes From Multiunit Data

📅 2026-01-22 📖 1 min 📝 250 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Communication and control in biological systems is mediated by the timing of discharges –spikes– from excitable cells such as neurons and muscle fibers. Each spike is associated to a characteristic waveform that can be captured by sensors. The waveform's characteristics depend on the cell's biophysi...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11361153

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📅 2026-01-29 📖 1 min 📝 1 words REVIEWS BIOMEDICAL ENGINEERING

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http://ieeexplore.ieee.org/document/11368645

Welcoming the Newly Elected EMBS AdCom Members for 2026–2028

👤 dziura 📅 2025-12-10 📖 1 min 📝 18 words EMBS

> **Summary:** The post Welcoming the Newly Elected EMBS AdCom Members for 2026–2028 appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/uncategorized/welcoming-the-newly-elected-embs-adcom-members-for-2026-2028/

Welcoming & Congratulating Our Newly Elected EMBS ExCom Members

👤 Nancy Zimmerman 📅 2025-12-14 📖 1 min 📝 18 words EMBS

> **Summary:** The post Welcoming & Congratulating Our Newly Elected EMBS ExCom Members appeared first on IEEE EMBS.

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https://www.embs.org/uncategorized/welcoming-congratulating-our-newly-elected-embs-excom-members/

EMBS AdCom Nominations Open

👤 Nancy Zimmerman 📅 2026-05-04 📖 1 min 📝 51 words EMBS

> **Summary:** Nominate volunteers for terms starting Jan 1, 2027: Asia Pacific Rep (3 yr), Technical Rep (3 yr), Student Rep (2 yr), Young Professional Rep (2 yr). Help shape the future… <a class="continue" href="https://www.embs.org/uncategorized/adcom-nominations-2027/">Continue Reading EMBS AdCo...

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IEEE Pulse Editor-in-Chief Search – Submit an Application!

👤 Deidre Artis 📅 2026-08-28 📖 1 min 📝 17 words EMBS

> **Summary:** The post IEEE Pulse Editor-in-Chief Search – Submit an Application! appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/ieee-pulse-editor-in-chief-search-submit-and-application/

NED-Net: an annotation-training loop for rodent EEG event detection

MIR, M. Y., Ledri, M.

2026-10-21 1 min 200 words BIORXIV NEUROSCIENCE

Summary: Long-term EEG monitoring is central to rodent epilepsy research, and manual review is laborious because recordings span hundreds to tens of thousands of animal-hours. Published automated detectors perform well on their development datasets but are typically fixed models that may not adapt to another...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.15.759362v1?rss=1

Repetitive suprathreshold paired-pulse transcranial magnetic stimulation alters corticospinal excitability

Suresh, S., Liao, W.-Y., Thatcher, T., Villalta, I., Prowse, V., Holden, M. M., Biabani, M., Opie, G. M., Goldsworthy, M. R., Rogasch, N. C.

2026-10-21 1 min 209 words BIORXIV NEUROSCIENCE

Summary: Background: Suprathreshold paired-pulse transcranial magnetic stimulation (TMS) at interstimulus intervals (ISIs) of 20-30 ms evokes cumulative online increases in corticospinal excitability. Whether these changes persist beyond stimulation remains unclear. This study investigated whether repetitive...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.14.759240v1?rss=1

Graph-Based Classification of Penetrating Arteries and Veins in Whole-Mouse Vasculature Datasets

Farina, S., Greco, L., Barra, G., Boyer, M., Renier, N., Schmid, F.

2026-10-21 1 min

246 words

BIORXIV NEUROSCIENCE

Summary: Advances in whole-brain imaging have produced large-scale vascular datasets, yet most lack artery-vein annotations. However, this is essential for quantitative analysis of whole brain vascular structure, blood flow simulation and translational cerebrovascular research. For unlabelled datasets, manua...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.15.758745v1?rss=1

A brain signature of vaping cue-elicited craving

Zhao, S., Yu, H., Koban, L., Wang, Y., Peng, Q., Ye, T., Amlung, M., MacKillop, J., Sweet, L., Liu, J.

2026-10-21 1 min 186 words

BIORXIV NEUROSCIENCE

Summary: E-cigarette use among young adults has grown rapidly, raising concerns about nicotine dependence and motivating a better understanding of the neurobiology of vaping. We used a cue-induced craving fMRI paradigm to develop and evaluate a brain signature of vaping cue-elicited craving, and to compare i...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.15.759290v1?rss=1

Temporal instability weakens entorhinal grid-like coding in older adults

👤 Segen, V., Callau Navarro, G., Bahrd, P., Belge Bernard, T., Schinke, J.-I., Wolbers, T.

📅 2026-10-21

📖 1 min 📝 151 words

BIORXIV NEUROSCIENCE

Summary: Entorhinal grid cell representations are thought to support spatial navigation and episodic memory, functions that decline with age. Their putative fMRI correlate, the hexadirectional signal, varies markedly among older adults, but the source of this variability is unknown. Using virtual navigation ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.15.759328v1?rss=1

LLM-assisted analytical review catches statistical errors that human researchers miss: documented evidence from two computational biology case studies

👤 Chhaya, O., Ananth, V.

📅 2026-10-21 📖 1 min 📝 206 words

BIORXIV NEUROSCIENCE

Summary: Large language models (LLMs) are increasingly used to assist with code generation and data analysis in computational biology, but their capacity to serve as analytical reviewers -- detecting statistical and methodological errors in completed analyses -- is less documented. We present two case studie...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.15.759333v1?rss=1

Social brain fingerprinting: unique functional connectomes underpin interpersonal behaviours in non-autistic and autistic adults

Ielo, A., Czekoova, K., Basile, G. A., Cacciola, A., Brazdil, M., Shaw, D., Amico, E.

2026-10-21

1 min

148 words

BIORXIV NEUROSCIENCE

Summary: Human social interaction requires continuous adaptation while preserving stable individual patterns of brain organisation. This study examined whether functional connectome fingerprinting captures person-specific organisation during reciprocal interaction and whether its spatial architecture relates...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.20.760343v1?rss=1

The human hippocampus retroactively binds experiences across time

He, X., Petzka, M., Khazali, M. F., Brandt, A., Kehl, M. S., Kunz, L., Schulze-Bonhage, A., Staresina, B. P.

2026-10-21 1 min 177 words BIORXIV NEUROSCIENCE

Summary: Creating memories requires linking events across seconds and minutes, but canonical mechanisms of associative plasticity operate over much shorter timescales. This poses a fundamental problem: how can the brain bind information that arrives at different times? Computational model and animal studies ...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.20.760394v1?rss=1

The marginal brain

Lokwani-Di Matteo, F., Sacco, P. L.

2026-10-21 1 min 180 words BIORXIV NEUROSCIENCE

Summary: Goal-directed cognition raises brain energy use only a few percent above resting use, a figure often read as evidence that cognition is metabolically cheap. Models of endogenous precision derive the precision of neural representations from a cost on encoding activity whose unit price is a free param...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.19.760208v1?rss=1

Differential Survival of Retinal Ganglion Cell Types Following Branch Retinal Vein Occlusion

Behboudi Tanourlouee, S., Ren, C., Santeford, A., McCracken, S., Zhao, M., Zhao, C., Apte, R. S., Zhou, C., Williams, P. R.

2026-10-21 1 min 262 words BIORXIV NEUROSCIENCE

Summary: Background. Branch retinal vein occlusion (BRVO) is a common cause of retinal ischemia, rapidly inducing neuronal death and often producing irreversible vison loss. Interventions such as thrombolysis restore perfusion but cannot recover neurons already lost, and most patients retain permanent visual...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.20.760463v1?rss=1

This Week in The Journal

McKeon, P.

2026-10-21 1 min 0 words JOURNAL NEUROSCIENCE THIS WEEK

🔗 **Read full article:**
http://www.jneurosci.org/cgi/content/short/46/42/etwij46422026?rss=1

Encoding of Natural Variability in a Dexterous Motor Skill over Multiple Days in the Cortex of Freely Behaving Mice

de Laittre, E. A., MacLean, J. N.
2026-10-21 1 min 250 words

JOURNAL NEUROSCIENCE CURRENT

Summary: <p>Skilled, goal-directed movements exhibit trial-to-trial variability even in experts, particularly in response to dynamic environmental conditions or when perfect repetition is not required for success. Identifying where, to what extent, and how stably this variability is encoded in the nervous sy...

🔗 **Read full article:**
http://www.jneurosci.org/cgi/content/short/46/42/e1829252026?rss=1

Motor Learning Driven by Sensory Prediction Errors Is Insensitive to Task Performance Feedback

Panthi, G., Mutha, P. K.
2026-10-21 1 min 249 words

JOURNAL NEUROSCIENCE CURRENT

Summary: <p>Accurate motor behavior relies on our ability to refine movements based on errors. Sensory prediction errors (SPEs), arising from discrepancies between expected and actual sensory feedback are a primary driver of motor adaptation. Task performance errors (TPEs), reflecting failures to achieve mov...

🔗 **Read full article:**
http://www.jneurosci.org/cgi/content/short/46/42/e1505252026?rss=1

Conflict-Dependent Gain Control in Area VIP during Visual-Vestibular Self-Motion Processing

Sun, H., Wang, R., Zeng, F., Chen, A.

2026-10-21 1 min 220 words

JOURNAL NEUROSCIENCE CURRENT

Summary: <p>Visual and vestibular signals are continuously integrated to estimate self-motion, yet these sensory cues are often inconsistent under natural conditions. How cortical circuits regulate multisensory interactions under cue conflict while preserving stable heading representations remains poorly und...

🔗 **Read full article:**
http://www.jneurosci.org/cgi/content/short/46/42/e0975262026?rss=1

Regulation of Protein Kinase Pathways in Primary Neuron Model of Anti-NMDA Receptor Encephalitis

Joo, Y., Ciryam, P., Zhang, T., Huang, W., Dean, C. A., Lawrence, A. P., Vu, T. H., Gooya, N., Dessain, S. K., Kane, M. A., Benavides, D. R.

2026-10-21 1 min 229 words

JOURNAL NEUROSCIENCE CURRENT

Summary: <p>The most common form of autoimmune encephalitis is associated with antibodies that target N-methyl-d-aspartic acid receptors (NMDARs). NMDARs play a pivotal role in neurotransmission and synaptic plasticity. Mounting evidence has shown that antibody targeting of the NMDAR GluN1 subunit, as...

🔗 **Read full article:**
http://www.jneurosci.org/cgi/content/short/46/42/e0708252026?rss=1

Molecular Disambiguation of Heart Rate Control by the Nucleus Ambiguus

Jalil, M., MacMillan, M. K., Coverdell, T. C., Gutierrez, V. A., Crook, M. E., Heinrichs, T. J., Dever, A., Cox, K. S., Stornetta, D. S., Wang, Y. B., Schwalbe, D. C., Singh, A., VanderVoort, L., Boychuk, C. R., Abbott, S. B. G., Campbell, J. N.

2026-10-21 1 min 175 words JOURNAL NEUROSCIENCE CURRENT

Summary: <p>The nucleus ambiguus (nAmb) provides parasympathetic control of cardiorespiratory functions as well as motor control of the upper airways and esophagus. A subset of nAmb neurons innervates the heart through the vagus nerve to control cardiac function at rest and during key autonomic reflexes such...

🔗 **Read full article:**
http://www.jneurosci.org/cgi/content/short/46/42/e0398262026?rss=1

Hippocampal-Dependent Navigation without Outcome-Sensitive Control

Wang, S., Grgurich, R., Blair, H. T.

2026-10-21 1 min 247 words JOURNAL NEUROSCIENCE CURRENT

Summary: <p>Goal-directed decisions depend on predictions of future outcomes and are thus sensitive to revaluation of expected outcomes, whereas habitual decisions depend on cached action values or stimulus–response associations and are insensitive to outcome revaluation. Hippocampal neurons encode fut...

🔗 **Read full article:**
http://www.jneurosci.org/cgi/content/short/46/42/e0389262026?rss=1

ASCL1 and OLIG2 Expression Dynamics Control Glial Cell Fate and Regional Diversity in the Dorsal Forebrain

👤 Paez-Beltran, L. E., Chen, H., Villicana, E., Liyanapathirana, M., Myers, B. L., Duan, J., Fletcher, D. E., Esparza, A. C., Riega, A., Vue, T. Y.

📅 2026-10-21 📖 1 min 📝 226 words JOURNAL NEUROSCIENCE CURRENT

Summary: <p>Gliogenesis is a multistep process in which glial progenitors are specified into migrating and proliferating precursor cells that subsequently differentiate into mature astrocytes and oligodendrocytes. How these developmental processes are coordinated to generate the diverse glial lineages across...

🔗 **Read full article:**
http://www.jneurosci.org/cgi/content/short/46/42/e0240262026?rss=1

Dopamine Modulates Striatal Activity When Controlling the Output of Working Memory

👤 Nielsen, N. M., Calcott, R. D., van den Bosch, R., Olraun, H. S., Spronkers, F. S., Verkes, R.-J., den Ouden, H. E. M., Cools, R.

📅 2026-10-21 📖 1 min 📝 178 words JOURNAL NEUROSCIENCE CURRENT

Summary: <p>Cognitive control depends on capacity-limited mechanisms that require selection. Working memory (WM) therefore relies on an input-gate to filter information into WM and an output-gate to transform current WM representations into a prioritized state to guide behavior. According to current theory, ...

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Calcium-Binding Protein Expression Cannot Serve as a General Classifier for GABAergic Neurons in the Macaque Cortex

👤 Brigande, A. M., Krueger, J., Park, C., Disney, A. A.
📅 2026-10-21 📖 1 min 📝 240 words

JOURNAL NEUROSCIENCE CURRENT

Summary: <p>Understanding neuron subclasses and their functional consequences can contribute to understanding brain circuits. A scheme long used to classify GABAergic neurons in the neocortex is based on expression of three calcium-binding proteins (CBPs): parvalbumin (PV), calbindin D-28K (CB), and calretin...

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This Week in The Journal

👤 McKeon, P.
📅 2026-10-21 📖 1 min 📝 0 words

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Improving performance in SNN-based deep reinforcement learning via transition-aware embeddings

Daisuke Miki 2026-10-05 1 min 211 words FRONTIERS NEUROROBOTICS

Summary: Deep Reinforcement Learning (DRL), which integrates reinforcement learning with deep neural networks (DNNs), has been extensively researched across diverse domains. Robotics, in particular, has seen significant advancements, as DRL enables agents to extract meaningful features from high-dimensional ...

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Made an accessibility linter for Django templates, is this useful?

/u/ Living_Dig_6828 2026-10-21 1 min 55 words REDDIT PYTHON

Summary: <!-- SC_OFF --> axe and Lighthouse only check pages you open in a browser, and generic HTML linters choke on <code>{% if %}</code>. What do you use, and what breaks most often - contrast, forms, ARIA? <!-- SC_ON -->   submitted by   <a href="https://www.re...

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https://www.reddit.com/r/Python/comments/1xbtqi2/made_an_accessibility_linter_for_django_templates/

Meta ordered to pay $25B penalty in New Mexico Cambridge Analytica case

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https://canarytokens.org/

Potential third critical PHP CVE

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GitLab Remote Code Execution - Integer Overflow issue

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Google's Rust Style Guide

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After automation: Solo founding will be the default

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Metformin alleviates neuronal injury in MK-801-Induced Schizophrenia-Like mice by inhibiting neuronal ferroptosis

📖 1 min 📝 25 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yanwen Lin, Gaoxia Wen, Xiaoyue Hu, Ran Hu, Yahong Duan, Na Wu, Peng Xiong, Xianlai Jin

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Distinctive patterns of right-hemisphere vulnerability in cerebral small vessel disease: a gray matter asymmetry study

📖 1 min 📝 15 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Jingdian Lu, Yanli Wu, Lei Gao

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N-methyl-D-aspartate receptors in the prelimbic cortex modulate chronic neuropathic pain and comorbid generalised anxiety in a rainbow boa snake threat paradigm

📖 1 min 📝 38 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Priscila Medeiros, Ana Carolina Medeiros, Ohana Turcato Macacare, Weverton Castro Coelho-Silva, Renata Moreira Acunha, Fernando René Bendaña-Córdoba, Farhad Ullah, Sérgio Henrique Ferreira, Renato L...

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Exploratory bone–brain transcriptomic associations and an ODE-based digital-twin framework for fracture healing

📖 1 min 📝 15 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Chao Ma, Wei Lu, Ruipeng Guo

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Resolving motion–position conflict: dorsal–ventral interactions during illusory jitter perception

📖 1 min 📝 16 words NEUROIMAGE

> **Summary:** Publication date: 1 November 2026 Source: NeuroImage, Volume 341 Author(s): Yujie Zhou, Ayumu Yamashita, Ryohei Nakayama, Kaoru Amano

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https://www.sciencedirect.com/science/article/pii/S1053811926005902?dgcid=rss_sd_all

Age-related compression of posterior cingulate-retrosplenial connectivity gradients and its association with allocentric navigation

📖 1 min 📝 22 words NEUROIMAGE

> **Summary:** Publication date: 1 November 2026 Source: NeuroImage, Volume 341 Author(s): Jiali Liu, Yao Zhang, Yingqi Xue, Yuxuan Yang, Haichao Zhao, Qinghua He, Congyuan Liao

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Achieving detailed medial temporal lobe segmentation with upsampled isotropic training from implicit neural representation

📖 1 min 📝 37 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Yue Li, Pulkit Khandelwal, Rohit Jena, Long Xie, Michael Tran Duong, Amanda E. Denning, Christopher A. Brown, Laura E.M. Wisse, Sandhitsu R. Das, David A. Wolk, Paul A. Yushkevich

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Sleep reactivation shapes memories of event and its temporal structure

👤 Zhang, Y., Chen, D., Xia, T., Di, T., Yao, Z., Yue, W. W. Y., Mu, Z., Hu, X. 📅 2026-10-21 📖 1 min

📝 200 words BIORXIV NEUROSCIENCE

Summary: Episodic memory provides a temporal scaffold that organizes past experiences according to their sequential occurrence. While memories are reactivated and consolidated during sleep, it remains unknown how exogenous reactivation reshapes memories of complex events, particularly the temporal structure ...

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https://www.biorxiv.org/content/10.64898/2026.10.20.760365v1?rss=1

Inverse covariation between the neocortex and the limbic system is concentrated at deep evolutionary divergences

Wu, B., Imam, N.
2026-10-21 1 min 199 words BIORXIV NEUROSCIENCE

Summary: The macroscale architecture of mammalian brains consists of a broadly conserved set of structures, but the relative sizes of these structures vary substantially across species. Much of this variation follows conserved scaling relationships, but particular systems can expand or contract relative to t...

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https://www.biorxiv.org/content/10.64898/2026.10.19.760202v1?rss=1

Daytime Napping and Associative-Memory Retrieval Speed: An Exploratory Behavioral and EEG Study

Yang, M., Wang, R., Guan, Y., Zhao, H., Wang, W., Ji, L., Liu, Y., Yan, L., Qu, C.
2026-10-21 1 min 167 words BIORXIV NEUROSCIENCE

Summary: Purpose: To test whether a daytime nap was associated with improved associative-recognition performance relative to monitored wakefulness and to determine where any response-time (RT) difference occurred in the RT distribution. Methods: Fifteen healthy students completed nap and wake sessions in a w...

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Left Temporoparietal Neural Synchrony Tracks Cooperative Performance Across Adult Dyads

Carollo, A., Cavallaro, R., Bizzego, A., Dimitriou, D., Hoehl, S., Esposito, G.

2026-10-21 1 min

268 words

BIORXIV NEUROSCIENCE

Summary: Cooperation requires interacting individuals to coordinate their actions toward shared goals, and interpersonal neural synchrony (INS) has been proposed as a neural correlate of this coordination. However, the relationship between INS and cooperative performance remains heterogeneous, and relatively...

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Dose-dependent myelin g-ratio alterations in the white matter of 16p11.2 copy number variations

Lu, W. D., Martin, C.-O., Jizi, K., Nelson, M. C., Draganski, B., Jacquemont, S., Tardif, C. L.

2026-10-21 1 min

227 words

BIORXIV NEUROSCIENCE

Summary: Copy number variations (CNVs) at the 16p11.2 locus are major genetic risk factors for neurodevelopmental conditions and drive reciprocal alterations in cerebral white matter structure and connectivity. However, whether these alterations reflect differences in the axonal or myelin compartments remain...

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Zero-Shot Generalization of Prospective Planning in Macaques

Seong Bak, M., Zhu, B., Minai, Y., Sakon, J., Okazawa, G., Kiani, R.

2026-10-21 1 min

181 words

BIORXIV NEUROSCIENCE

Summary: Achieving a goal in natural environments often requires a sequence of actions whose consequences extend beyond their immediate outcomes. Because actions within a sequence can hinder or facilitate subsequent actions, successful behavior often demands prospective planning: selecting current actions ba...

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Representational continuity through cognitive reconfiguration of magnitude networks in the human brain

Sirmatel Bakriyanik, B., Balci, F., Ustun, S., Cicek, M.

2026-10-21 1 min 147 words

BIORXIV NEUROSCIENCE

Summary: Whether space, time, and numerosity rely on a common magnitude system remains debated. This functional MRI study offers a resolution by showing that the overlap of their neural substrates depends heavily on the cognitive phase. Forty-five adults performed spatial, temporal, numerical, and color (con...

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PINK1 agonist MTK458 attenuates cerebral edema by protecting blood-brain barrier integrity in intracerebral hemorrhage mice

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Persistent neuronal stress signalling underlies Angelman syndrome phenotypes

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CD51 delineates functionally distinct astrocyte subtypes in neural circuits

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Threshold-free neural network models for swim bout detection of larval zebrafish

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Closed-loop EEG feedback pacing alters task load during digital learning

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Functional architecture of the human insula

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Multiple sources contribute to predictive tactile attenuation in the mouse whisker system

👤 Kalpana GuptaShubhodeep ChakrabartiCornelius Schwarzahttps://ror.org/03a1kwz48Systems Neurophysiology, Werner Reichardt Center for Integrative Neuroscience, Eberhard Karls University Tübingen, Tübingen 72076, Germanybhttps://ror.org/04zzwzx41Systems Neurophysiology, Hertie Institute for Clinical Brain Research, Eberhard Karls University Tübingen, Tübingen 72076, Germanychttps://ror.org/03a1kwz48Graduate Training Center for Neuroscience, Eberhard Karls University Tübingen, Tübingen 72076, Germany

📅 2026-10-20 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceThe brain has evolved multiple interacting neuronal circuits that predict varying aspects of the consequences of one’s own actions. The predicted consequences range from sensory consequences of mov...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2617276123?af=R

Connexins are essential for the contribution of latent progenitors to healing after spinal cord injury

María Victoria FalcoGabriela FabbianiDaniel PrietoMateo VidalMilagros BenítezMaría Inés RehermannMaría Constanza SilveraRenata SimeoneFederico F. TrigoRaúl E. Russoahttps://ror.org/05b50ej63Departamento de Neurofisiología Celular y Molecular, División Neurociencias, Instituto de Investigaciones Biológicas Clemente Estable, Montevideo CP11600, Uruguaybhttps://ror.org/05b50ej63Departamento de Señalización Neuronal y Glial, División Neurociencias, Instituto de Investigaciones Biológicas Clemente Estable, Montevideo CP11600, Uruguay

2026-10-20 1 min 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceEpendymal cells of the spinal cord are latent progenitors whose manipulation is regarded as a potential strategy to improve healing after injury. Therefore, a better understanding of the mechanisms...

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https://www.pnas.org/doi/abs/10.1073/pnas.2605647123?af=R

CX3CL1 (fractalkine)–CX3CR1 interface drives CD8⁺ T cell–induced neurodegeneration in human brain organoids

👤 Woo Sub YangMin Sun ShinMu Seog ChoeFerdi Ridvan KiralJumee KimHong-Jai ParkMario FelixLais OsmaniIn-Hyun ParkInsoo KangaDepartment of Genetics, Interdepartmental Neuroscience Program, Yale Stem Cell Center, Yale Child Study Center, Wu Tsai Institute, Yale School of Medicine, New Haven, CT 06520bDepartment of Internal Medicine, Section of Rheumatology, Allergy & Immunology, Yale School of Medicine, New Haven, CT 06520chttps://ror.org/00vvvt117College of Pharmacy, Sookmyung Women's University, Seoul 04310, Republic of KoreadDepartment of Pathology, Yale School of Medicine, New Haven, CT 06520

📅 2026-10-20 📖 1 min 📝 45 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceNeurodegenerative diseases involve not only intrinsic neuronal pathology but also dysregulated immune responses. However, how peripheral adaptive immune cells directly influence human neurons under...

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